Event Horizon Fund — Independent Deep Dive

Analysis date: November 8, 2026. Price cutoff: November 6, 2026, regular-session close. Investment endpoint: December 31, 2030.

Research revision: November 9, 2026 — AI 2027 / AI Futures / AI 2040 deep audit. The price snapshot and company valuation inputs remain at the original cutoff. This revision adds source-workbook formula checks, linked forecast datasets, updated forecast distributions, an upstream-scarcity portfolio challenge, and explicit regime-risk sensitivity. It changes the interpretation and monitoring requirements materially, but not the recommended stock weights. The pre-revision report is preserved in inputs/research/forecast_audit/pre_revision/.

Independent research proposal prepared from ANALYSIS_PROMPT_TEMPLATE.md, the September 10, 2025 V7.3 foundation, the May 10, 2026 analysis, and fresh public research. This is not an adopted portfolio version, a brokerage statement, or an instruction that has been executed. The active allocation and README have not been changed.

1. Executive verdict

Recommendation: concentrate in cash-generating infrastructure at prices that compensate for the cycle

Choose the **eight-stock concentrated architecture**: NVDA 15%, AVGO 15%, VRT 15%, CLS 12%, NVT 12%, NRG 12%, VST 11%, COHR 8%. Total **100.0%**. Keep Event Horizon at approximately 50% of net investments, leave Foundation and Continuity unchanged, and keep Moonshot at 0%. No options, leverage, shorts, private securities, cryptocurrencies, or ETFs are recommended.

The decisive change is not a prediction that AGI arrives on a particular date. It is a move from owning an inventory of strategic bottlenecks to owning a smaller set of favorable business-price combinations. There is no requirement to hold a foundry, lithography vendor, defense contractor, vaccine developer, uranium miner, or security platform merely because its industry matters. Nineteen of the 24 baseline holdings receive a zero target; three new names enter. Five existing positions remain, all at increased target weights.

The model implies approximately **135% expected cumulative total return**, or **22.8% annualized**, over 4.151 years **conditional on functioning capital markets and the modeled commercial regimes**. The current allocation produces approximately 61.5% and 12.2%; the 15-stock core-and-satellite alternative produces approximately 113.7% and 20.1%. These are subjective scenario calculations, not consensus forecasts, backtested alpha, or assurances of the highest realized return. They do not comprehensively price loss of human control, confiscation, or destruction of enforceable shareholder claims. Section 7 now shows that adding a separate illustrative 10% zero-recovery regime reduces the selected CAGR to 19.8%; 20% reduces it to 16.4%. Those probabilities are sensitivity inputs, not asserted catastrophe forecasts.

Revised thesis judgment. My first report gave the technical forecasting work insufficient weight by grouping it under “stale background.” Its methods and updates deserve serious consideration. The August 2026 AI Futures forecasts make rapid research automation within this investment horizon a material planning case; the authors' track record is directionally informative, but not a verified probability of the whole race narrative. Importantly, software-led takeoff can arrive before another huge hardware buildout, and the fastest technological outcome need not maximize infrastructure equity returns. Sections 3.3–3.7 document the evidence; section 5.6 tests the resulting allocation challenge. Taking the forecasts seriously strengthens both the opportunity case and the case against treating 22.8% as an unconditional return estimate.

The concentrated portfolio's simultaneous bear-state terminal loss is approximately **57%**. A severe interim drawdown could be worse. Taiwan interruption, systemic credit stress, an AI investment bust, or a major failure at one of the largest positions could permanently impair capital. The 15% stock limit means an individual total loss costs 15% of Event Horizon, approximately 7.5% of total net investments before overlapping Continuity exposure. Continuity is another equity portfolio, not a capital guarantee.

What changes the investment decision

1. **Price and fundamentals have diverged.** The hypothetical May baseline lost approximately 12.27% through November 6, despite substantial operating progress at many constituents. VRT is a particularly clear example: a lower price and higher earnings guidance materially improve its valuation. Buying it now is a different decision from paying May's multiple.

2. **The low-multiple challengers deserve real consideration.** CLS combines an ambitious new 2027 earnings outlook with an unusually low multiple; NRG offers a much lower earnings entry price than the nuclear-premium alternatives; NVT provides electrical distribution exposure below several better-known equipment names. Their discounts have identifiable risks, not simply market irrationality.

3. **Memory profits are extraordinary, not demonstrably permanent.** MU's current earnings power is much higher than in May, but the model explicitly normalizes margins and reserves substantial cash. A low peak-cycle P/E is not enough to displace the preferred eight.

4. **The strongest past winner was not evidence for the stated hedge.** MRNA's enormous gain coincided with oncology progress. That does not validate the claim that a small vaccine holding reliably insures against an engineered pandemic. At the new price, the required commercial success is materially greater.

5. **Financing is now a bottleneck alongside power and chips.** Growth capex, GPU replacement, customer concentration, equity issuance, acquisition liabilities, and deposits complicate headline earnings. The neoclouds remain investable common stocks, but none passes the final capital-allocation test here.

Scope, evidence quality, and limits

All 24 existing positions were freshly researched. The numerical model covers **56 stocks**, including every stock in any architecture, at least ten excluded challengers, and additional discounted or strategically relevant names. Section 4 provides at least two alternatives for each of the 18 sleeves, a cross-sleeve screen, complete numerical scenario assumptions, and individual underwriting notes. Section 8 reanswers all 15 strategic questions.

Prices are split-adjusted regular-session closes in USD; historical total returns use adjusted-close ratios as a gross distribution-reinvestment proxy. The public Yahoo chart responses are saved locally, with timestamps and source URLs. Selected important quotes were cross-checked against other public histories, including CLS, NVDA, MRNA's issuer quote, and NBIS. Price observations are November 6, not November 8 weekend prices or overnight quotes.

Unavailable: actual account holdings, transactions, cost bases, tax lots, liquidity needs, archived point-in-time analyst consensus for every ticker, a complete historical valuation-percentile database, independently verified customer-level AI economics, and objectively established probabilities of AGI, war, or a pandemic. Published subjective AI probability distributions do exist and are now explicitly analyzed; absence of objective calibration is not a reason to ignore them. Nothing below substitutes invented consensus for those gaps. Management guidance, annualized quarterly run rates, and this report's own forecasts are labeled separately. Any source published before October 9 is marked **STALE (>30 days)** when used as a current operating anchor; the latest available quarterly filing can still be the correct as-of source. No November 9 or later earnings are used.

The expected-return ranking is conditional on the assumptions, universe, and risk limits used here. It is not a mathematical proof of global portfolio optimality. Taxes and implementation costs can materially alter a taxable investor's best transition; the deployment plan requires those checks before orders.

2. Prior-cycle accountability and what changed since last cycle

2.1 Measurement: a reproducible model portfolio, not claimed personal performance

May 10 and November 8 were Sundays. Use May 8 and November 6 closes. Assume an investment of $100 at May 8, with the supplied V9.1 weights, no contributions, no withdrawals, and no rebalancing. For holding i, contribution in percentage points equals its initial weight times its adjusted total return. Ending weights drift with returns. This convention deliberately evaluates the stated baseline, not the obsolete intermediate V8.3/V9.0 tables embedded in the May document.

The assumption that V9.1 existed at the May 8 close is a counterfactual implementation convention: the May report was written afterward. It is useful for accountability but is not an investable same-day backtest. A real May 11 execution would produce different numbers. Dividend withholding, fees, ADR charges, taxes, and investor cash-flow timing are not included.

Ticker	May weight %	May 8 close $	Nov 6 close $	Price return %	Total return %	Contribution pp	Drift weight %
TSM	11	411.68	375.44	-8.80	-8.34	-0.92	11.49
NVDA	6	215.20	186.64	-13.27	-13.07	-0.78	5.95
AVGO	3	430.00	293.69	-31.70	-31.47	-0.94	2.34
MU	6	746.81	690.00	-7.61	-7.58	-0.45	6.32
ASML	8	1592.02	1424.75	-10.51	-10.27	-0.82	8.18
AMAT	4	435.44	332.72	-23.59	-23.41	-0.94	3.49
SNPS	5	516.48	450.46	-12.78	-12.78	-0.64	4.97
ANET	4	141.77	170.80	+20.48	+20.48	+0.82	5.49
VRT	6	339.97	182.86	-46.21	-46.19	-2.77	3.68
DELL	2	260.46	328.53	+26.13	+26.52	+0.53	2.88
CEG	9	303.63	249.95	-17.68	-17.42	-1.57	8.47
VST	6	147.72	124.80	-15.52	-15.26	-0.92	5.80
PWR	4	745.00	540.90	-27.40	-27.37	-1.09	3.31
CRWD	3	131.94	216.90	+64.39	+64.39	+1.93	5.62
PLTR	2	137.80	160.38	+16.39	+16.39	+0.33	2.65
AVAV	1	168.29	112.76	-33.00	-33.00	-0.33	0.76
KTOS	1	57.89	35.10	-39.37	-39.37	-0.39	0.69
MRNA	1	54.35	254.19	+367.69	+367.69	+3.68	5.33
MP	3	67.43	35.41	-47.49	-47.49	-1.42	1.80
INTC	3	124.92	56.28	-54.95	-54.95	-1.65	1.54
GEV	5	1040.15	774.11	-25.58	-25.50	-1.27	4.25

Ticker	May weight %	May 8 close $	Nov 6 close $	Price return %	Total return %	Contribution pp	Drift weight %
CCJ	2	116.75	71.30	-38.93	-38.93	-0.78	1.39
BE	3	261.03	173.24	-33.63	-33.63	-1.01	2.27
COHR	2	335.26	192.81	-42.49	-42.49	-0.85	1.31

Checksum: starting weights 100%; unrounded contributions sum to **−12.2714 percentage points**; ending modeled wealth $87.7286 per initial $100. Rounded displayed contributions need not sum perfectly. A price-only version is slightly worse because it excludes distributions.

For the same dates, the adjusted-close benchmark returns are approximately SPY **+0.19%**, QQQ **−7.55%**, SMH **−18.11%**, XLU **−2.23%**, ITA **−7.23%**, and IBB **+25.73%**. Thus the baseline lagged the broad market by approximately 12.46 points and QQQ by 4.72 points, while beating the semiconductor ETF by 5.84 points. The benchmark comparison does not turn a loss into success or establish risk-adjusted alpha. Sources and raw data are in research/market_snapshot.json and the individual chart files.

MRNA contributed +3.68 points and CRWD +1.93; together they offset 5.61 points of weakness elsewhere. VRT cost 2.77 points; INTC, CEG, MP, and GEV were other large detractors. The small initial MRNA position would have drifted to approximately 5.33%, illustrating why stale target weights must not be mistaken for current account weights.

2.2 Operating revisions, valuation changes, and decision quality — every holding

“Revision” in this table means a verified management-guidance or reported-operating update, **not** an unavailable archived consensus revision. May anchors are the prior report's recorded information, not retroactively substituted current data. Changes in fiscal period, acquisition scope, currency, and share count prevent a clean numerical attribution for many names.

Ticker	New evidence available by Nov 8; source publication date	Valuation change / return explanation	Was May right for the right reasons?
TSM	Q3 ADR EPS $4.60; 2nm has entered the revenue mix. Oct 15 release.	Price fell 8.8% despite operating growth. Roughly 20.4× annualized Q3 EPS is a run-rate proxy, not next-year consensus.	Foundry scarcity was right; treating indispensability as adequate valuation analysis was not. Taiwan risk remains uncompensated by a guaranteed hedge.
NVDA	Q2 FY27 revenue $96.2bn; Q3 revenue guide $108bn. Aug 26 release — STALE.	Price fell 13.3%; about 21× annualized Q2 adjusted EPS, versus much weaker operating scale in May. Fundamental growth did not produce a positive stock return.	Correct business thesis, insufficient allocation to the best improving business-price combination now. No claim that the May trim was irrational with May information.
AVGO	Q3 revenue $29.6bn, AI revenue $16.7bn, adjusted EPS $3.32. Sep 2 — STALE.	Approximately 22.1× annualized quarterly adjusted EPS; a 31.7% price fall alongside faster AI sales indicates a large valuation reset.	Custom silicon confirmed; the old 3% weight now has a high opportunity cost. Customer concentration still matters.
MU	Q4 revenue $54.2bn; next-quarter adjusted EPS guide $38.15. Sep 30 — STALE.	About 4.5× annualized next-quarter guided EPS versus about 9.7× the May quarter-guidance annualization. These are different quarters, not a same-year consensus revision.	HBM scarcity confirmed spectacularly; “decyclicalized” remains unsupported. The normalization question dominates today's apparent cheapness.
ASML	FY sales guide €44–45bn versus May €36–40bn. Oct 14.	Price fell 10.5% while the revenue guide rose. Exact historical normalized P/E unavailable; euro earnings require FX conversion.	Lithography moat correct; even stronger demand does not eliminate premium-price risk.
AMAT	Q3 adjusted EPS $3.50 and revenue $9.115bn. Aug 13 — STALE.	About 23.8× annualized quarterly adjusted EPS after a 23.6% decline; demand and price moved in opposite directions.	Broad process intensity correct, but LRCX now offers a stronger modeled return.
SNPS	FY adjusted EPS midpoint about $15.07 versus May $14.42. Aug 26 — STALE.	Same-fiscal-year guide +4.5%; guide-based P/E approximately 35.8× to 29.9×. Acquisition growth is not all organic growth.	EDA durability confirmed; the valuation reset is real, but not large enough to win final capital.
ANET	Q3 revenue $3.658bn, +58.5% year over year. Nov 3.	Price +20.5%; significant fundamental growth supports some appreciation. Exact fundamental/multiple decomposition unavailable without matched estimates.	Ethernet thesis confirmed. Any blanket inference that supply pressure meant structural deterioration was too pessimistic.

Ticker	New evidence available by Nov 8; source publication date	Valuation change / return explanation	Was May right for the right reasons?
VRT	FY adjusted EPS midpoint $7.15 versus $6.35 in May. Oct 28.	EPS guide +12.6%; guide P/E approximately 53.5× to 25.6×. Price −46.2% was overwhelmingly de-rating relative to this guide.	The May valuation warning was correct; retaining a substantial position still hurt. Today's entry economics differ markedly.
DELL	FY27 revenue guide $192bn and adjusted EPS $25.50; large AI backlog. Sep 1 — STALE.	Price +26.1%, now 12.9× guided EPS. Strong earnings expansion, not simply thematic re-rating. Working capital weakens the earnings-to-cash bridge.	May's structural integrator caution was reasonable; its low weight missed a strong operating/stock outcome. Do not dismiss it without underwriting.
CEG	FY adjusted EPS $11.50–12.50 versus $11–12 in May. Aug 6 — STALE.	Same-year midpoint +4.3%; guide P/E approximately 26.4× to 20.8×. Next report is Nov 9, outside this cutoff.	Existing nuclear's scarcity remains; premium concentration was excessive relative to cheaper power alternatives.
VST	FY26 EBITDA midpoint $7.3bn; initial FY27 midpoint $8.1bn. Nov 6.	Lower stock price, higher operating outlook; raw GAAP P/E is distorted by hedge marks. Acquisition scope changed.	Directional preference for VST over CEG was justified; the stock still lost money. Use after-growth cash and net debt, not EBITDA alone.
PWR	FY adjusted EPS midpoint $18.10 versus $13.90; backlog $63.7bn. Oct 29.	Guide +30.2%; guide P/E approximately 53.6× to 29.9× despite a 27.4% price decline.	Construction bottleneck correct; May paid too much for it. Not an argument that PWR's business failed.
CRWD	Q2 FY27 revenue growth about 26%; positive but small GAAP profit; 4-for-1 split. Aug 26 — STALE.	Split-adjusted return +64.4%, much faster than revenue growth. Expectation expansion is material; exact matched-multiple attribution unavailable.	Security demand correct, but appreciation does not establish attractive future returns at the new market capitalization.
PLTR	FY revenue guide approximately $8.532bn versus May $7.65bn. Nov 2 SEC release.	Revenue guide +11.5%, price +16.4%; improved economics explain part, not all, of the stock move. SBC remains an ownership cost.	Enterprise/government adoption confirmed. Extraordinary execution still needs an extraordinary terminal franchise to support today's price.
AVAV	FY27 adjusted EPS guide $3.02–3.34; Q1 revenue +6%, funded backlog $1.5bn. Sep 9 — STALE.	About 35.5× current fiscal-year guide after the fall. Purchase accounting, integration, and dilution complicate comparability.	Military autonomy demand was right; integration economics and price discipline were not adequately protected.
KTOS	FY26 FCF use $110–120m despite revenue growth; FY27 EBITDA guide $230–240m. Nov 5.	Price −39.4%; still dependent on future margin and cash conversion. Lower price is not automatically a discount to owner earnings.	Contract potential confirmed, cash-generation thesis not yet. Unmanned revenue weakness prevents a blanket “CCA win” conclusion.
MRNA	Positive phase-3 oncology results since May; Q3 vaccine revenues and liquidity update. Nov 6 earnings, Aug 19 trial release — STALE.	Price +367.7% reflects a transformed oncology opportunity, not a demonstrated pandemic insurance payoff. Financing contributes to cash balances.	Excellent realized contribution, substantially different reason from the proposed hedge. Re-underwrite rather than extrapolate.
MP	Q3 revenue $147.7m, adjusted EPS $0.04; customer magnet qualification advanced. Nov 5.	Price −47.5%; low current earnings, policy support, one-off sales, and future magnet margins dominate valuation.	Domestic rare-earth chain thesis progressed. It was never a direct gallium/germanium hedge or a guaranteed equity floor.
INTC	Q3 adjusted EPS $0.52; Q4 guide $0.25; 14A PDK progress. Oct 29.	Price −54.9% but roughly 54× a Q4 annualized earnings guide remains demanding. Foundry spend and dilution are decisive.	The “Taiwan insurance” framing failed: US support does not guarantee profitable external foundry manufacturing or equity returns.
GEV	Backlog $197bn; FY FCF guide $16–17bn. Oct 28 SEC release.	Price −25.6%; customer deposits make headline FCF an unsafe recurring-earnings denominator.	Turbine and grid scarcity confirmed; “monopoly” was incorrect and deposit cash must not be capitalized indefinitely.

Ticker	New evidence available by Nov 8; source publication date	Valuation change / return explanation	Was May right for the right reasons?
CCJ	Production range narrowed to 19.5–20.5m pounds; contracted deliveries remain important. Oct 30.	Price −38.9%; spot uranium is not equivalent to realized contract revenue. Canadian reporting also creates FX comparisons.	Fuel security remains real; near-term supply/outage and valuation risk were understated.
BE	FY revenue guide $4.4–4.6bn versus May $3.4–3.8bn. Oct 22.	Guide midpoint +25%; price −33.6%. Still a high price for the diluted earnings and cash ultimately available to common owners.	Faster deployment thesis confirmed; economics do not establish permit-free, fuel-independent, or low-risk power.
COHR	Q1 FY27 revenue $2.37bn; adjusted EPS $2.09 versus May's $1.41 quarter. Nov 4 SEC release.	Annualized-quarter adjusted P/E approximately 59.4× to 23.1×; different fiscal quarters, not a same-period consensus comparison.	Optical demand confirmed, May entry valuation failed. New capex and cash-consumption risks remain despite the improved price.

For the four reasonably comparable annual-guide examples, use the identity P1/P0 = (EPS1/EPS0) × (PE1/PE0). VRT's 0.538 price ratio is approximately 1.126 times 0.478; PWR's 0.726 is approximately 1.302 times 0.557; SNPS's 0.872 is approximately 1.045 times 0.834; CEG's 0.823 is approximately 1.043 times 0.789. These are guide-based decompositions, not claims about the market's unobservable exact valuation mechanism. For the other names, “mostly fundamentals” or “mostly expectations” is a reasoned interpretation with lower precision.

2.3 New evidence versus changed interpretation

New evidence includes the subsequent earnings releases, changing price denominators, CLS's 2027 outlook and equity raise, NVT's October acquisition completion, the proposed MRVL/LITE combination, MOD's spin-off, MRNA's oncology readout, the latest power-company guidance, and the evolving financing needs of cloud builders. The exact allocation changes are in Section 6.

The criticisms of automatic bottleneck ownership, MRNA-as-insurance, and INTC-as-Taiwan-insurance are **analytical corrections**, not newly discovered events. Likewise, buyer firms were never disqualified by being AI customers. Continuity ownership creates overlap to measure; it is not a reason to exclude MSFT, GOOGL, AMZN, or META from competition.

Do not score May decisions by hindsight alone. The right accountability question is whether the decision used a defensible estimate of future owner cash at the price then paid. A thesis can be operationally right and financially wrong, and a poorly specified thesis can make money by luck or through a different catalyst.

3. Zero-based portfolio test and bottleneck map

3.1 Start without sleeve entitlements

The blank-slate process is: identify scarce inputs; determine who can retain economic rents; translate operating scenarios into fully diluted per-share value; compare expected terminal wealth at today's price; then apply permanent-loss constraints. Only afterward compare the result with V9.1. The initial universe included the prior watchlists, hyperscalers, equipment competitors, electrical suppliers, optics specialists, neoclouds, data/identity platforms, and non-AI alternatives.

The candidate ranking is not based on recent winners or on each stock's old portfolio weight. The selected eight would be the same starting from cash under these assumptions. In particular, TSM and ASML receive zero despite exceptional strategic positions; MRNA receives zero despite being the largest contributor; and VRT receives the maximum default weight despite being the largest detractor. These outcomes distinguish zero-based capital competition from thesis-preserving rebalancing.

3.2 Complete dependency chain: scarcity is not the same as shareholder rent

Layer	What actually constrains delivery	Rent-capture test and substitutes	Public candidates / investment conclusion
Customer economics and finance	Paying workloads, counterparty credit, financing commitments, utilization	Are end users paying more than compute's full replacement cost? Deposits and backlog are not independent proof of profitable demand.	Hyperscalers; CRWV, NBIS, IREN, CORZ, APLD. Underwrite financing before MW. No selected neocloud.
Site and permissions	Usable land, zoning, water, air permits, local acceptance	A speculative queue position is not an energized site. Curtailable access and geographic relocation can substitute for new firm capacity.	DLR, EQIX, infrastructure builders. Scarcity real; equity return depends on cost of capital.
Firm electrical service	Interconnection, transmission, substations, switchgear, protection, transformers	Local capacity and delivery date matter more than national annual energy totals. Regulation can transfer rents to customers.	NVT, VST, NRG selected; ETN, GEV, PWR, HUBB, POWL alternatives.
Generation and fuel	Existing dispatchable plants, gas delivery, outages, turbine slots, nuclear licensing	Existing firm assets can monetize sooner than unbuilt reactors. Gas, renewables plus storage, efficiency, and flexible load are partial substitutes.	VST/NRG selected. CEG, TLN, BE, CCJ remain competitive or conditional, not mandatory.
Leading-edge wafers	Qualified process capacity, yields, equipment installation	Almost no near-term replacement for a qualified leading node, but customers and governments can limit economics.	TSM; GFS and INTC are not equal substitutes. No direct foundry holding at current modeled returns.
Semiconductor equipment	EUV, deposition, etch, inspection, metrology, materials	Installed base and service differ from new-system cyclicality; customer capex and export exposure remain linked.	ASML, AMAT, LRCX, KLAC, ASM, ONTO, ENTG. Best excluded WFE challenger: LRCX.
Packaging, substrates, test	CoWoS-class advanced packaging, bonding, ABF substrates, thermal materials, test capacity	A single delayed component can strand costly GPUs; capacity expansion can rapidly compress scarcity premiums.	TSM, AMKR, BESI, KLIC, Ajinomoto; no automatic small sleeve.
HBM and conventional memory	Qualified HBM stacks, yields, DRAM wafer allocation, bandwidth	Qualification creates switching costs; high margins invite capacity and architecture changes.	MU, SK hynix, Samsung. MU fully modeled with margin normalization and accumulated cash.
Accelerator systems and software	GPU/custom-ASIC supply, software compatibility, networking and racks	A chip's theoretical FLOPS are not useful throughput. Software, developer adoption, reliability, and time-to-production support rents.	NVDA and AVGO selected; AMD, ARM, MRVL, hyperscaler ASICs are real competitors.
Scale-up / scale-out networks	Switch silicon, NICs, retimers, cabling, optical links, congestion and failure rates	Copper, active electrical cables, pluggables and co-packaged optics solve different distances and power budgets.	AVGO and COHR selected; ANET, LITE/MRVL, CRDO, ALAB, FN, GLW, CIEN screened or modeled.
Rack integration and thermal management	Qualified power delivery, CDUs, liquid loops, pumps, controls, heat rejection, commissioning	Cooling capacity without qualified deployment/service is insufficient. Competing suppliers and customer redesign can erode margins.	VRT and CLS selected; DELL, HPE, MOD, FLEX, SMCI assessed.
Design and verification	EDA, multiphysics, IP, design closure, process qualification	AI design tools may increase output but also reduce seat-based revenue; verification liability protects some incumbent value.	SNPS/CDNS; durable businesses, not highest current expected returns.
People and execution	Electrical trades, project management, procurement, safety, field maintenance	Backlog converts to shareholder value only after labor, working capital, rework and guarantees.	PWR, MYRG, MTZ, service networks. PWR modeled but zero target.
Data, trust and operations	Rights-cleared data, reliable agents, access control, audit, security, enterprise change	Generic information is not a monopoly. Proprietary updates, permissioning, liability and distribution matter.	RELX, VRSK, MSFT, PANW, CRWD, PLTR, NOW. None selected solely to complete the map.

Layer	What actually constrains delivery	Rent-capture test and substitutes	Public candidates / investment conclusion
Resilience and policy	Critical-material processing, trade rules, physical security, biosafety and sovereign capability	A social need need not produce investable excess returns; state support can favor customers over shareholders.	MP, defense names, MRNA, INTC. Risk monitored without pretending these stocks are comprehensive insurance.

3.3 Revised research judgment: serious forecasting work, with distinct conditional claims

What was inadequate before. The first version reviewed the headline background sources, but did not audit the linked compute workbook or trace the evolving forecast distributions deeply enough. Calling the whole body of work "stale background" obscured substantive quantitative methods and post-May updates. This revision corrects that. An old forecast remains valuable as an ex-ante prediction even when it is too old to serve as a current operating observation. Conversely, a recently edited web page is not necessarily a new forecast.

The expanded review covers the requested race narrative, summary, compute, timelines, takeoff, goals and security supplements; the shared compute workbook; AI 2040's main argument, full scenario PDF, compute and economics supplements; and the trillion-dollar-cluster argument. Material side links include the authors' January/February, April and August 2026 updates; two grading workbooks; the underlying August forecast comparison table; METR measurement/productivity work and raw time-horizon data; and Epoch chip-sales and company-revenue datasets. This is a targeted audit of investment-relevant dependencies, not a claim to have independently replicated every simulation or verified every linked anecdote.

Four propositions must be evaluated separately:

Proposition	Revised assessment	What it does not establish
AI capabilities and useful coding agents can advance rapidly	Substantial empirical support; material acceleration belongs inside planning for 2027–2030	Full autonomous research, reliable general agency, or a particular takeoff date
AI-assisted research can create an unusually fast feedback loop	A technically argued, consequential hypothesis; not dismissible merely because factories take years	That every form of progress scales with installed hardware or electricity
Demand can fund a very large physical buildout	Increasingly supported by spending, revenues and supplier results, but financing and utilization remain constraints	That announced capacity is commissioned, or that all spending is additive
Today's public suppliers capture durable excess profits	Must be established company by company at today's price	That social importance, GDP growth or a correct AI timeline guarantees stock outperformance

The first three strengthen the investment opportunity. They do not dispense with the fourth. The appropriate response is to underwrite rapid progress, changing bottlenecks and adverse governance outcomes explicitly—not to assume that "fast AI" means every incumbent supplier deserves a higher multiple.

3.4 Track record and current forecasts: what has actually been tested?

A meaningful but uneven record. The authors' February 12 scorecard reported quantitative progress around 65% of the scenario's pace, subsequently noting roughly 75% in July. Their August 16 reassessment reports approximately **70–90%**, depending on metrics and aggregation. This supports taking their direction and timescale seriously. It is not a 70–90% accuracy rate, calibrated probability of AGI, or probability that the race ending occurs. The August comparison also changes the set of predictions being graded, so the improvement is not a clean fixed-cohort longitudinal score. February grading and methodological explanation, August update.

I read the August grading workbook's populated Aug 2026 grades!A1:K27, including formulas, rather than relying only on its chart. Its category results and the investment interpretation are:

Category	Workbook pace multiplier	Evidence quality / limitation	Investment reading
Coding time horizon	0.885 category mean; 1.05 against corrected central trajectory, 0.72 against the originally displayed erroneous curve	Same observed capability measured against two trajectories, not two independent observations	A strong reason not to dismiss rapid coding progress; retain both comparisons rather than silently replacing the original
AI software R&D uplift	0.335	Internal/public uplift is estimated using judgment and the authors' model; not an independently observed whole-lab productivity series	The feedback loop's most important link is less well verified than the compute buildout
Economic value	1.065	Revenue pace 1.43; valuation pace 0.70; private marks and annualized revenue are imperfect measures	Commercial demand progressed strongly, without an equally strong claim about shareholder valuations
Compute	1.081	Mixes estimated training FLOP, spending, stock and lab budgets; some inputs use secondary sources and sparse interpolation	Strong buildout evidence, but not four independent measurements of intelligence
Public salience	0.10	Survey threshold, small changes and baseline choices make this a noisy proxy	A weak direct input to supplier earnings; potentially important to policy timing
All categories	Mean 0.693; median 0.885	Weighting and sample composition change the headline	These reproduce the authors' roughly 70–90% summary; neither is a formal forecasting skill score

Source: August grading workbook. Own recomputation is saved in forecast_audit/audit_results.json.

Two formula observations matter to the audit, although neither overturns that summary. Cell I7 displays about $352bn from an apparent tenfold scaling discrepancy; the narrative's interpolated quantity would be about $35.2bn. C7 instead uses F7/G7, so the displayed discrepancy does **not** drive the revenue pace score. F12 interpolates a lab compute budget using 3.2E+28, whereas I12 uses 3.2E+27 and the surrounding checkpoints indicate the latter. Using the latter consistently changes that row's pace from 0.84 to about 1.02 and the overall category mean from 0.693 to about 0.702. These are analyst consistency checks, not edits to the source or author-confirmed corrections. Some resolution notes explicitly acknowledge limited data and model-assisted interpolation. That warrants verification, not wholesale dismissal.

Current distributions, not just the title year. The underlying August forecast comparison table distinguishes a most-likely year from a median and from model output at median parameters. It reports these *all-things-considered* judgments, conditional on technically feasible progress without an imposed slowdown:

Forecaster / milestone	April 2025 publication-era view	August 2026 view	Interpretation
Daniel: probability of TED-AI by end-2027	40%	23%	Serious early-arrival probability, not a majority forecast of that exact year
Eli: probability of TED-AI by end-2027	Approximately 20% in the publication's broader-author comparison	9%	Author disagreement is material; do not present one consensus probability
Daniel: TED-AI median	2028	2028	Still within this portfolio horizon
Eli: TED-AI median	2031	2032	A materially slower view remains inside the forecasting team

Forecaster / milestone	April 2025 publication-era view	August 2026 view	Interpretation
Daniel / Eli: ASI median	Daniel approximately 2028–29; precise original median not recorded	2029 / 2033	Faster and slower paths must both be considered
Daniel / Eli: superhuman-coder median	2028 / 2030	2028 / 2031	Stronger than ordinary coding assistance and not identical to the newer AC milestone

Source: native values in forecast comparison workbook, Aug 2026!A1:D6. Some table hyperlinks still point to older comparisons; the displayed August values are quoted here as the table's stated judgments, not silently substituted with those older links. TED-AI means top-expert-level or better performance across virtually all cognitive tasks, not “an impressive chatbot.”

The updated model's AC medians are December 2027 for Daniel, July 2028 for Brendan and March 2029 for Eli; these are **model medians**, not interchangeable with the table's broader judgments. The newer model adds coding-uplift and revenue anchors and explicitly accounts for training time when incorporating algorithmic improvements. Three agreeing extrapolations are encouraging, but share assumptions and data; they are not three independent demonstrations. The no-slowdown conditioning is crucial for investment use. Model and limitations, August methodological changes.

Independent cross-checks and dates. METR's January 29 TH1.1 revision found a faster post-2023 trend than its earlier suite, but also documents task-composition sensitivity and sparse measured human baselines for long tasks. Its raw file gives Mythos Preview a roughly 186-minute 80%-success horizon and 1,045-minute 50%-success horizon, with wide intervals; these are not autonomous weeks of dependable employment. The dashboard was last updated September 8 and is no longer actively maintained. A later retrieval is not a November capability measurement. TH1.1 methodology, raw measurements, dashboard limitations.

Do not use METR's early-2025 slowdown result as if it settled late-2026 productivity. Its February 2026 follow-up explains selection problems as developers became less willing to work without AI. Its May 11 survey reports median self-assessed value gains of 1.4–2× among 349 technical workers, while warning about self-report and sampling biases. This is genuinely post-May-cycle evidence of perceived usefulness, not proof of a 2× acceleration of the entire research process. Follow-up experiment, May survey. All these observations are **STALE (>30 days)** as current capability anchors, but relevant to evaluating the trajectory.

My assessment: the user is right that the forecasts deserve substantially more analytical weight than the first version gave them. The evidence supports a serious rapid-progress case. It does **not** validate every original calendar event, every internal-productivity estimate, or the complete 2027–2030 narrative. No independent, comprehensive proper-scoring-rule evaluation of this entire scenario was established in this audit. Forecaster reputation is a reason to examine the work closely, not a substitute for checking its load-bearing claims.

3.5 Deep audit of the shared AI chip compute workbook

Scope and vintage. The user-linked workbook is “[shared 2024-10-02] AI chip compute forecast,” Clean version, gid 1866551567. I read its metadata, displayed values through row 120, and formulas/cached values/notes/links across B5:O78; the populated model ends at row 78 in those reads. There is one exposed tab. Drive metadata reports modification on April 23, 2025, so the title's October 2024 date does not certify that every current cell was frozen then. The workbook explicitly conditions on **high demand**. Its outputs are capacity scenarios, not observed future spending or a probability distribution. Source snapshots are preserved; the online sheet was not modified.

How its supply calculation actually works. Leading-edge wafers, HBM supply and advanced packaging are converted into chip-equivalent capacities. Production is their minimum; chip performance then converts units into annual compute and cumulative available compute. Revenue is separately inferred from unit production and pricing assumptions. The global system is not modeled as “all bottlenecks grow independently and then multiply together.” Crucially, recomputation shows **HBM is the minimum in every year 2024–2030**, with logic nearly binding by 2030. More packaging alone does not increase output while HBM remains the lower ceiling.

Quantity / row	2026	2027	2030	Meaning
Produced accelerator units, row 46	11.89m	20.85m	92.18m	Model's package/chip units; not automatically comparable across generations
Implied chip market, row 47	$396bn	$695bn	$3,073bn	Annual chip-market dollars, not all datacenter capex
Implied HBM market, row 24	$20.6bn	$40.7bn	$306.4bn	Sensitive to quantities, yields, stack assumptions and price compounding
Cumulative raw FP16 H100-equivalent capacity, derived from row 51	55.7m	180.8m	1,842.4m	Annual compute capacity divided by seconds/year and 10^15 FLOP/s; no retirement
Global peak power, row 54 with unit conversion	36.5 GW	77.0 GW	434.7 GW	A model output, not a current grid connection survey
One leading lab's largest-training-run power, row 71	5.65 GW	15.88 GW	89.60 GW	Different scope from global AI power
Palo Verde comparisons, row 75	1.59	4.47	25.24	Training-run energy / half-year Palo Verde generation; not global reactor requirements

Numbers are own extraction/recalculation from the named cells, rounded for presentation. The separate published compute supplement has different estimates; combining its narrative with this workbook's totals without reconciling versions creates false precision.

Audit issue	Exact dependency	Consequence and treatment
High-demand conditioning	D1 and production minimum in row 46	Supply that could be produced is not proof customers will finance and use it. Add demand, financing and commissioning tests outside this workbook.
HBM wafer expansion dominates	G18 onward multiplies prior-year wafer volume by 1.6; row 22 applies yield, units/wafer and stacks/chip	Lowering only wafer growth from 60% to 30% annually cuts 2030 output from 92.18m to 26.52m units, about 71% lower. This is sensitivity, not a new central forecast.
HBM price escalation is stronger than a flat 3%	G24 uses prior TAM × capacity growth × 1.03^2; H24 uses 1.03^3, continuing through 1.03^7	Because these multiply the previous year's result, exponents accumulate. Keeping quantities and the 2024 anchor but using 3% price growth each year yields about **$164.7bn**, not $306.4bn, in 2030. Increasing price inflation could be intentional; the distinction must be explicit.
Packaging AI share is applied twice	Row 37 already equals row 35 × row 36; row 46 again uses row 37 × row 36	A mechanical double discount. It does not change current outputs because HBM binds, but can distort packaging sensitivities.
Chip / reticle definition changes	F46 includes 814/F5; later columns omit an equivalent size normalization, while N46 describes H100-sized reticles	Later values behave like assumed multi-die packages. Do not use row 46 as a uniform physical-die series without resolving the definition.
Fixed HBM vendor shares	Rows 25–27 hold SK Hynix/Samsung/Micron at 45/45/10 from 2025 onward	Not a current market-share finding and not a Micron earnings forecast. MU requires its own share, pricing, capital intensity and dilution analysis.
Stock converted at newest-generation efficiency	Row 54 divides cumulative compute by the current column's efficiency	Older chips do not automatically acquire newer power efficiency. A shipment-cohort sum using the sheet's own per-unit watts gives about 92.6 GW in 2027 and 528.0 GW in 2030 for 2023-onward shipments, versus 77.0/434.7 GW in row 54. This comparison excludes pre-2023 equipment, retirement and added facility overhead; it isolates the vintage issue, not a validated power forecast.
Cumulative stock never retires	Row 51 adds prior stock and current production	Useful for a short aggressive expansion case, increasingly problematic by 2030. Retirement and stranded/uncommissioned hardware can lower usable stock. This partly offsets, but does not repair, the power-efficiency issue.
Stock/flow mismatch in apparent value for money	Row 55 divides cumulative compute by current-year chip spending	Not the price-performance of newly purchased hardware. Use current-year production FLOP/s divided by current-year dollars for that comparison.
Unit and utilization labels	Rows 54/71 contain kW numerically but display GW; row 72's formula is per kW despite a per-W label; row 64 uses 10% utilization while a note references 40%	Normalize units and use actual formulas. Row 74 compares a training load with US nameplate capacity, not annual US electricity production. Do not turn it into an energy-share claim.

The old HBM anchor also needs updating; formula checks are not all bearish. Row F24 starts at approximately $6.69bn for 2024, whereas Micron's December 18, 2024 prepared remarks described a $16bn 2024 market. Thus the $164.7bn flat-price-escalation sensitivity is emphatically **not** a current 2030 TAM estimate. Updating the starting market, product mix, prices and vendor shares could raise the result even while correcting compounding. Micron's September 30, 2026 prepared remarks — STALE report substantial 2027 HBM price increases and 26 strategic agreements estimated to cover over 35% of revenue through 2030. Three-quarters of that agreement revenue has a defined pricing framework, often with floors and ceilings. This is stronger evidence for durability than the old sheet's fixed 10% Micron share, but neither management's demand outlook nor contracts with variable pricing guarantee a permanent peak margin. The upstream operating sensitivity in Section 5.6 takes the possibility of stronger rents

seriously.

The power comparison needs a correction in inherited thesis framing. This workbook's greater-than-nine-Palo-Verde figure occurs in **2028**, not 2027, and concerns one modeled lab's frontier training run. The 2027 number is about 4.47. Neither is a count of nuclear plants that must actually be built, and Palo Verde is a multi-reactor station. These distinctions change interpretation even though the underlying power requirement is still very large.

What survives the audit. HBM, memory bandwidth, advanced packaging, leading-edge manufacturing and power deserve explicit treatment as coupled constraints. The workbook exposes exactly why a shallow “more compute” extrapolation is insufficient. Its 2030 leading-edge wafer claim reaches about 99.5% under its own definitions, signaling a severe supply-allocation challenge rather than a comfortably unconstrained forecast. The appropriate use is to stress input growth, substitutions and bottleneck migration—not to import its $3.1tn chip market or 435 GW directly into stock targets. None of these formula issues by itself invalidates the broader AI 2027 research program.

3.6 Reconcile the three compute/buildout narratives before valuing stocks

Published AI 2027 compute forecast, April 2025. Its end-2027 global stock is approximately 100m H100-equivalents, versus 40m at end-2026; annual datacenter investment is roughly $1tn in 2027 and $600bn in 2026; global peak power is about 60–62 GW in 2027. The approximately $2tn cost attached to active compute is a **stock/ownership-cost measure**, not another $2tn of annual capex. Later sections about compute allocation are more exploratory. The forecast also shifts research toward post-training, experiments and synthetic data; one gigantic pretraining run is not the whole demand story. These figures are **STALE forecasts**, not observations. Published compute supplement.

My investment inference is that inference, memory and networking deserve at least as much attention as peak training FLOP. Compute owned by a cloud provider and rented by a lab must not be counted twice as new hardware demand. A smaller, more efficient model can still be expensive to run at enormous scale; equally, efficient algorithms can permit rapid capability gains on an existing fleet. These alternatives require separate quantities and margins in a valuation model.

AI 2040 / Plan A compute supplement, July 2026 scenario vintage. Before its hypothetical policy agreement, the model projects approximately $2.4tn of annual AI capex in 2028, with about 38% funded by external debt/equity, and roughly 155m new H100-equivalents that year. Its EUV constraint includes logic **and HBM**, with AI claiming approximately 65% of effective EUV passes by 2028 and 77% by 2029. A hand-chosen scarcity surcharge feeds back into affordability. Under Plan A, a temporary loss of external-financing-led capex lowers 2029 investment even though inference continues. These are conditional modeled quantities; its total-spending proxy, financing multiples, yields, resource-equivalence factors and surcharge are uncertain. Compute in Plan A.

The crucial upgrade relative to a simple leading-edge-wafer forecast is **exposure intensity**: a memory wafer and a logic wafer consume different scanner time, and better chips do not translate one-for-one into greater usable throughput if bandwidth or networking binds. Conversely, assumed scanner utilization, effective passes per year and yield can move the ceiling substantially. ASML's own 2025 report confirms 48 EUV systems recognized in revenue, versus 44 in 2024; those are sales units, not an instantaneous installed productive fleet. Its October 14, 2026 release says planned 2027–2028 supply will not fully meet customer demand and that further 2028 capacity expansion is under examination. This is stronger evidence for an upstream scarcity case than a generic “chips matter” argument. It is still not a dollar-for-dollar forecast of ASML revenue or profit. ASML 2025 report, October results and capacity commentary.

Situational Awareness, June 2024. Its approximately 100 GW / trillion-dollar cluster near 2030 is an aggressive individual-cluster scenario, alongside a wider industrial mobilization argument. It emphasizes power and permits, natural gas, packaging and HBM, with strategic competition potentially sustaining spending before conventional near-term ROI is demonstrated. The essay's historical power-plant cost and construction assumptions must not be reused as November 2026 project quotes. Trillion-dollar-cluster chapter.

My revised treatment is to carry that scale as an explicit upside/stress branch, not reject it because it exceeds a conventional energy projection. But “100 GW global AI,” “100 GW incremental US IT,” and “one 100 GW campus” are different bets. The last adds exceptional geographic and interconnection concentration. It is possible to be right about an enormous aggregate buildout and wrong about the feasibility or investment economics of a single site.

Data cross-check, not a demand guarantee. Summing the downloaded Epoch chip-sales median estimates produces approximately 5.09m H100-equivalents shipped in 2024, 13.61m in 2025 and 11.31m in the first half of 2026. The completed 2022–H1 2026 rows sum to about 31.60m before retirement/commissioning adjustments. Rows marked incomplete, including Q3 2026, are excluded from that comparison. These are estimated shipments in Epoch's equivalence convention, not verified live capacity, and individual medians do not produce a confidence interval for the sum. The underlying completed observations are **STALE**, even though retrieved November 9. Epoch dataset, downloaded series.

Epoch's November 6 company-revenue dataset lists Anthropic at an estimated $81bn annualized run rate as of October 31, and OpenAI at $50bn as of September 30; both are marked “Likely” and rely on media reports, not audited financial statements. The dataset explicitly revises an earlier approximately $70bn OpenAI figure that had grossed up partner revenue. I do not sum supplier, cloud and lab revenues into independent end-user demand or equate ARR with cash earnings. These observations support a stronger monetization case, while their reporting uncertainty prevents treating them as verified customer-level ROI. Epoch company data and sourcing, revenue records.

3.7 What taking rapid takeoff seriously changes for this portfolio

A software-led intelligence explosion can outrun physical construction. The original takeoff supplement conditions on superhuman coding in March 2027 and models subsequent progress with training compute held fixed. Its approximate one-year median transition is therefore not refuted merely by multi-year fab or turbine lead times. It relies heavily on judgments about researcher quality, experiment selection and automation, with very broad uncertainty. The August model's training-lag treatment is a meaningful refinement, not a reason to ignore the original mechanism. Takeoff forecast, linked software-explosion analysis.

My resulting valuation discipline is to separate three clocks: **capability progress**, **physical production/commissioning**, and **shareholder cash realization**. Rapid capability progress can raise the willingness to pay for existing chips and power immediately; it can also improve utilization, invent substitutes and shorten product lives before the next generation of infrastructure earns its expected return. Thus the current infrastructure bull cases cannot double as a complete ASI model.

AI goals are not automatically shareholder goals. The goals supplement analyzes specified intentions, imperfectly learned objectives, reward seeking, proxies and other possibilities, including context-dependent mixtures. Its elicited percentages overlap and are not a probability distribution over mutually exclusive extinction outcomes. The investment implication is an additional agency/control question, not an instruction to model every advanced system as an obedient profit maximizer. AI goals forecast.

Security is distinct from buying cybersecurity equities. The security supplement separates model-weight protection from protection of algorithmic information and considers loss of control. Its forecasts are conditional on the race trajectory and use uncertain expert judgments; a benchmarked capability is not a demonstrated successful real-world event. My implication is to monitor customer security, controlled deployment, independent evaluations and operational continuity across the entire basket. Generic endpoint-security growth does not prove that CRWD or PANW is insurance against these scenarios. Security forecast.

Plan A is a recommended intervention, not an AGI forecast pushed out to 2040. Its story restrains a much earlier default trajectory through an international agreement, transparency, verification and a later pause. Continued inference and economic automation are compatible with slowed frontier research. A new binding permit regime could shift scarcity rents from manufacturers or generators toward governments and permit holders. The economics supplement also contemplates very rapid abundance and changing relative prices; it explicitly cautions that its explorer is highly uncertain. AI 2040, full scenario PDF, economics supplement.

For equity valuation, these are separate branches: ordinary commercial expansion; a rapid software-led transition; governed slowdown with continuing inference; and severe loss of control or enforceable property rights. A boom in real output can coexist with competitive deflation, rapid obsolescence, forced reinvestment, redistribution or impaired minority-shareholder claims. This is why I strengthen the rapid-progress case **without automatically raising all terminal multiples**. Section 5.6 tests upstream ownership; section 7.2 exposes the limitation of conventional expected-return arithmetic; the targeted answers below now incorporate these mechanisms directly.

4. Sleeve-by-sleeve review, alternatives, and forward-return analysis

4.1 Stage-one sleeve competition

Ratings are **valuation-adjusted judgments versus the named current holding or basket**, not declarations that one company is universally better. “Comparable” means differences are too assumption-sensitive to establish a confident preference. Two alternatives are explicitly ranked in every sleeve; additional candidates follow. A better alternative can still receive zero because it loses cross-portfolio competition.

Sleeve / baseline	Alternative 1: rating and reason	Alternative 2: rating and reason	Sleeve decision
Foundry / TSM	GFS — Worse: mature/specialty process exposure does not replace leading-edge economics.	Samsung 005930.KS — Worse: qualified leading-node execution and conglomerate exposure dilute the substitute.	TSM remains best direct foundry, but 0%. INTC separately fails the external-foundry test.
Accelerators/ custom / NVDA, AVGO	AMD — Worse: strong ramp, but much more growth required at $449.04 and warrant/dilution risk.	MRVL — Worse: promising custom/connectivity exposure, now complicated by LITE merger dilution.	NVDA 15%, AVGO 15%. No assumption of permanent GPU monopoly.
Memory / MU	SK hynix 000660.KS — Comparable: excellent HBM position; local-market access and a verified comparable valuation are required.	Samsung 005930.KS — Worse: broader business and qualification/execution uncertainty.	MU 0%; no claim that a complete comparable foreign-stock valuation was obtained.
WFE / ASML, AMAT	LRCX — Better versus AMAT: attractive etch/deposition earnings torque and modeled 16.1% return.	KLAC — Comparable: process-control quality, but 10.6% modeled return and still a premium.	All 0% in final; ASML remains technologically unique, financially less attractive.
EDA / SNPS	CDNS — Worse at current prices: excellent franchise but lower modeled return.	Siemens SIE.DE — Worse as a targeted replacement: EDA is diluted inside a conglomerate.	SNPS wins the direct comparison but gets 0%.
Datacenter fabric / ANET	AVGO — Better: switch/custom-silicon earnings exposure at a stronger entry valuation.	HPE — Comparable: Juniper and server integration at a low earnings multiple, offset by acquisition and legacy risk.	ANET 0%; AVGO serves this need as well as custom compute.
Cooling/integration / VRT, DELL	CLS — Better versus DELL: stronger modeled per-share growth for the paid multiple, with customer/margin risk.	MOD — Comparable: attractive continuing-business upside but negative cash conversion and spin-off comparability risk.	VRT 15%, CLS 12%; DELL 0%. NVT complements electrical/thermal exposure.
Generation/grid / CEG, VST, PWR	NRG — Better versus CEG: cheaper owner-earnings exposure; gas/retail mix and leverage are the price of admission.	TLN — Comparable: strong contracted nuclear economics, higher asset and geographic concentration.	VST 11%, NRG 12%; CEG/PWR 0%.
AI security / CRWD	PANW — Better on relative valuation, but still unattractive in absolute expected return after SBC.	ZS — Comparable: focused zero-trust exposure; no verified valuation advantage sufficient to justify a full replacement.	Entire dedicated sleeve 0%. Microsoft's bundled offerings also constrain pricing.
Operate/control / PLTR	MSFT — Better: multiple distribution channels and stronger valuation-adjusted expected return.	NOW — Comparable: workflow distribution is valuable but seat disruption, SBC and consumption economics need care.	PLTR 0%; MSFT earns a place only in the alternative architecture.
Military / AVAV, KTOS	LHX — Comparable: established programs and execution diversify risk but lower growth torque.	CACI — Comparable: government cyber/IT and services exposure with budget and contract risk.	Dedicated defense 0%; NOC, RTX, LMT, GD, DRS, TDY fail to show a sufficiently superior AI-linked return.

Sleeve / baseline	Alternative 1: rating and reason	Alternative 2: rating and reason	Sleeve decision
Bio-risk / MRNA	BNTX — Comparable: another mRNA/oncology platform, not interchangeable catastrophe insurance.	ABT — Comparable: diagnostics and diversified cash generation, weaker direct rapid-response upside.	MRNA 0%; no reliable liquid-stock pandemic hedge established.
Materials / MP	FCX — Better as broad industrial scarcity exposure, not as a rare-earth substitute; copper cyclicality remains.	Lynas LYC.AX — Comparable: established non-China rare-earth processing, but separate product mix, costs and market access.	MP 0%; no mineral basket added merely for strategic completeness.
Domestic foundry / INTC	GFS — Better for demonstrated external foundry operations, worse as a leading-node substitute.	TSM — Better for execution and earnings; US fabs do not remove Taiwan-wide risk.	INTC 0%. No single public stock replicates sovereign semiconductor resilience.
Electrical / GEV	NVT — Better at this price: electrical distribution growth with a more favorable modeled return.	ETN — Better on valuation-adjusted return, though below NVT in this model.	NVT 12%; GEV 0%. Schneider and Siemens Energy remain real competitors.
Nuclear fuel / CCJ	Kazatomprom KAP — Comparable: resource/cost advantages offset jurisdiction, transport and instrument-access risks.	LEU — Worse for this portfolio: enrichment scarcity but concentrated execution, policy and financing risk.	CCJ 0%; UEC/UUUU are higher-risk operating/development exposures.
Grid bypass / BE	CMI — Comparable: engine/genset execution and diversification, but capacity, emissions and fuel risks persist.	CAT — Comparable: distributed/backup generation plus industrial exposure, not a pure prime-power replacement.	BE 0%; NRG's customer-backed generation offers a different, less technology-specific route.
Optical / COHR	LITE — Worse as a simple standalone substitute: the announced exchange-ratio merger changes the investment.	FN — Comparable: cleaner manufacturing model, but lower modeled return and customer concentration.	COHR 8%; GLW, CIEN, AAOI, CRDO and ALAB do not displace it.

Foreign ordinary shares are legally within the stated public-common-stock universe, but actual brokerage access, foreign exchange, withholding and liquidity must be verified. This report does not pretend that an OTC ticker always offers a liquid ADR. The foreign candidates not fully modeled failed an access/comparability or relative-advantage screen; their current multiples are **unavailable**, not estimated by analogy.

4.2 Cross-sleeve candidate register and prior-watchlist disposition

This table closes the thematic gaps left by the 18-sleeve structure. Ratings are against the nearest relevant incumbent/selected use of capital. All “N” decisions mean no final allocation, not that the company is bad. Names grouped in a row share the stated screen result; they are not treated as a synthetic security.

Ticker	Candidate Theme	Why Considered	Better/Comparable/Worse	Add? (Y/N)	Why/Why not
CLS	Compute/network manufacturing	New earnings outlook against a much lower price	Better	Y	12%; fully modeled, lower margins and customer concentration explicitly penalized.
NVT	Electrical distribution and enclosure	Grid-to-rack investment plus new power platform	Better	Y	12%; integration, leverage and component competition cap sizing.
NRG	Dispatchable power and retail	Cheap earnings plus customer-backed buildout	Better	Y	12%; no purchase before the Nov 9 results are checked.
MSFT	Cloud, agents, identity, distribution	Potential monetization winner rather than just a buyer	Better versus PLTR	N	17.6% modeled return; strong fallback but below concentrated holdings.
META	AI-enabled advertising and distribution	Profitable core can monetize efficiency without selling GPUs	Comparable	N	18.6% modeled return; only slightly below COHR, so a serious replacement candidate. Legal costs and capex matter.
GOOGL	Search, cloud, TPU and data	Search resilience plus custom compute	Better versus PLTR	N	New financing/dilution and cash conversion reduce apparent cheapness; 10.2% modeled return.
AMZN	AWS, retail/logistics and custom silicon	AI demand and retail efficiency	Comparable	N	AWS growth is strong, but full capex and replacement costs leave a less attractive owner-return bridge.
ORCL, IBM	Enterprise/cloud distribution	Existing customer access and enterprise workflows	Comparable	N	Oracle's financing/counterparty concentration and IBM's slower growth do not establish an advantage over the fully modeled platforms. No price target asserted.
NOW, SNOW	Enterprise workflows and data cloud	Explicit September foundation watchlist	Comparable versus PLTR; Worse owner-cash evidence than MSFT	N	Growth is real, but stock compensation, acquisition effects and rights to customer data limit the inference from adjusted margins. Dated screen below; no invented full valuation.
CSCO	Networking and security platform	Explicit original optical/networking watchlist	Comparable versus ANET; Worse direct AI-growth capture than AVGO	N	Established cash generation and AI orders warrant monitoring; broad legacy mix and no verified superior price-to-owner-growth edge keep it screened out.
SPXC	Cooling towers, HVAC and detection	Original grid/site watchlist	Comparable versus VRT	N	Growth is strong, but higher revenue guidance accompanied slightly lower EBITDA/EPS ranges; no demonstrated advantage over fully modeled VRT/NVT/MOD.

Ticker	Candidate Theme	Why Considered	Better/Comparable/Worse	Add? (Y/N)	Why/Why not
Thales HO.PA / THLEF	Identity, encryption and defense	Original identity/compliance watchlist	Comparable versus CRWD/PLTR as a business; no verified stock-value ranking	N	Diversified defense growth is not proof of a fast-growing identity toll; instrument-level valuation remains a research gap.
CORZ, IREN, APLD	Energized land and cloud/colocation	Large declines and contracted-capacity potential	Worse	N	Fully modeled with debt, preferred/NCI claims, dilution and zero-equity bear states. Backlog is not distributable equity value.
CRWV, NBIS	AI cloud platforms	Strong contracted growth and post-peak price declines	Comparable strategically; Worse risk-adjusted	N	Fully modeled; financing and GPU residual-value uncertainty make arithmetic upside insufficient.
RELX, VRSK	Rights-cleared proprietary information	Data refresh, workflow and liability moats	Better versus generic “data” exposure	N	Fully modeled, roughly 13–14% returns; neither is dismissed because data are intangible.
SPGI, MCO, LSEG, EXPN/EXPGY, FICO	Financial/credit/identity information	Recurring data and decision workflows	Comparable	N	Strong incumbents but no demonstrated discount exceeding the RELX/VRSK opportunity; SPGI corporate actions require care in historical comparisons.
PANW, OKTA, ZS	Identity and zero trust	Machine identities and enterprise permissioning	Better/Comparable versus CRWD	N	PANW fully modeled; OKTA/ZS do not receive a default “security hedge” allocation. CYBR is no longer an independent US-listed replacement.
MSI, AXON	Public safety platforms	Installed base, subscriptions, evidence and dispatch	Comparable	N	Both fully modeled; public-sector economics, SBC and civil-liberties/regulatory risks are part of the investment case.
GEN	Privacy, VPN and consumer security	Low headline earnings multiple	Comparable	N	Fully modeled; mature consumer products, leverage and commoditization constrain growth.
CLBT, CGNT, ARLO, YOU	Digital evidence, intelligence, home security, identity	Prior targeted-question watchlists	Worse	N	Narrow procurement/channel exposure and no demonstrated valuation-adjusted advantage. Do not equate surveillance demand with universal shareholder rents.
ZFOX; CLGX/CoreLogic/Cotality; Flock	External threat intelligence / property data / public safety	Historical watchlists	Worse: unavailable as the requested public common equity	N	ZeroFox went private; CoreLogic became private Cotality; Flock is not an eligible listed common-stock recommendation.

Ticker	Candidate Theme	Why Considered	Better/Comparable/Worse	Add? (Y/N)	Why/Why not
ASMI/ASMIY, BESI/BESIY	Deposition and advanced bonding	Process intensity and advanced packaging	Comparable	N	Real bottlenecks, but verified current instrument-level valuation/liquidity unavailable; LRCX/KLAC already provide strong modeled comparators.
AMKR, KLIC, ENTG, ONTO	Packaging, materials and metrology	Capacity/yield constraints	Comparable	N	Lower moat breadth, greater cycle sensitivity or insufficient verified valuation edge. ONTO is not a substitute for EUV.
ARM	CPU/IP licensing	Broad architecture participation	Worse	N	Less direct near-term AI profit exposure per valuation than NVDA/AVGO; royalties are not accelerator-system revenue.
Ajinomoto 2802.T	ABF substrate materials	Substrate bottleneck	Comparable	N	Economic exposure diluted by unrelated businesses; no verified superior stock-level expected return.
HPE, FN, POWL	Integration, optical assembly, switchgear	Credible apparent bargains	Comparable	N	All fully modeled; split-adjusted POWL and normalized margins remove much of the apparent upside.
MOD	Dedicated cooling after spin-off	Large apparent decline and fast datacenter growth	Comparable	N	Fully modeled; 19.7% arithmetic return but confidence 2, cash use and changing perimeter. Do not call its raw price change a shareholder total return.
FLEX, SMCI	Integration and manufacturing	AI server volumes	Worse versus CLS/VRT	N	Volume without durable owner margins is insufficient; customer/product mix and execution risk need a larger verified discount.
ETN, HUBB, APH, TEL	Electrical and connectivity	Broad distribution, connector and grid content	Comparable	N	ETN fully modeled; the others fail the incremental edge test, not the quality test. APH's split must be reflected.
Schneider SU.PA/SBGSY, ABB /ABBN.SW, ENR.DE/SMNEY, Hitachi 6501.T, MHI 7011.T	Global grid, controls and turbines	Challenge the claim that GEV is a monopoly	Comparable	N	Genuine global competitors; FX, business mix and instrument-level data gaps prevent claiming superior returns.
MYRG, MTZ, XYL	Construction, engineering and water	Trades, grid and site constraints	Comparable	N	Contract execution, working capital and cyclicality remain; no established advantage over NVT/VRT at the portfolio margin.

Ticker	Candidate Theme	Why Considered	Better/Comparable/Worse	Add? (Y/N)	Why/Why not
CAT, CMI, GNRC, EQT	Engines, distributed power and gas	Firm power or upstream fuel	Comparable	N	The producer, turbine maker and gas supplier capture different economics. No automatic pass-through of an AI scarcity premium.
NEE, CWEN	Renewables/storage and regulated infrastructure	Fast energy additions and contracted cash flow	Comparable	N	Important deployment route, but capital intensity, rate regulation and financing reduce equity upside versus selected merchant power.
EQIX, DLR	Datacenter landlords	Space, interconnection and capital access	Comparable	N	Lease economics are credible; capital intensity and financing make a sufficient return advantage unproven.
GLW, CIEN, AAOI	Fiber and optical systems	Bandwidth and connection demand	Comparable/Worse versus COHR	N	GLW/CIEN Comparable strategic exposure; AAOI Worse execution/customer risk. No unverified same-multiple comparisons.
ALAB, CRDO	Scale-up switches, retimers and active cables	New architecture bottlenecks	Comparable businesses; Worse stock returns	N	Both fully modeled; strong current growth does not justify ignoring dilution or terminal competition.
LHX, CACI, TDY, DRS, LMT, NOC, RTX, GD	Defense and intelligence	Procurement, autonomous systems and resilience	Comparable	N	No allocation solely because conflict risk exists. Established contractors are not a guaranteed hedge against a Taiwan-driven market loss.
FCX, SCCO, Lynas, KAP, LEU, UEC, UUUU, CLF, POSCO	Copper, rare earths, fuel, steel	Processing and industrial supply dependencies	Comparable/Worse	N	FCX/Lynas/KAP Comparable; development-heavy uranium and broad steel Worse as targeted AI rent capture. MP is not interchangeable with these commodities.
BNTX, PFE, ABT, QDEL, ZM	Bio-response, diagnostics, remote work	Potential pandemic resilience	Comparable/Worse	N	BNTX/ABT Comparable research candidates; PFE/QDEL/ZM weaker targeted insurance. No reliable joint payoff in an unspecified crisis.
ON, STM, IFX, NVTS	Power semiconductors	Conversion efficiency and high-voltage architectures	Comparable/Worse	N	ON/STM/IFX Comparable industrial exposures; NVTS Worse execution/financing risk. Revenue exposure and profitable design wins matter more than voltage headlines.

Ticker	Candidate Theme	Why Considered	Better/Comparable/Worse	Add? (Y/N)	Why/Why not
ROK, FANUC, Yaskawa, SYM; KUKA	Robotics and industrial automation	Physical-world intelligence adoption	Comparable	N	Deployment economics and customer concentration do not yet beat selected infrastructure; KUKA is not an eligible independent listed recommendation.
SPY, QQQ, SMH/SOXX, XLU, ITA, IBB/XBI, URA	Benchmarks and fallback funds	Opportunity-cost and implementation tests	Comparable fallback	N	None demonstrated superior expected return to the selected basket under this model; broad funds remain a valid fallback if active underwriting is not trusted.
PHYS / physical gold; BTC	Historical non-core macro hedge ideas	Original F8 watchlist	Worse fit for this mandate	N	Gold vehicles are not superior demonstrated AI cash-flow investments; direct BTC and derivative hedges are outside the authorized universe. No Moonshot allocation.

The private-status checks are supported by ZeroFox's acquisition completion notice and Cotality's rebranding notice. These are **STALE structural facts**, checked during this review, not new developments since May. Small or foreign screened names without current verified valuation data are explicitly research gaps; no precision is claimed and none enters an architecture.

Additional original-watchlist evidence. NOW's October 28 release reports 20.5% revenue growth, but 9.5% GAAP operating margin versus 32.5% adjusted, with $633m of stock compensation in the operating reconciliation. At the saved November 6 price of $101.97, approximately 25.5× annualized quarterly adjusted EPS is not the same as a cheap owner-cash multiple. SNOW's September 2 release — STALE reports 37% product growth and 126% retention, but a 17% GAAP operating loss margin and compensation-related adjustments equal to about 29% of revenue; its saved price is $275.88. These are credible businesses, not dismissed as obsolete software, but they fail the evidence threshold for displacing the stronger after-compensation cash cases. No unsourced 2030 price target is presented for screened-out names.

CSCO's August 12 release — STALE reports $17.3bn quarterly revenue, $1.22 adjusted EPS and $5.4bn operating cash flow. That establishes a real cash alternative, not a superior valuation without a matched current price/forecast bridge. SPXC's October 29 release shows revenue growth of 20.5%, but its revised $8.20–8.50 adjusted EPS range is slightly below the prior $8.20–8.60 despite a higher revenue outlook. Thales's October 22 issuer-distributed release reports strong group orders while Cyber & Digital order intake declined organically. These observations support qualified screening, not a claim that every excluded company is inferior at every price. Reopen the funnel if better verified valuation evidence changes that judgment.

4.3 Valuation conventions — reproducible, not black-box price targets

All terminal scenarios are this report's **own November 8 assumptions**, not company guidance or consensus. The endpoint is December 31, 2030. The elapsed period from the November 6 close is 4.150582 years. Revenue, profits and share counts are in USD billions unless stated otherwise. ASML and CCJ are modeled in USD; the base translation conventions are $1.10 per euro and $0.73 per Canadian dollar, **model assumptions rather than claimed spot rates**. RELX's model is USD per ADR, with one ordinary-share-equivalent unit; any execution requires confirming the ADR terms and FX.

RELX's one-for-one ADS ratio is supported by its 2025 security description filed in 2026, a **STALE structural reference** checked for instrument consistency. The model does not confuse RELX with the unrelated RLX Technology ticker.

For earnings models:

terminal stock value = 2030 revenue × sustainable after-tax common margin ÷ diluted shares × terminal P/E

terminal total wealth = terminal stock value + cumulative 2027–2030 dividends + explicitly modeled excess cash

For EV models:

terminal stock value = max(0, EBITDA × EV/EBITDA − net debt − preferred/NCI claims) ÷ diluted shares

For CRWD/PANW, the margin is **owner FCF after an economic SBC charge**, and the multiple is P/owner-FCF. Residual share growth is also allowed for; repurchase-funded compensation is not treated as free money. For other earnings models, sustainable margins are after interest and taxes, and after a normalized compensation burden. Acquisition amortization is normalized only where replacement investment and cash economics warrant it; reported adjusted EPS is not blindly capitalized.

expected terminal wealth = p_bear × TV_bear + p_base × TV_base + p_bull × TV_bull

implied expected CAGR = (expected terminal wealth ÷ current price)^(1/4.150582) − 1

Scenario probabilities sum to 100% for each stock. **Scenario CAGRs are never averaged.** This CAGR is the annualized growth rate to expected terminal wealth; it is not the expectation of the investor's realized CAGR, which is a different nonlinear quantity. Dividends are accumulated without reinvestment, making forward total-wealth calculations slightly conservative relative to identical successful reinvestment. Historical adjusted-close returns use

a reinvestment proxy, so the two conventions are disclosed rather than silently mixed.

Terminal multiples are normative underwriting ranges, not asserted historical medians. Their lower end allows ordinary cyclical/commodity economics; the upper end requires durable moats and long remaining growth. The May-versus-November valuations above provide verified within-cycle history where possible. **Reliable complete five- or ten-year multiple percentiles are unavailable**; a two-date comparison is not a substitute for that database. Peer comparisons use like economic measures as far as possible and explicitly flag adjusted earnings, lease/capex intensity, merger perimeter and FX differences.

The model uses representative bear/base/bull values inside much wider plausible ranges. Display rounding is not confidence. Bear probabilities rise to 30–40% for fragile, capital-intensive or technology-transition businesses; bull probabilities fall to 20%. These are judgmental, not frequencies estimated from a large historical sample. Catastrophic total loss remains possible even for companies with a positive modeled bear value.

4.4 Complete underwriting ledger: assumptions and terminal outcomes

Read this ledger together with the per-company notes in 4.5–4.6. It supplies the revenue, earnings/FCF margin, diluted shares, valuation, distributions and explicit balance-sheet claims for **all 56 fully underwritten names**. These are scenario assumptions, not claims that management has guided to these figures.

In the claims column, EV rows show net financial claims in **$bn**. Earnings rows show additional terminal excess cash in **$/share**; zero means no separate addition, not a debt-free company. INTC's $6 bear entry is a speculative residual equity/recovery allowance, not cash or guaranteed liquidation value. MU's additions reserve part of accumulated cycle cash; they are not simultaneously distributed as dividends or used for the same buybacks. No dividend is assumed for nonpayers unless explicitly shown.

Ticker / method	2030 revenue $bn B/B/B	Margin % B/B/B	Diluted shares bn B/B/B	Exit multiple B/B/B	Cumulative div $ B/B/B	Net claims $bn OR excess cash $/share B/B/B
TSM / Earnings	140 / 310 / 430	32.0 / 45.0 / 47.0	5.300 / 5.200 / 5.150	15 / 23 / 28	15 / 20 / 25	0 / 0 / 0
NVDA / Earnings	350 / 850 / 1400	32.0 / 46.0 / 50.0	24.000 / 23.500 / 23.000	18 / 25 / 30	4 / 6 / 8	0 / 0 / 0
AVGO / Earnings	100 / 250 / 400	38.0 / 50.0 / 52.0	5.000 / 4.900 / 4.800	18 / 24 / 28	10 / 13 / 16	0 / 0 / 0
MU / Earnings	120 / 240 / 360	20.0 / 40.0 / 50.0	1.200 / 1.200 / 1.200	12 / 12 / 15	4 / 8 / 12	40 / 150 / 250
ASML / Earnings	40 / 70 / 100	24.0 / 34.0 / 37.0	0.385 / 0.375 / 0.365	20 / 30 / 35	30 / 40 / 50	0 / 0 / 0
AMAT / Earnings	25 / 50 / 70	20.0 / 28.0 / 31.0	0.780 / 0.720 / 0.700	16 / 24 / 28	8 / 10 / 12	0 / 0 / 0
SNPS / Earnings	10 / 16 / 23	18.0 / 30.0 / 34.0	0.205 / 0.200 / 0.205	23 / 30 / 35	0 / 0 / 0	0 / 0 / 0
ANET / Earnings	13 / 30 / 46	28.0 / 35.0 / 38.0	1.300 / 1.300 / 1.320	22 / 30 / 35	0 / 0 / 0	0 / 0 / 0
VRT / Earnings	12 / 30 / 48	10.0 / 20.0 / 23.0	0.390 / 0.380 / 0.375	18 / 26 / 30	1 / 2 / 3	0 / 0 / 0
DELL / Earnings	130 / 300 / 450	3.0 / 5.5 / 6.5	0.650 / 0.600 / 0.570	10 / 17 / 20	10 / 12 / 14	0 / 0 / 0
CEG / Earnings	35 / 50 / 65	7.0 / 12.0 / 15.0	0.360 / 0.350 / 0.340	14 / 22 / 26	9 / 12 / 15	0 / 0 / 0
PWR / Earnings	32 / 60 / 80	5.0 / 8.0 / 9.5	0.150 / 0.150 / 0.150	18 / 25 / 30	2 / 3 / 4	0 / 0 / 0
CRWD / Owner FCF	9 / 16 / 24	10.0 / 23.0 / 28.0	1.120 / 1.100 / 1.100	20 / 35 / 45	0 / 0 / 0	0 / 0 / 0
PLTR / Earnings	12 / 28 / 45	22.0 / 38.0 / 45.0	2.800 / 2.750 / 2.700	22 / 40 / 50	0 / 0 / 0	0 / 0 / 0
AVAV / Earnings	2.3 / 4 / 6	4.0 / 10.0 / 14.0	0.060 / 0.060 / 0.062	16 / 24 / 28	0 / 0 / 0	0 / 0 / 0
KTOS / Earnings	2.1 / 3.8 / 6	2.0 / 7.0 / 10.0	0.220 / 0.230 / 0.240	18 / 28 / 35	0 / 0 / 0	0 / 0 / 0
MRNA / Earnings	4 / 12 / 25	0.0 / 25.0 / 35.0	0.440 / 0.450 / 0.460	15 / 25 / 30	0 / 0 / 0	5 / 10 / 15
MP / Earnings	0.6 / 2 / 3.5	2.0 / 22.0 / 28.0	0.200 / 0.210 / 0.220	16 / 24 / 30	0 / 0 / 0	0 / 0 / 0
INTC / Earnings	55 / 85 / 130	0.0 / 10.0 / 17.0	6.000 / 5.800 / 6.000	12 / 18 / 22	0 / 0 / 0	6 / 0 / 0
GEV / Earnings	40 / 75 / 105	5.0 / 13.0 / 16.0	0.270 / 0.265 / 0.260	18 / 28 / 32	8 / 12 / 16	0 / 0 / 0

Ticker / method	2030 revenue $bn B/B/B	Margin % B/B/B	Diluted shares bn B/B/B	Exit multiple B/B/B	Cumulative div $ B/B/B	Net claims $bn OR excess cash $/share B/B/B
CCJ / Earnings	2.5 / 5 / 8	12.0 / 25.0 / 30.0	0.440 / 0.440 / 0.445	18 / 25 / 30	2 / 3 / 4	0 / 0 / 0
BE / Earnings	3 / 10 / 18	2.0 / 15.0 / 19.0	0.350 / 0.360 / 0.380	18 / 28 / 35	0 / 0 / 0	0 / 0 / 0
COHR / Earnings	7 / 20 / 30	8.0 / 20.0 / 23.0	0.220 / 0.220 / 0.230	15 / 24 / 28	0 / 0 / 0	0 / 0 / 0
CLS / Earnings	25 / 65 / 100	3.5 / 6.5 / 8.0	0.140 / 0.140 / 0.145	10 / 17 / 20	0 / 0 / 0	0 / 0 / 0
NVT / Earnings	5.5 / 10 / 15	10.0 / 17.0 / 20.0	0.160 / 0.160 / 0.160	16 / 22 / 26	3 / 4 / 5	0 / 0 / 0
MSFT / Earnings	380 / 680 / 950	30.0 / 37.0 / 40.0	7.500 / 7.300 / 7.200	22 / 29 / 33	16 / 20 / 24	0 / 0 / 0
GOOGL / Earnings	430 / 800 / 1150	18.0 / 27.0 / 32.0	13.300 / 13.100 / 13.000	17 / 24 / 28	4 / 5 / 6	0 / 0 / 0
AMZN / Earnings	850 / 1350 / 1800	5.0 / 10.0 / 13.0	11.300 / 11.400 / 11.500	22 / 30 / 34	0 / 0 / 0	0 / 0 / 0
META / Earnings	230 / 400 / 600	20.0 / 32.0 / 38.0	2.650 / 2.600 / 2.600	16 / 23 / 27	9 / 12 / 15	0 / 0 / 0
LRCX / Earnings	25 / 50 / 70	24.0 / 35.0 / 38.0	1.250 / 1.200 / 1.180	16 / 25 / 29	5 / 7 / 9	0 / 0 / 0
KLAC / Earnings	13 / 25 / 34	30.0 / 38.0 / 40.0	1.330 / 1.280 / 1.250	20 / 28 / 32	4 / 5 / 6	0 / 0 / 0
CDNS / Earnings	6.5 / 10 / 14	22.0 / 34.0 / 37.0	0.280 / 0.270 / 0.270	25 / 35 / 40	0 / 0 / 0	0 / 0 / 0
ETN / Earnings	29 / 48 / 65	12.0 / 18.0 / 21.0	0.390 / 0.380 / 0.380	18 / 25 / 29	18 / 22 / 26	0 / 0 / 0
MOD / Earnings	2.8 / 5.5 / 8.5	4.0 / 9.0 / 13.0	0.056 / 0.056 / 0.058	13 / 22 / 26	0 / 0 / 0	0 / 0 / 0
PANW / Owner FCF	16 / 27 / 39	16.0 / 26.0 / 31.0	0.930 / 0.920 / 0.930	22 / 32 / 38	0 / 0 / 0	0 / 0 / 0
RELX / Earnings	12 / 17 / 22	20.0 / 26.0 / 29.0	1.800 / 1.750 / 1.730	18 / 24 / 28	4 / 5 / 6	0 / 0 / 0
VRSK / Earnings	3.4 / 4.5 / 5.7	25.0 / 33.0 / 37.0	0.130 / 0.120 / 0.118	20 / 28 / 32	8 / 9 / 10	0 / 0 / 0
MSI / Earnings	13 / 19 / 25	16.0 / 22.0 / 25.0	0.170 / 0.160 / 0.160	19 / 26 / 30	20 / 24 / 28	0 / 0 / 0
AXON / Earnings	4.5 / 8 / 12	6.0 / 16.0 / 22.0	0.090 / 0.095 / 0.100	25 / 35 / 42	0 / 0 / 0	0 / 0 / 0
GEN / Earnings	4.5 / 6 / 7.5	17.0 / 24.0 / 28.0	0.640 / 0.610 / 0.600	10 / 14 / 17	2 / 2.5 / 3	0 / 0 / 0

Ticker / method	2030 revenue $bn B/B/B	Margin % B/B/B	Diluted shares bn B/B/B	Exit multiple B/B/B	Cumulative div $ B/B/B	Net claims $bn OR excess cash $/share B/B/B
AMD / Earnings	45 / 125 / 230	13.0 / 25.0 / 30.0	1.800 / 1.900 / 2.000	18 / 28 / 33	0 / 0 / 0	0 / 0 / 0
ALAB / Earnings	1.5 / 5 / 9	12.0 / 26.0 / 30.0	0.200 / 0.210 / 0.220	20 / 32 / 40	0 / 0 / 0	0 / 0 / 0
CRDO / Earnings	2 / 5 / 9	15.0 / 28.0 / 32.0	0.210 / 0.220 / 0.230	18 / 28 / 35	0 / 0 / 0	0 / 0 / 0
HPE / Earnings	40 / 70 / 100	5.0 / 8.5 / 11.0	1.400 / 1.350 / 1.350	9 / 13 / 16	2.3 / 2.8 / 3.2	0 / 0 / 0
FN / Earnings	5 / 10 / 16	6.0 / 9.0 / 11.0	0.037 / 0.037 / 0.038	13 / 20 / 24	0 / 0 / 0	0 / 0 / 0
POWL / Earnings	1.1 / 2.2 / 3.3	8.0 / 14.0 / 17.0	0.038 / 0.038 / 0.038	12 / 20 / 24	1 / 1.5 / 2	0 / 0 / 0
VST / EV/EBITDA	20 / 28 / 36	30.0 / 39.3 / 41.7	0.340 / 0.300 / 0.280	7 / 9 / 10	4 / 5 / 6	26 / 20 / 15
NRG / EV/EBITDA	25 / 32 / 40	18.0 / 21.9 / 22.5	0.220 / 0.200 / 0.190	6.5 / 8 / 9	7 / 8 / 9	18 / 14 / 10
TLN / EV/EBITDA	5 / 8 / 11	35.0 / 45.0 / 50.0	0.050 / 0.045 / 0.043	7 / 9 / 10	0 / 0 / 0	5 / 4 / 3
CORZ / EV/EBITDA	0.7 / 2.2 / 3.5	35.0 / 55.0 / 60.0	0.500 / 0.550 / 0.600	8 / 12 / 14	0 / 0 / 0	5 / 5 / 5
IREN / EV/EBITDA	2 / 7 / 12	25.0 / 45.0 / 50.0	0.500 / 0.550 / 0.650	6 / 9 / 11	0 / 0 / 0	12 / 13 / 15
CRWV / EV/EBITDA	12 / 32 / 50	40.0 / 55.0 / 60.0	0.650 / 0.750 / 0.850	6 / 8 / 10	0 / 0 / 0	45 / 65 / 80
APLD / EV/EBITDA	1 / 3 / 5	35.0 / 55.0 / 60.0	0.400 / 0.450 / 0.500	8 / 12 / 14	0 / 0 / 0	6 / 10 / 15
NBIS / EV/EBITDA	6 / 25 / 45	35.0 / 50.0 / 55.0	0.400 / 0.500 / 0.600	6 / 9 / 11	0 / 0 / 0	18 / 45 / 70
MRVL / Merger earnings	9 / 30 / 47	15.0 / 30.0 / 34.0	0.900 / 1.380 / 1.420	18 / 25 / 30	1 / 1 / 1	0 / 0 / 0
LITE / Merger conversion	4.5 / 30 / 47	18.0 / 30.0 / 34.0	0.110 / 1.380 / 1.420	18 / 25 / 30	0 / 4.5 / 4.5	0 / 0 / 0

For MRVL/LITE, base and bull describe the combined business, not two independent companies: a LITE common share receives 4.5 MRVL shares if the announced deal closes. The bear states assume transaction failure and standalone normalization. Closing probability is simplified into the scenario weights; this is **not** a riskless arbitrage recommendation. Preferred conversion and incremental common shares are included in the combined share assumptions. Nov 2 merger filing.

Ticker	Nov 6 price $	2030 total wealth $ B/B/B	Probabilities % B/B/B	Expected wealth $	Expected total return %	Implied CAGR %	Status / confidence
TSM	375.44	142 / 637 / 1124	25 / 50 / 25	634.9	+69.1	13.5	Fair / 4
NVDA	186.64	88 / 422 / 921	25 / 50 / 25	463.2	+148.2	24.5	Attractive / 4
AVGO	293.69	147 / 625 / 1229	25 / 50 / 25	656.7	+123.6	21.4	Attractive / 4
MU	690.00	284 / 1118 / 2512	30 / 50 / 20	1146.6	+66.2	13.0	Fair / 3
ASML	1424.75	529 / 1944 / 3598	25 / 50 / 25	2003.7	+40.6	8.6	Fair / 4
AMAT	332.72	111 / 477 / 880	25 / 50 / 25	486.0	+46.1	9.6	Fair / 3
SNPS	450.46	202 / 720 / 1335	25 / 50 / 25	744.3	+65.2	12.9	Fair / 3
ANET	170.80	62 / 242 / 463	25 / 50 / 25	252.4	+47.8	9.9	Fair / 4
VRT	182.86	56 / 413 / 886	25 / 50 / 25	441.9	+141.7	23.7	Attractive / 4
DELL	328.53	70 / 480 / 1040	25 / 50 / 25	517.3	+57.5	11.6	Fair / 3
CEG	249.95	104 / 389 / 761	25 / 50 / 25	410.8	+64.3	12.7	Fair / 3
PWR	540.90	194 / 803 / 1524	25 / 50 / 25	831.0	+53.6	10.9	Fair / 4
CRWD	216.90	16 / 117 / 275	25 / 50 / 25	131.3	-39.5	-11.4	Expensive / 3
PLTR	160.38	21 / 155 / 375	25 / 50 / 25	176.3	+9.9	2.3	Expensive / 3
AVAV	112.76	25 / 160 / 379	30 / 50 / 20	163.2	+44.8	9.3	Speculative / 2
KTOS	35.10	3 / 32 / 88	30 / 50 / 20	34.7	-1.1	-0.3	Speculative / 2
MRNA	254.19	5 / 177 / 586	35 / 45 / 20	198.4	-22.0	-5.8	Speculative / 2
MP	35.41	1 / 50 / 134	35 / 45 / 20	49.7	+40.3	8.5	Speculative / 2
INTC	56.28	6 / 26 / 81	35 / 45 / 20	30.2	-46.4	-13.9	Speculative / 2
GEV	774.11	141 / 1042 / 2084	25 / 50 / 25	1077.4	+39.2	8.3	Fair / 3
CCJ	71.30	14 / 74 / 166	25 / 50 / 25	82.0	+15.0	3.4	Expensive / 3

Ticker	Nov 6 price $	2030 total wealth $ B/B/B	Probabilities % B/B/B	Expected wealth $	Expected total return %	Implied CAGR %	Status / confidence
BE	173.24	3 / 117 / 315	30 / 50 / 20	122.3	-29.4	-8.1	Speculative / 2
COHR	192.81	38 / 436 / 840	30 / 50 / 20	397.6	+106.2	19.1	Attractive / 3
CLS	206.68	62 / 513 / 1103	30 / 50 / 20	496.0	+140.0	23.5	Deep Discount / 3
NVT	106.22	58 / 238 / 492	25 / 50 / 25	256.5	+141.5	23.7	Attractive / 3
MSFT	530.61	350 / 1020 / 1766	25 / 50 / 25	1038.8	+95.8	17.6	Fair / 4
GOOGL	284.78	103 / 401 / 799	25 / 50 / 25	425.7	+49.5	10.2	Fair / 3
AMZN	259.30	83 / 355 / 692	25 / 50 / 25	371.3	+43.2	9.0	Fair / 3
META	610.15	287 / 1144 / 2383	25 / 50 / 25	1239.5	+103.1	18.6	Fair / 3
LRCX	200.44	82 / 372 / 663	25 / 50 / 25	371.9	+85.6	16.1	Fair / 4
KLAC	138.49	63 / 213 / 354	25 / 50 / 25	210.6	+52.1	10.6	Fair / 4
CDNS	319.92	128 / 441 / 767	25 / 50 / 25	444.1	+38.8	8.2	Fair / 3
ETN	341.72	179 / 590 / 1068	25 / 50 / 25	606.8	+77.6	14.8	Fair / 4
MOD	96.61	26 / 194 / 495	30 / 50 / 20	204.1	+111.3	19.7	Speculative / 2
PANW	313.84	61 / 244 / 494	25 / 50 / 25	260.7	-16.9	-4.4	Expensive / 3
RELX	40.17	28 / 66 / 109	25 / 50 / 25	67.1	+67.1	13.2	Fair / 4
VRSK	204.59	139 / 356 / 582	25 / 50 / 25	357.9	+74.9	14.4	Fair / 4
MSI	514.51	252 / 703 / 1200	25 / 50 / 25	714.7	+38.9	8.2	Fair / 4
AXON	313.00	75 / 472 / 1109	30 / 50 / 20	480.0	+53.4	10.9	Speculative / 2
GEN	29.79	14 / 36 / 63	25 / 50 / 25	36.9	+23.8	5.3	Expensive / 3
AMD	449.04	58 / 461 / 1138	30 / 50 / 20	475.5	+5.9	1.4	Speculative / 2
ALAB	190.71	18 / 198 / 491	30 / 50 / 20	202.6	+6.3	1.5	Speculative / 2

Ticker	Nov 6 price $	2030 total wealth $ B/B/B	Probabilities % B/B/B	Expected wealth $	Expected total return %	Implied CAGR %	Status / confidence
CRDO	120.51	26 / 178 / 438	30 / 50 / 20	184.5	+53.1	10.8	Speculative / 2
HPE	44.77	15 / 60 / 134	25 / 50 / 25	67.2	+50.2	10.3	Fair / 3
FN	327.13	105 / 486 / 1112	30 / 50 / 20	497.2	+52.0	10.6	Fair / 3
POWL	133.12	29 / 164 / 356	30 / 50 / 20	161.7	+21.5	4.8	Expensive / 3
VST	124.80	51 / 268 / 488	25 / 50 / 25	269.0	+115.5	20.3	Attractive / 4
NRG	88.29	58 / 218 / 383	25 / 50 / 25	219.2	+148.3	24.5	Deep Discount / 3
TLN	338.64	145 / 631 / 1209	25 / 50 / 25	654.1	+93.2	17.2	Fair / 3
CORZ	11.76	0 / 17 / 41	35 / 45 / 20	15.9	+35.4	7.6	Speculative / 2
IREN	21.64	0 / 28 / 78	40 / 40 / 20	26.9	+24.1	5.3	Speculative / 2
CRWV	46.40	0 / 101 / 259	40 / 40 / 20	92.2	+98.7	18.0	Speculative / 2
APLD	15.55	0 / 22 / 54	40 / 40 / 20	19.5	+25.5	5.6	Speculative / 2
NBIS	129.49	0 / 135 / 337	40 / 40 / 20	121.4	-6.2	-1.5	Speculative / 2
MRVL	141.20	28 / 164 / 339	30 / 50 / 20	158.1	+12.0	2.8	Speculative / 2
LITE	538.59	133 / 738 / 1524	30 / 50 / 20	713.6	+32.5	7.0	Speculative / 2

Scenario total returns (not annualized), including modeled distributions:

Ticker	Bear total return %	Base total return %	Bull total return %
TSM	-62.2	+69.7	+199.3
NVDA	-52.9	+126.1	+393.5
AVGO	-50.0	+112.9	+318.6
MU	-58.8	+62.0	+264.1
ASML	-62.9	+36.4	+152.5
AMAT	-66.8	+43.3	+164.5
SNPS	-55.2	+59.8	+196.4
ANET	-63.9	+41.9	+171.4
VRT	-69.2	+125.6	+384.6
DELL	-78.7	+46.0	+216.7
CEG	-58.3	+55.7	+204.3
PWR	-64.1	+48.5	+181.8
CRWD	-92.6	-46.0	+26.7
PLTR	-87.1	-3.5	+133.8
AVAV	-78.2	+41.9	+236.4
KTOS	-90.2	-7.7	+149.3
MRNA	-98.0	-30.5	+130.4
MP	-97.3	+42.0	+277.4
INTC	-89.3	-53.1	+44.0
GEV	-81.7	+34.6	+169.2
CCJ	-80.0	+3.8	+132.5
BE	-98.2	-32.7	+81.8
COHR	-80.2	+126.3	+335.7
CLS	-69.8	+148.2	+433.9
NVT	-45.4	+123.8	+363.7
MSFT	-34.0	+92.1	+232.8
GOOGL	-63.9	+40.7	+180.4
AMZN	-68.1	+37.0	+166.8
META	-53.0	+87.5	+290.5
LRCX	-59.2	+85.4	+230.6
KLAC	-54.8	+53.7	+155.7

Ticker	Bear total return %	Base total return %	Bull total return %
CDNS	-60.1	+37.8	+139.9
ETN	-47.7	+72.8	+212.5
MOD	-73.1	+101.3	+412.7
PANW	-80.7	-22.2	+57.4
RELX	-30.3	+63.3	+172.0
VRSK	-32.2	+73.8	+184.4
MSI	-50.9	+36.7	+133.2
AXON	-76.0	+50.7	+254.2
GEN	-53.2	+19.3	+109.8
AMD	-87.0	+2.6	+153.5
ALAB	-90.6	+3.9	+157.4
CRDO	-78.7	+47.9	+263.7
HPE	-66.1	+34.2	+198.3
FN	-67.8	+48.7	+239.8
POWL	-78.4	+22.9	+167.7
VST	-59.1	+115.0	+291.1
NRG	-34.2	+146.9	+333.4
TLN	-57.2	+86.4	+257.1
CORZ	-100.0	+47.2	+245.8
IREN	-100.0	+29.0	+262.6
CRWV	-100.0	+117.8	+457.8
APLD	-100.0	+40.1	+247.3
NBIS	-100.0	+4.3	+160.3
MRVL	-80.2	+16.2	+139.8
LITE	-75.4	+37.1	+182.9

FCF cross-check: cash conversion below is a normalization assumption, not reported FCF. It prevents a high earnings terminal from being justified with no thought to replacement capex, working capital or compensation. The EV businesses are handled separately because EBITDA is farther from owner cash.

Ticker	Base common profit / owner FCF $bn	Base owner FCF $bn	Cash conversion	Interpretation
TSM	139.50	97.65	70%	After normalized investment; not extra terminal cash
NVDA	391.00	371.45	95%	After normalized investment; not extra terminal cash
AVGO	125.00	118.75	95%	After normalized investment; not extra terminal cash
MU	96.00	62.40	65%	After normalized investment; not extra terminal cash
ASML	23.80	21.42	90%	After normalized investment; not extra terminal cash
AMAT	14.00	13.30	95%	After normalized investment; not extra terminal cash
SNPS	4.80	4.08	85%	After normalized investment; not extra terminal cash
ANET	10.50	9.97	95%	After normalized investment; not extra terminal cash
VRT	6.00	5.10	85%	After normalized investment; not extra terminal cash
DELL	16.50	10.72	65%	After normalized investment; not extra terminal cash
CEG	6.00	4.20	70%	After normalized investment; not extra terminal cash
PWR	4.80	3.84	80%	After normalized investment; not extra terminal cash
CRWD	3.68	3.68	100%	Owner FCF is valuation basis
PLTR	10.64	9.04	85%	After normalized investment; not extra terminal cash
AVAV	0.40	0.26	65%	After normalized investment; not extra terminal cash
KTOS	0.27	0.15	55%	After normalized investment; not extra terminal cash
MRNA	3.00	2.40	80%	After normalized investment; not extra terminal cash
MP	0.44	0.29	65%	After normalized investment; not extra terminal cash
INTC	8.50	2.12	25%	After normalized investment; not extra terminal cash
GEV	9.75	7.80	80%	After normalized investment; not extra terminal cash
CCJ	1.25	0.94	75%	After normalized investment; not extra terminal cash

Ticker	Base common profit / owner FCF $bn	Base owner FCF $bn	Cash conversion	Interpretation
BE	1.50	0.98	65%	After normalized investment; not extra terminal cash
COHR	4.00	3.00	75%	After normalized investment; not extra terminal cash
CLS	4.23	2.75	65%	After normalized investment; not extra terminal cash
NVT	1.70	1.45	85%	After normalized investment; not extra terminal cash
MSFT	251.60	176.12	70%	After normalized investment; not extra terminal cash
GOOGL	216.00	140.40	65%	After normalized investment; not extra terminal cash
AMZN	135.00	81.00	60%	After normalized investment; not extra terminal cash
META	128.00	89.60	70%	After normalized investment; not extra terminal cash
LRCX	17.50	16.62	95%	After normalized investment; not extra terminal cash
KLAC	9.50	9.03	95%	After normalized investment; not extra terminal cash
CDNS	3.40	2.89	85%	After normalized investment; not extra terminal cash
ETN	8.64	7.78	90%	After normalized investment; not extra terminal cash
MOD	0.49	0.32	65%	After normalized investment; not extra terminal cash
PANW	7.02	7.02	100%	Owner FCF is valuation basis
RELX	4.42	3.98	90%	After normalized investment; not extra terminal cash
VRSK	1.49	1.34	90%	After normalized investment; not extra terminal cash
MSI	4.18	3.76	90%	After normalized investment; not extra terminal cash
AXON	1.28	0.90	70%	After normalized investment; not extra terminal cash
GEN	1.44	1.37	95%	After normalized investment; not extra terminal cash
AMD	31.25	25.00	80%	After normalized investment; not extra terminal cash
ALAB	1.30	1.04	80%	After normalized investment; not extra terminal cash

Ticker	Base common profit / owner FCF $bn	Base owner FCF $bn	Cash conversion	Interpretation
CRDO	1.40	1.19	85%	After normalized investment; not extra terminal cash
HPE	5.95	4.76	80%	After normalized investment; not extra terminal cash
FN	0.90	0.81	90%	After normalized investment; not extra terminal cash
POWL	0.31	0.26	85%	After normalized investment; not extra terminal cash
MRVL	9.00	7.20	80%	Combined MRVL/LITE business; do not add twice
LITE	9.00	7.20	80%	Combined MRVL/LITE business; do not add twice

For non-EV cases, the displayed base cash-conversion ratio is also the explicit bear/bull conversion assumption. Applying it to each scenario produces the following cash checks. No separate FCF multiple or incremental asset is added to the earnings valuation. Merger cases change perimeter; LITE base/bull represent the combined business, not standalone LITE.

Ticker	Bear / base / bull owner FCF $bn
TSM	31.36 / 97.65 / 141.47
NVDA	106.40 / 371.45 / 665.00
AVGO	36.10 / 118.75 / 197.60
MU	15.60 / 62.40 / 117.00
ASML	8.64 / 21.42 / 33.30
AMAT	4.75 / 13.30 / 20.61
SNPS	1.53 / 4.08 / 6.65
ANET	3.46 / 9.97 / 16.61
VRT	1.02 / 5.10 / 9.38
DELL	2.54 / 10.72 / 19.01
CEG	1.71 / 4.20 / 6.82
PWR	1.28 / 3.84 / 6.08
CRWD	0.90 / 3.68 / 6.72
PLTR	2.24 / 9.04 / 17.21
AVAV	0.06 / 0.26 / 0.55
KTOS	0.02 / 0.15 / 0.33
MRNA	0.00 / 2.40 / 7.00
MP	0.01 / 0.29 / 0.64
INTC	0.00 / 2.12 / 5.53
GEV	1.60 / 7.80 / 13.44
CCJ	0.22 / 0.94 / 1.80
BE	0.04 / 0.98 / 2.22
COHR	0.42 / 3.00 / 5.18
CLS	0.57 / 2.75 / 5.20
NVT	0.47 / 1.45 / 2.55
MSFT	79.80 / 176.12 / 266.00
GOOGL	50.31 / 140.40 / 239.20
AMZN	25.50 / 81.00 / 140.40
META	32.20 / 89.60 / 159.60

Ticker	Bear / base / bull owner FCF $bn
LRCX	5.70 / 16.62 / 25.27
KLAC	3.71 / 9.03 / 12.92
CDNS	1.22 / 2.89 / 4.40
ETN	3.13 / 7.78 / 12.29
MOD	0.07 / 0.32 / 0.72
PANW	2.56 / 7.02 / 12.09
RELX	2.16 / 3.98 / 5.74
VRSK	0.76 / 1.34 / 1.90
MSI	1.87 / 3.76 / 5.62
AXON	0.19 / 0.90 / 1.85
GEN	0.73 / 1.37 / 1.99
AMD	4.68 / 25.00 / 55.20
ALAB	0.14 / 1.04 / 2.16
CRDO	0.26 / 1.19 / 2.45
HPE	1.60 / 4.76 / 8.80
FN	0.27 / 0.81 / 1.58
POWL	0.07 / 0.26 / 0.48
MRVL	1.08 / 7.20 / 12.78
LITE	0.65 / 7.20 / 12.78

For VST's base case, $11bn EBITDA must cover approximately $1.5bn cash interest, $1.1bn cash taxes, $2.2bn maintenance/environmental capex and $0.7bn other recurring obligations, leaving about $5.5bn before discretionary expansion. For NRG, $7bn EBITDA less approximately $1.0bn interest, $0.5bn tax, $0.7bn maintenance and $1.0bn recurring contract/customer costs leaves about $3.8bn. These **2030 assumptions** are not current guidance. The debt/share trajectories assume retained cash funds deleveraging and repurchases before optional expansion; they cannot also fund unlimited new plants. TLN's corresponding normalized owner cash is about $1.7–2.0bn in the base scenario.

For GPU lessors, EBITDA margins exclude the most important economic cost: replacement hardware. Base CRWV/NBIS/IREN cases require post-interest, post-tax, post-replacement owner cash margins of roughly 5–15%, after funding the modeled debt and dilution. Their EV multiples are therefore capped well below software multiples. CORZ/APLD, which are more colocation-oriented, require approximately 15–25% owner cash margins after recurring facility investment and financing; tenant credits, preferred equity and noncontrolling claims still belong ahead of or beside common owners. These are feasibility constraints on the terminal scenarios, not additional uncounted cash to add to equity value.

4.5 Current-holding underwriting notes

Each note addresses why an apparent discount exists, whether it is likely temporary or structural, the relevant peer/history context, a catalyst, and a falsifier. Numerical revenue, earnings, dilution, dividends and terminal returns are in the preceding ledger; the following judgments explain why those assumptions were chosen.

TSM — exceptional business, fair stock, 0%. Relative to its May price it is cheaper, but the discount cannot be separated from geographic concentration and extraordinary ongoing capital requirements. The base assumes roughly $310bn revenue, a 45% common margin and 5.2bn ADR-equivalent diluted shares by 2030; owner FCF is below earnings because fabs require reinvestment. A 23× terminal multiple gives meaningful growth credit without assuming geopolitical risk vanishes. GFS/Samsung/INTC do not match its leading-node economics. Catalysts are successful node/packaging ramps and durable customer commitments. Falsifiers are sustained yield or utilization deterioration, a materially worse geographic risk regime, or overseas expansion earning persistently inadequate returns. The stock is not a value trap in the ordinary operating sense; it is a risk-bearing asset whose expected return loses to customers and suppliers at these prices.

NVDA — attractive, 15%. The discount reflects fears about AI return on investment, custom ASIC substitution, export restrictions, customer financing and peak margins. Some concerns are structural; the portfolio does not assume they disappear. The base reduces the terminal earnings multiple to 25× and supports approximately $850bn revenue with a 46% normalized common margin, not perpetual 100% growth. Net share count falls only modestly; cash buys back stock rather than becoming a second terminal asset. FCF conversion is assumed near earnings because it is fabless, subject to inventory and ecosystem investment needs. AMD is a credible technical rival but has a less forgiving valuation; AVGO is complementary. Catalyst: useful workload demand and per-rack delivery become recurring cash. Falsifiers: gross margin below 65% for two quarters without a temporary ramp explanation, material uncollectible receivables, or structurally worsening price/performance competitiveness.

The August 27 Q2 10-Q, **STALE**, makes the financing risk concrete: supply commitments reached $279bn and a new class of AI-cloud agreements involved $36bn of commitments, generally over six years, reduced as capacity is consumed. These are not all immediate debt and must not be subtracted blindly from equity value; they are also not costless demand creation. The model assumes adequate utilization and ordinary cash conversion. A material cash call or unused committed capacity would invalidate that assumption and require lower margins/cash conversion, potentially lower weight. The 15% target is a cap, not a declaration that fabless means capital-risk-free.

AVGO — attractive, 15%. The market discounts a small set of major AI customers, project timing, possible customer insourcing, acquisition accounting and leverage. The base's 50% common margin is aggressive and only credible if software and high-value silicon retain substantial economics; it is not a claim that all hardware companies deserve that margin. The normalized measure charges recurring compensation and maintenance economics, rather than using every management add-back. Revenue of $250bn and a 24× multiple are enough to support the modeled return without an unlimited re-rating. FCF services debt, dividends and modest buybacks, so the same cash is not counted twice. Catalyst: diversified custom-silicon ramps plus software cash realization. Falsifiers: an anchor program loss, AI growth below 20% across two years, materially rising net debt despite strong reported cash flow, or persistent common-margin compression below the mid-40s.

MU — seemingly cheap peak earnings, fair normalized opportunity, 0%. The market is not obliged to put a software multiple on a memory supercycle. The base allows $240bn revenue but normalizes net margin to 40%; bear margin is 20%, bull 50%. Even these levels are strong and need HBM mix and discipline. The ledger separately reserves $150/share of terminal excess cash in the base, avoiding the opposite mistake of ignoring extraordinary current cash generation. Capital expansion consumes much of the rest; the share count does not simultaneously assume heroic repurchases. SK hynix is an important competitor, not a reason to model global memory as a monopoly. Catalyst: cash accumulation persists after meaningful new supply arrives. Falsifiers: HBM pricing/qualification weakens, inventory rises while lead times shrink, or industry additions force rapid margin reversion. A durable cash-return plan could make MU a future replacement; “sold out” does not establish that outcome through 2030.

ASML — excellent scarcity, fair-to-rich valuation, 0%. Its moat is qualitatively stronger than most holdings, but the stock already prices a very substantial earnings increase. The base implies about $70bn revenue and 34% net margin in USD, versus a terminal multiple of 30×. That is not a pessimistic view. Current euro financials and the ADR price must be translated consistently; no supposed cheapness from mixing currencies is accepted. The peer comparison is process-equipment cyclicality, not a claim that another vendor can replace EUV. Customer capacity planning and installation schedules can move revenue across years. Catalyst: shipment/service growth with strong cash realization. Falsifiers: prolonged customer capex cuts, export restrictions impairing service, or a technical

roadmap that reduces expected lithography intensity. Its 8.6% modeled return is insufficient for this mandate despite its strategic rank.

AMAT — fair, 0%. Lower price and stronger operating earnings improve the setup, but export exposure, customer capex and competitive process share are structural risks. Base revenue $50bn, 28% common margin, modest buybacks and a 24× terminal multiple are already constructive. FCF conversion is high after ordinary equipment-business reinvestment, not customer-fab spending. LRCX's faster earnings scaling is the better current equipment challenger; KLAC offers inspection quality. Catalyst: sustained process intensity and broad equipment/service demand. Falsifiers: share loss in key process steps, weak services resilience, or margin erosion despite rising system volumes. No need to own both AMAT and ASML for narrative completeness.

SNPS — fair, 0%. The de-rating creates a credible opportunity, but Ansys integration, debt, issuance and the gap between reported and adjusted earnings warrant discipline. The August balance sheet showed approximately $10bn debt and material goodwill/intangibles; the model does not value acquisition revenue as free organic growth. Base diluted shares remain around 200m, allowing more ownership than a pre-acquisition share count. Owner FCF is discounted for integration and compensation. CDNS is cleaner in some respects but more expensive against this model. Catalyst: debt reduction and cross-selling with sustained renewal strength. Falsifiers: organic bookings weakness, poor integration returns, repeated dilution without commensurate per-share growth, or AI tools undermining monetization faster than they expand demand. A better business than many selected companies can still be a worse stock at the margin.

ANET — fair, 0%. Strong recent operating execution justifies revising overly bearish competitive claims. The reason for excluding it is valuation-adjusted opportunity cost, not a prediction that Ethernet loses. Base revenue $30bn and 35% net margin with a 30× terminal P/E represent a very successful business. No extra terminal cash windfall is assumed on top of an already premium multiple; share dilution is modest. NVDA networking, custom switch silicon, open systems and customer concentration all limit unconstrained pricing. Catalyst: large-scale Ethernet deployments with stable economics. Falsifiers: structural share loss, chronic gross-margin pressure or a customer concentration event. AVGO supplies broader earnings exposure at the stronger modeled return.

VRT — attractive after a major reset, 15%. The May valuation concern has substantially dissipated without an equivalent decline in the earnings outlook. That is the strongest kind of price decline to investigate, but orders can include duplication and customers can defer delivery. Base revenue $30bn, 20% common margin and 26× earnings require operational maturity rather than a permanent shortage multiple. Normalized owner FCF is 85% of earnings after capacity/working-capital needs. Modine and nVent are real competitors; VRT's service, qualification and power/thermal integration support a premium. Catalyst: backlog converts into cash at stable margins. Falsifiers: two quarters of book-to-bill below one alongside cancellations, a persistent adjusted operating margin below 20%, or deteriorating cash conversion not explained by a demonstrably profitable ramp.

DELL — attractive headline multiple, less attractive normalized return, 0%. The low guided P/E is genuine, but the appropriate question is how much AI hardware revenue becomes sustainable common earnings and cash. The base assumes $300bn revenue but only 5.5% net margin, versus much stronger short-term reported earnings; component pass-through and mix can reverse. FCF conversion is modeled at 65% of earnings because working capital is economically important. Buybacks reduce shares only gradually and must compete with debt and inventory funding. HPE is a viable comparable; CLS's modeled growth-to-price ratio is better. Catalyst: repeat AI orders with sustained post-working-capital cash. Falsifier: AI growth expands receivables/inventory much faster than cash profits or produces margin compression. May's “integrator” label was not enough analysis; the new zero is an explicit valuation decision.

CEG — fair, 0%. Existing nuclear capacity is valuable, but this is not simply an untouched nuclear asset with riskless AI pricing. Acquisition perimeter, integration, financing, outages, hedging and political limits to ratepayer costs must be considered. The base common margin and 22× terminal P/E deliberately exclude a perpetual scarcity premium. The November 9 report is a near-term data gate, not information available here. VST and NRG offer better modeled entry economics; TLN is the closest concentrated nuclear challenger. Catalyst: contracted output converts to higher durable earnings without adverse policy or operating surprises. Falsifiers: outages or integration costs consume expected gains, or political/contracting arrangements prevent rent capture. Nuclear strategic

importance is not disputed by a zero weight.

VST — attractive, 11%. The EV model recognizes both valuable generating assets and substantial financial claims. It does not capitalize derivative mark-to-market gains. Base EBITDA reaches $11bn, net claims fall to $20bn, and diluted shares to 300m; these outcomes must be financed from cash after recurring investment, not from unlimited leverage. The 9× terminal EV multiple is deliberately ordinary. Lower realized margins and weaker deleveraging produce a large bear loss despite owning physical assets. Catalyst: contracted cash, Cogentrix integration and disciplined capital returns. Falsifiers: net leverage stays above roughly 3.5× through a strong operating period, acquisitions outrun cash generation, or post-growth cash falls persistently below the underwriting bridge. NRG is cheaper; VST supplies a different fleet/retail mix and reduces dependence on one power company's execution.

PWR — fair, 0%. It is a high-quality way to own the people and execution needed for infrastructure. Yet the new lower price still buys a large future margin and scale improvement. Base revenue $60bn and 8% net margin, plus 25× earnings, yield only about 10.9% modeled CAGR. Booked work is not free cash: labor, claims, customer terms, acquisition costs and equipment investment matter. MYRG and MTZ offer related exposure, not riskless cheaper substitutes. Catalyst: backlog converts with stable project economics and cash. Falsifiers: margin deterioration, project losses or working capital rising structurally faster than revenue. The final portfolio deliberately accepts less direct EPC exposure because electrical and cooling equipment provide stronger return estimates.

CRWD — expensive, 0%. The split-adjusted price is essential: using a pre-split EPS with a post-split price would fabricate cheapness. The thesis of rising security demand is intact, but owner FCF after compensation is much less than an unadjusted FCF headline. The base allows $16bn revenue, a 23% owner-FCF margin and a generous 35× cash multiple; it still does not support the current price. PANW is relatively less expensive but also fails the final return hurdle. Catalyst for reconsideration: materially faster durable revenue and lower dilution, or a substantially lower price. Falsifier of the zero: recurring owner cash per share compounds well above this model without an offsetting competitive or reliability cost. No “insurance” label can justify ignoring valuation.

PLTR — expensive, 0%. The operating improvement is real and substantial. The base nevertheless requires $28bn revenue, a 38% common margin and 40× terminal earnings; the bull requires $45bn and 45% margin. Those are already favorable assumptions. Client data remain client data, not a proprietary asset that Palantir can freely resell. SBC and the fully diluted share denominator cannot be omitted. MSFT is the stronger diversified platform challenger. Catalyst for reconsideration: durable large contracts and cash economics that exceed the model, with dilution constrained. Falsifiers: deal growth stalls, implementation burdens rise, government concentration increases, or the high-margin growth narrative fails to reach owners. Conversely, exceptional current growth does not prove every valuation is rational.

AVAV — speculative, 0%. The stock is cheaper after a difficult period, but the discount reflects integration, accounting complexity and cash conversion as well as sentiment. Base revenue $4bn, 10% common margin and 60m diluted shares allow a successful business without assuming all announced military demand becomes high-margin orders. LHX/CACI offer more established economics; none has shown a sufficiently superior AI-linked return to warrant a sleeve. Catalyst: funded backlog, delivery execution and ordinary free cash flow. Falsifiers: repeated integration charges, weak funded conversion, further dilution or a major contract/program loss. A diversified defense budget is not the same as a diversified equity cash flow.

KTOS — speculative, 0%. Program announcements and engineering credibility precede production revenue and profits, sometimes by years. The base requires $3.8bn revenue, a 7% net margin and a materially larger share count. Even a 28× multiple produces little expected advantage at today's price. The cash-use outlook makes the discount partly structural rather than simply temporary market pessimism. Catalyst: recurring funded production at attractive unit economics and positive FCF. Falsifiers: program delays, cancellations, repeated capital raises, or unmanned revenue failing to scale. The model does not assume a particular military outcome or procurement winner.

MRNA — speculative at a transformed price, 0%. Positive oncology evidence genuinely increases business value, but rapid stock appreciation also increases the revenue and margin hurdle. The base assumes $12bn 2030 revenue, a 25% margin and 450m diluted shares; bull $25bn and 35% margin, with separate limited residual cash. That valuation must respect partner economics, manufacturing capacity, reimbursement, launches, R&D and

continued financing. BNTX and diversified diagnostics challenge the assertion of a unique hedge. Catalyst: approvals and repeatable commercial economics, not a hypothetical catastrophe. Falsifiers: delayed or narrower approvals, poor uptake, manufacturing difficulty, cash burn or dilution beyond the model. Selling is not denying the scientific progress; it is rejecting a forward return and insurance claim at $254.19.

MP — speculative, 0%. The apparent bargain after a 47% fall rests on magnet ramp economics and policy-supported processing, not current common profit. The base assumes $2bn revenue and 22% net margin, allowing substantial success; it also allows dilution. Government support may reduce project risk without eliminating common-share risk, and one-off materials sales should not be annualized as recurring economics. FCX is a broad copper investment; Lynas a different rare-earth exposure. Neither solves gallium by owning MP. Catalyst: qualified magnet sales scale with positive cash after expansion. Falsifiers: poor yields/customer acceptance, policy changes, or sustained dependence on external funding. A floor under product pricing is not a floor under the stock.

INTC — speculative/value-trap risk, 0%. A much lower price is not enough when leading-edge foundry economics remain unproven and capital consumption is substantial. The base allows $85bn revenue and 10% common margin but also 5.8bn shares; the bull allows a genuine recovery. The bear $6 is a judgmental residual value, not a state guarantee. TSM is the operational comparator; GFS the more proven external-foundry model at different nodes. Catalyst: externally funded, profitable customer volume with credible process yields. Falsifiers: customer announcements never become revenue, capital requirements force repeated dilution, or internal transfers are mistaken for external demand. A Taiwan crisis could damage Intel's ecosystem and customers too; the purported hedge is unreliable.

GEV — strategically strong, fair-to-rich normalized stock, 0%. The discount reflects doubts about how much backlog and deposit-rich cash flow becomes sustainable profit. Deposits fund future obligations and cannot all be distributed without affecting delivery. Base earnings use $75bn revenue and 13% net margin, with normalized owner cash at 80% of earnings, rather than capitalizing the current exceptional FCF guide forever. Siemens Energy and Mitsubishi are real turbine competitors; equipment scarcity is not a literal monopoly. Catalyst: margins and delivery execution catch up with bookings. Falsifiers: warranty/service costs, customer cancellations, deposit reversals or weak cash conversion after the order wave. NVT and ETN offer stronger modeled valuation-adjusted alternatives.

CCJ — expensive on normalized owner economics, 0%. The stock is not a direct spot-uranium instrument. Contract timing, realized pricing, production costs, purchases and Westinghouse earnings create a complicated bridge to cash. The model converts the economics to USD and allows $5bn revenue and 25% net margin in the base; these are constructive assumptions. Kazatomprom adds jurisdiction/transport issues; LEU adds enrichment execution and policy risk. Catalyst: production reliability and higher-priced contracts translate to distributable cash. Falsifiers: outages force expensive purchases, cost inflation absorbs pricing, or new-reactor expectations do not become orders. Fuel scarcity alone does not justify the entry multiple.

BE — speculative, 0%. Revenue and margin progress are important, yet the common share price still requires large success. The base allows $10bn revenue and 15% net margin but 360m shares; the bull allows much more. Working capital, capex, service obligations and financing prevent adjusted operating earnings from equaling distributable cash. Gas connections, air permits, fuel price and equipment reliability remain relevant. CMI/CAT and grid-delivered generation are substitutes with different deployment constraints. Catalyst: commissioned customer capacity at profitable lifetime economics. Falsifiers: warranty degradation, delayed acceptance, adverse fuel economics, permitting problems or repeated capital raises. Announced multi-GW frameworks are not the same as delivered, paid, high-margin systems.

COHR — attractive but cash-conversion dependent, 8%. A lower multiple now accompanies much stronger optical demand. The discount also has a concrete reason: the latest quarter's operating cash flow was negative while capex was heavy, with additional borrowing. The base assumes $20bn revenue, 20% normalized common margin and 220m diluted shares; it requires manufacturing scale, yields and cash to improve. No separate cash windfall or dividend is added. FN offers a cleaner assembly model; LITE's proposed merger changes its exposure. Catalyst: 1.6T/advanced optical demand converts to cash after the capacity build. Falsifiers: two successive quarters of major cash burn without a credible funded ramp, net leverage rising above roughly 3× normalized

EBITDA, or structural margin pressure from architecture changes. This is the smallest selected weight because its return edge and confidence are weaker than the core positions.

4.6 Challenger underwriting — including the ten strongest excluded alternatives

The **top ten excluded challengers by modeled expected return**, not counting existing holdings, are MOD, META, CRWV, MSFT, TLN, LRCX, ETN, VRSK, RELX and AXON. Their exclusion is not concealed by only discussing weak alternatives. MOD and CRWV are particularly important: their arithmetic estimates are competitive, but the fragility of those estimates is materially higher. Additional fully modeled challengers address the possibility that a large discount, technological development or capital-structure change could alter the decision.

CLS — add 12%; deep discount with an execution price. October 26 guidance put 2027 revenue at $40bn and adjusted EPS at $21.50, after a $3.45bn equity offering. At $206.68, that is approximately **9.6× management's next-year EPS outlook**, versus 17.9× its current-year $11.55 outlook. Neither is a normalized mid-cycle multiple. Issuer results, Oct 26. The prior May price was $375.55; the decline reflects financing/dilution, customer concentration and fear that an assembler cannot retain AI economics. The base allows $65bn revenue in 2030, 6.5% common margin and 140m shares, explicitly below an assumption that current exceptional growth persists indefinitely. Owner FCF is only 65% of earnings after inventory/capacity needs. DELL/HPE have more diversified franchises, but lower modeled upside. Catalyst: funded orders and post-working-capital cash validate 2027 guidance. Falsifiers: adjusted operating margin below 7%, major customer insourcing, unfinanced inventory growth or another large dilutive raise without comparable value. The 12% cap recognizes limited pricing power relative to NVDA/AVGO.

NVT — add 12%; attractive, not a monopoly. The October 30 release raised 2026 adjusted EPS to $5.36–5.39 and reported 44% organic sales growth; Maverick Power closed October 1. Price $106.22 is about **19.8× that guide**. Q3 SEC release, Oct 30. Its May quote was $169.95; the discount is partly a growth/multiple reset and partly acquisition/capital-allocation risk. Base revenue $10bn and common margin 17% require sustained demand but not another four years at 44% growth. The model allows 160m shares and adequate retained cash for integration/debt reduction; it does not add the acquisition for free. ETN is the higher-quality diversified comparator, MOD a narrower thermal comparator. Catalyst: power-distribution capacity and new products convert orders at good cash margins. Falsifiers: organic order decline, leverage above about 3× for a prolonged period, or acquisition returns failing to cover the cost of capital.

NRG — add 12%; deep discount, stale earnings anchor requires an immediate gate. The August 4 guidance was $7.90–9.90 adjusted EPS, $5.325–5.825bn EBITDA and $2.8–3.3bn FCF before growth. The November 6 price is approximately **9.9× guided EPS midpoint**, materially below CEG's approximately 20.8×. Q2 results, Aug 4 — STALE. The discount prices gas/retail exposure, acquisition leverage and less glamorous assets, so part is structural. The base uses only $7bn EBITDA by 2030, an 8× EV multiple, $14bn net claims and 200m shares; a large re-rating is unnecessary. Growth projects must be customer-supported and leave room for deleveraging. Catalyst: contracted power economics and disciplined buybacks at low multiples. Falsifiers: materially weaker underlying cash, debt-funded expansion outrunning returns or retail/generation hedges failing. **Do not buy on this stale guidance before reviewing the scheduled November 9 results.** Issuer earnings-date notice, Oct 26.

MOD — excluded challenger #1; speculative despite 19.7% modeled return. November 3 continuing-business guidance is $3.1–3.2bn sales and $470–500m adjusted EBITDA after the October 1 spin-off. Datacenter sales grew rapidly but their gross margin fell, and first-half FCF was negative. Q2 FY27 results, Nov 3. The historical raw quote decline is **not** a clean shareholder-return or valuation comparison across the spin-off. The model uses continuing operations, $5.5bn base revenue, 9% net margin and 56m shares. Debt, working capital and capacity spending limit FCF conversion. Catalyst: a clean post-spin cost base and the November 18 investor update. Falsifier: continued growth without owner cash or persistent materials/tariff-driven margin erosion. It exceeds COHR's central estimate by about 0.7 points annually, but that difference is smaller than the uncertainty around perimeter and margins. VRT is the stronger cash/quality choice; NVT has a broader product route.

META — excluded challenger #2; serious alternative, not mechanically banned as a buyer. October 21 results showed 26% revenue growth and a large litigation charge; capex exceeded quarterly operating cash flow. Q3

results, Oct 21. The price was nearly unchanged from May despite business growth; litigation, heavy investment and uncertain AI monetization explain the discount. Base revenue $400bn, margin 32%, 2.6bn shares and 23× earnings produce 18.6% expected CAGR. Compensation, debt, legal payments and capex prevent simply annualizing a charge-adjusted quarter as owner cash. Catalyst: AI improves advertising conversion and new products monetize without equivalent incremental capital. Falsifiers: ad economics weaken, costs absorb growth, or governance/capital allocation dilutes common owners. It is the preferred non-hardware replacement if COHR's cash-conversion thesis fails; a 0.4-point model gap is not statistically meaningful.

CRWV — excluded challenger #3; speculative capital structure, not cheap software. August 11 Q2 figures included $2.575bn revenue, $1.510bn adjusted EBITDA, a GAAP operating loss and $640m quarterly net interest expense. Q2 release — STALE. Price declined roughly 59% from May. That is not proof of undervaluation when equity sits behind a growing debt/lease stack and hardware ages rapidly. The base allows $32bn revenue, 55% EBITDA margin, an 8× EV multiple, $65bn net financial claims and 750m shares; bear common equity is zero. Replacement capex must be deducted before judging economic profitability. Catalyst: capacity delivery, customer diversification and durable funding terms. Falsifiers: failed refinancing, tenant concentration stress, shortened GPU useful lives or dilution beyond the model. Its 18.0% arithmetic CAGR comes with 40% assigned zero-equity bear probability, making it a poor substitute for similarly priced mature earnings streams.

MSFT — excluded challenger #4; the best broad enterprise fallback. October 28 results included revenue $92.3bn, adjusted EPS $4.89, Azure growth of 48% and cash property/equipment additions of $39.4bn. FY27 Q1 results, Oct 28. At $530.61, annualized quarterly adjusted EPS implies about 27.1×; seasonality and leases make this a rough proxy. The stock rose approximately 28% in total return since May, so not all improved economics remain unpriced. Base $680bn revenue, 37% net margin and 29× earnings imply 17.6% expected CAGR, with FCF below earnings while infrastructure expands. Catalyst: agent/cloud monetization turns commitments into owner cash. Falsifiers: customer spending fails to earn returns, persistent capex escalation without cash gains, or bundling/reliability erodes economics. Strong alternative architecture holding, but lower upside than the selected core.

TLN — excluded challenger #5; fair, 17.2% modeled return. November 3 guidance put 2027 EBITDA at $2.5–2.85bn and adjusted FCF at $1.5–1.75bn, with a large repurchase program. Q3 results, Nov 3. The stock's smaller decline than other power names leaves a less compelling entry against VST/NRG. The model gives it $3.6bn base EBITDA, 9× EV, $4bn net claims and 45m shares, with no assumed dividend. That is meaningful buyback credit and must be financed. Catalyst: contracted nuclear economics and disciplined capital return. Falsifiers: outage/concentration risk, adverse contracting or regulatory outcomes, or financial claims rising rather than falling. It is a credible substitute if a selected power thesis fails, not a necessary third power stock.

LRCX — excluded challenger #6; better WFE value than the incumbents. October 21 reported revenue of $8.22bn and adjusted EPS $2.20, about 22.8× annualized quarter EPS at the cutoff price. September-quarter results, Oct 21. May-to-November price fell about 32%; the discount mixes temporary order timing with structural memory/export cyclicality. Base $50bn revenue, 35% net margin and 25× earnings yield 16.1%. Cash conversion is strong, and modest buybacks are allowed. Catalyst: etch/deposition intensity plus service growth across more complex devices. Falsifiers: memory capex retrenchment, process share loss or weaker China-access/service economics. It would be the first WFE re-entry, but not ahead of higher-return selected stocks at current prices.

ETN — excluded challenger #7; excellent quality, lower prospective return. November 3 outlook was $13.70–13.80 adjusted EPS; price is about 24.9× midpoint, versus NVT's approximately 19.8×. Q3 issuer release. The May price was higher, yet the quality premium remains. Base $48bn revenue, 18% net margin, 380m shares and 25× earnings yield 14.8%. The model respects acquisition financing and ordinary dividends; it does not call electrical order growth free cash. Catalyst: backlog conversion with sustained segment margins. Falsifiers: new capacity compresses returns, integration dilutes cash economics, or electrical demand proves duplicated. ETN is the preferred quality substitute for NVT if its integration risk becomes unacceptable.

VRSK — excluded challenger #8; genuine data moat, fair stock. November 5 results showed organic constant-currency revenue growth of 7.4%, adjusted EPS $1.96 and weaker quarterly FCF partly from timing. Q3 release. Price $204.59 is about 26.1× annualized adjusted EPS and higher than May, so it is not a distressed data

bargain. Base $4.5bn revenue, 33% common margin, 120m shares and 28× earnings imply 14.4%. Leverage and buybacks must be evaluated together; the smaller share count is not costless. Catalyst: insurers pay for verified proprietary updates and workflow integration. Falsifiers: customer consortium substitution, reduced data rights, pricing pressure or an AI-induced decline in information intermediation. It provides real economic diversification, but the expected-return sacrifice is too large for the chosen mandate.

RELX — excluded challenger #9; credible data/identity exposure. The October 22 update reported 7% underlying group growth and faster legal growth. Trading update, Oct 22. The ADR price rose since May as concerns about AI displacement eased. The discount question is structural: does agentic software increase demand for licensed trusted content or bypass the incumbent workflow? Base USD revenue $17bn, 26% common margin, 1.75bn share-equivalents and 24× earnings give 13.2%. Strong cash conversion funds dividends and modest buybacks while servicing debt. Catalyst: paid adoption of AI-enabled legal, scientific and risk tools. Falsifiers: commoditized content, rights litigation, deteriorating renewals or a gap between usage and monetization. Both a proprietary-data and identity candidate, not an automatic holding in either sleeve.

AXON — excluded challenger #10; speculative execution and ownership economics. November 5 results reported $909m revenue, delayed federal deployments and substantial expected annual SBC; an investment in Skydio adds another capital claim. Q3 shareholder letter, Nov 5. The price is lower than May, but growth spending and compensation make an EBITDA valuation flattering. Base $8bn revenue, 16% common margin and 95m shares generate 10.9% expected CAGR. Cash conversion is below earnings while hardware, inventory and capacity grow. Catalyst: subscriptions and integrated public-safety workflows become recurring owner cash. Falsifiers: procurement delay, legal/privacy constraints, service reliability failures, weak acquisition returns or continued dilution. MSI is more mature; neither deserves capital merely because surveillance expands.

GOOGL — fair, 0%. October 28 results reported $127.6bn revenue and $2.12 EPS available to common, plus rapid cloud growth. The balance sheet now includes substantial debt and mandatory-convertible preferred financing. Q3 release, Oct 28. The roughly 29% price decline since May is meaningful, but annualized common EPS gives about 33.6×, not an automatically low multiple. The base $800bn revenue/27% margin case assumes dilution to 13.1bn shares and owner cash below earnings; preferred conversion must not be ignored. Catalyst: Search and cloud/TPU economics overcome the financing and replacement-capex burden. Falsifiers: lower monetization per query, ongoing capital issuance without per-share accretion, or legal remedies that impair distribution. This is a real contender, rejected on valuation assumptions rather than buyer status.

AMZN — fair, 0%. October 29 results showed strong AWS growth and operating leverage, but trailing FCF was negative after large infrastructure outlays. Q3 release, Oct 29. A near-flat price since May is not evidence that all the growth is free: capex commitments and hardware replacement are real. The base $1.35tn revenue, 10% net margin, 11.4bn shares and 30× earnings are ambitious, yet imply only 9.0% expected CAGR. Owner FCF is 60% of earnings during continued expansion. Catalyst: AWS returns and retail productivity improve enough to fund capex internally. Falsifiers: continued capital intensity with weak pricing, excess capacity or material dilution. MSFT/META offer better modeled platform exposure.

KLAC — fair, 0%. October 28 EPS was $1.22 adjusted, on the **post-10-for-1-split** basis; price $138.49 is about 28.4× annualized quarterly EPS. FY27 Q1 release. Lower price since May does not eliminate a quality premium. Base $25bn revenue, 38% net margin and 28× earnings support 10.6% expected CAGR; strong cash conversion funds dividends/buybacks. Catalyst: inspection intensity rises as process complexity increases. Falsifiers: customer capex cuts, lost process-control share or weaker service retention. Better business defensibility than many challengers, but LRCX offers more modeled upside.

CDNS — fair-to-rich, 0%. October 26 full-year adjusted EPS guidance was $8.37–8.43, making the price approximately 38.1× midpoint, versus SNPS near 29.9× its guide. CFO commentary, Oct 26. The discount from May remains only relative to a richer former price. Base $10bn revenue and 34% common margin, with 35× earnings, imply 8.2% expected CAGR. Owner FCF is below earnings after compensation/reinvestment. Catalyst: AI-enabled design expands paid workloads and recurring revenue. Falsifiers: seat/product monetization breaks, export restrictions or integration demands impair returns. SNPS is the preferred EDA value, though both receive zero.

PANW — expensive in absolute owner-cash terms, 0%. September 1 guidance called for FY27 adjusted EPS $4.16–4.19 and a 38% adjusted FCF margin; the $313.84 price is approximately 75× adjusted EPS. FY26 results — STALE. Acquisition perimeter and share count changed with CyberArk, so raw year-over-year growth is not all organic. Base owner FCF is only 26% of $27bn revenue after compensation and economic adjustments, valued at 32×. Catalyst: integrated identity/security platforms produce cash per share rather than just consolidated ARR. Falsifiers: dilution, integration complexity, cloud bundling or weak retention. The relative rating versus CRWD does not justify a negative modeled expected return.

MSI — fair-to-rich, 0%. November 4 guidance raised adjusted EPS to $17.94–17.99, roughly 28.6× at $514.51. Q3 release. A roughly 35% total return since May has absorbed some operating improvement. Base $19bn revenue, 22% margin, 160m shares and 26× earnings yield 8.2%. Recurring service cash supports debt, dividends and buybacks; acquisition spending is not ignored. Catalyst: installed-base upgrades and software/services growth. Falsifiers: public-sector budget pressure, interoperability-driven pricing loss, legal/privacy restrictions or weak acquisition returns. Mature mission-critical systems are valuable, not unlimited-return assets.

GEN — low multiple, but insufficient normalized return, 0%. At $29.79, the stock is approximately 10.1× the $2.93–2.98 FY27 adjusted EPS guide. Q2 revenue was $1.369bn, up 12%; GAAP EPS $0.48 versus adjusted $0.74; year-to-date reported FCF $753m. October 28 release. The discount deserves investigation, not dismissal: management's current earnings basis is stronger than the conservative common-earnings basis used here. The model's base $6bn revenue, 24% common margin, 610m shares and 14× earnings only produces 5.3%, because it normalizes compensation/acquisition economics and assumes mature consumer growth rather than capitalizing all adjusted profit. A complete matched historical owner-cash valuation series is unavailable. Relative to RELX/VRSK, lower pricing power, leverage and competitive bundling justify a lower multiple. Catalyst: retention, cross-selling and deleveraging. Negative falsifiers: churn, lower pricing or acquisitions consuming cash; positive falsifier of exclusion: durable after-compensation cash validates the higher adjusted earnings trajectory. A privacy theme alone does not settle that accounting and growth debate.

AMD — speculative at the price, 0%. November 3 results included $13.4bn revenue, adjusted EPS $2.09 and a $17bn Q4 revenue outlook. Q3 results. At $449.04 it trades around 53.7× annualized quarterly adjusted EPS, materially above NVDA's comparable run-rate proxy. The base assumes $125bn revenue and 25% common margin, but 1.9bn diluted shares reflect financing/warrant risk. The bull is very successful; the expected return remains low because the entry price requires success. Catalyst: qualified rack-scale systems gain profitable volume. Falsifiers: weak software adoption, unfavorable customer terms, inability to meet supply commitments or dilution outpacing earnings. Strong competition is relevant to NVDA risk without making AMD the better stock.

ALAB — speculative, 0%. November 3 results showed quarterly revenue $584.9m and rapid scale-up switch adoption. Q3 release. At $190.71, the forward expectations remain high despite a lower quote than May. Base $5bn revenue, 26% net margin and 210m shares produce little expected return even at 32× earnings. Compensation, customer warrants and architectural competition are real costs. Catalyst: multiple high-volume design wins and broadening customers. Falsifiers: vendor integration bypasses discrete products, a concentrated customer changes architecture, or dilution exceeds the revenue benefit. It is a strong technological exposure with an insufficient stock-price cushion.

CRDO — speculative, 0%. September 1 quarterly revenue was $479m and adjusted EPS $1.20; current price is about 25.1× that annualized EPS, but GAAP earnings are lower and compensation is large. FY27 Q1 release — STALE. The decline since May improves valuation, yet concentrated customers and the copper-to-optics transition make the discount partly structural. Base $5bn revenue, 28% common margin and 220m shares yield 10.8%. Catalyst: qualified connectivity revenue broadens with good cash per share. Falsifiers: design displacement, reduced content per rack, price pressure or excessive dilution. AECs can thrive at some distances without making this the preferred optical-fabric investment.

HPE — fair, 0%. September 2 guidance raised FY26 adjusted EPS to $3.75–3.85 and FY27 growth expectations, with substantial FCF targets. Q3 release — STALE. The price is about 11.8× current-year adjusted EPS, but Juniper acquisition leverage, integration, legacy exposure and normalization explain much of the low multiple. Base $70bn revenue and 8.5% common margin, with 1.35bn shares, imply 10.3%. Dividends, debt and customer

financing compete for cash. Catalyst: networking synergies and profitable AI systems. Falsifiers: acquisition returns disappoint, commodity hardware margins erode, or reported profits do not become cash. It was fully tested, not discarded as “just servers.”

FN — fair, 0%. November 2 adjusted EPS was $4.34, while GAAP EPS was $2.86; the difference matters for ownership economics. FY27 Q1 results. Price $327.13 is approximately 18.8× the adjusted-quarter annualization after a large decline from May. Base $10bn revenue, 9% common margin and 37m shares avoid capitalizing all adjusted profit as durable owner earnings. Cash conversion is strong relative to many builders, but customer concentration and limited component IP constrain the multiple. Catalyst: high-value optical programs scale without proportionate capital demands. Falsifiers: customer insourcing, design change, regional disruption or compensation eroding per-share economics. COHR has more upside but more manufacturing/cash risk; that is an explicit tradeoff.

POWL — expensive on normalized assumptions, 0%. August 3 reported EPS $1.42 and backlog $2.4bn, including a large datacenter order. The April three-for-one split means **post-split shares and earnings** must be used. Q3 results — STALE, split filing — STALE. Price $133.12 is about 23.4× annualized-quarter EPS, not a single-digit P/E obtained with old shares. Base revenue $2.2bn and net margin 14% allow growth but normalize project mix; 38m shares and 20× earnings yield 4.8%. Catalyst: backlog converts with discipline. Falsifiers: project concentration, lower margins or stretched working capital. The apparently spectacular discount was not sufficient after correct units and normalization.

CORZ — speculative, 0%. November 5 reported 485MW billing capacity and rapidly higher colocation revenue; GAAP profit benefited from noncash warrant revaluation, while capex was nearly $1bn for the quarter. Q3 results. The stock fell sharply since May, but a warrant gain is not recurring earnings. Base EV assumes $2.2bn revenue, 55% EBITDA margin, 12× EV and $5bn net claims against 550m shares. Tenant concentration, construction costs and dilution remain central. Catalyst: contracted capacity earns reliable post-investment cash. Falsifiers: delayed billing, failed financing, tenant stress or adverse conversion/warrant dilution. The zero-equity bear case is real, not rhetorical.

IREN — speculative, 0%. November 5 results showed $200m AI-cloud quarterly revenue but a net loss; management identified $6bn of remaining targeted financing sources that were not yet committed for broader planned capex. FY27 Q1 release. “Fully funded” applied to a defined contracted program, not every future GW. The price decline is therefore partly a rational financing discount. Base $7bn revenue, 45% EBITDA margin, $13bn net claims and 550m shares require a successful buildout but still yield only 5.3%. Catalyst: commissioning and customer prepayments arrive on enforceable terms. Falsifiers: further delays, unavailable financing, expensive hardware replacement or dilution. Announced land/power pipeline is not a balance-sheet-equivalent asset at face value.

APLD — speculative, 0%. October 7 results showed $341.9m revenue but a large loss attributable to common holders. FY27 Q1 release — STALE by 32 days. Its lower share price must be considered against project financing, preferred/noncontrolling economics, tenant obligations and share issuance. The base uses $3bn revenue, 55% EBITDA margin, 12× EV and $10bn combined net financial/other equity claims against 450m shares. It is a possible successful infrastructure business with a less favorable common-equity payoff. Catalyst: energized leased capacity and distributable project cash. Falsifiers: delayed delivery, financing gaps, higher senior claims or weak tenant economics. No dividend is assumed while expansion consumes capital.

NBIS — speculative, 0%. August 12 shareholder disclosures described rising cloud revenue, $5.7bn quarterly capex, equity issuance and a shift to five-year server/network useful lives. Q2 shareholder letter — STALE. November 6 close was $129.49, not an overnight or tokenized-stock quote. The base allows $25bn revenue, 50% EBITDA margin, 9× EV and $45bn net claims, diluted to 500m shares. That produces slightly negative expected CAGR with a 40% zero-equity bear probability. Catalyst: asset-light partnerships and contracted usage reduce the need for fresh capital. Falsifiers: shorter economic GPU life than accounting life, capital-market closure, customer concentration or cash obligations exceeding nominal backlog. An asset-light initiative is promising evidence, not permission to value the entire business as software today.

MRVL — speculative, 0%. The November 2 proposed LITE combination adds optical scale and integration potential, but also a large new common-share denominator. The price declined from May; a standalone pre-merger

earnings multiple is no longer the right primary comparison. Base combined revenue $30bn, margin 30%, shares 1.38bn and 25× earnings generate about $164 per MRVL share plus the modeled dividend component. Catalyst: approval, integration and profitable combined design wins. Falsifiers: costly concessions, financing/convertible effects, failed closing or customer overlaps reducing growth. The bull is meaningful, but weighted CAGR is only 2.8%. The combined model is an independent approximation, not management pro-forma guidance. Nov 2 merger filing.

LITE — speculative transaction exposure, 0%. November 2 results showed $1.26bn quarterly revenue and adjusted EPS $4.57, but an investor at $538.59 must now evaluate the proposed 4.5-share exchange, not simply extrapolate standalone growth. FY27 Q1 results, Nov 2. The merger spread compensates for price movement and closing risk; no hedge is authorized or recommended. The base/bull use converted MRVL value, while bear assumes standalone earnings normalization. Preferred conversion, acquisition cash needs and dilution complicate an apparently cheap optical growth story. Catalyst: a value-accretive completion. Falsifiers: deal failure, poor terms, customer loss or margin deterioration. Its 7.0% modeled return does not beat COHR, and holding both acquirer and target would not create two independent risks.

4.7 Required recommended-holding valuation table

Terminal values below are **2030 stock value plus cumulative dividends**, not present values. The current quote date is November 6, 2026 for every row. “Deep Discount” is reserved for CLS and NRG because current management earnings anchors and conservative terminal multiples support the label; a large historical price decline alone does not. “Attractive” means the modeled return exceeds the portfolio hurdle with identifiable risks. Confidence assesses the underwriting, not certainty of positive returns.

Ticker	Current Price + Date	Valuation Status	Bear/Base/Bull Terminal Value	Scenario Probabilities	Expected Terminal Value	Implied Expected CAGR	Confidence	Key Assumptions
NVDA	$186.64; 2026-11-06	Attractive	$ 88 / 422 / 921	25 / 50 / 25%	$463.2	24.5%	4/5	Base revenue $850bn; margin 46%; 25x; shares 23.5bn
AVGO	$293.69; 2026-11-06	Attractive	$ 147 / 625 / 1229	25 / 50 / 25%	$656.7	21.4%	4/5	Base revenue $250bn; margin 50%; 24x; shares 4.9bn
VRT	$182.86; 2026-11-06	Attractive	$ 56 / 413 / 886	25 / 50 / 25%	$441.9	23.7%	4/5	Base revenue $30bn; margin 20%; 26x; shares 0.38bn
CLS	$206.68; 2026-11-06	Deep Discount	$ 62 / 513 / 1103	30 / 50 / 20%	$496.0	23.5%	3/5	Base revenue $65bn; margin 6.5%; 17x; shares 0.14bn
NVT	$106.22; 2026-11-06	Attractive	$ 58 / 238 / 492	25 / 50 / 25%	$256.5	23.7%	3/5	Base revenue $10bn; margin 17%; 22x; shares 0.16bn
NRG	$88.29; 2026-11-06	Deep Discount	$ 58 / 218 / 383	25 / 50 / 25%	$219.2	24.5%	3/5	$7bn base EBITDA; 8x; $14bn net claims; 0.200bn shares
VST	$124.80; 2026-11-06	Attractive	$ 51 / 268 / 488	25 / 50 / 25%	$269.0	20.3%	4/5	$11bn base EBITDA; 9x; $20bn net claims; 0.300bn shares
COHR	$192.81; 2026-11-06	Attractive	$ 38 / 436 / 840	30 / 50 / 20%	$397.6	19.1%	3/5	Base revenue $20bn; margin 20%; 24x; shares 0.22bn

The most debatable selection is COHR versus MOD/META. MOD's point estimate is higher but relies on a newly separated cost structure and improved cash conversion; META is only slightly lower and offers less direct hardware exposure. COHR's inclusion is a deliberate preference for qualified optical-component economics with a capped 8% weight, **not a claim of a statistically significant 0.4–0.7-point advantage**. If an investor cannot underwrite COHR's cash bridge, META is the cleaner replacement to investigate; the report does not force an optics sleeve at any price.

5. Portfolio architecture comparison and cross-portfolio capital competition

5.1 Three fully specified architectures

Every architecture is long-only, unlevered, sums to 100%, and stays at or below 15% per stock. “Current” means the supplied target allocation re-established at today's prices, not the drifted May portfolio. No ETFs or temporary implementation reserves enter these strategic weights.

Ticker	Current %	Concentrated %	Core-and-satellite %
TSM	11	0	5
NVDA	6	15	12
AVGO	3	15	10
MU	6	0	3
ASML	8	0	4
AMAT	4	0	0
SNPS	5	0	3
ANET	4	0	0
VRT	6	15	10
DELL	2	0	0
CEG	9	0	0
VST	6	11	8
PWR	4	0	3
CRWD	3	0	0
PLTR	2	0	0
AVAV	1	0	0
KTOS	1	0	0
MRNA	1	0	0
MP	3	0	0
INTC	3	0	0
GEV	5	0	0
CCJ	2	0	0
BE	3	0	0
COHR	2	8	6
CLS	0	12	8
NVT	0	12	8

Ticker	Current %	Concentrated %	Core-and-satellite %
NRG	0	12	8
MSFT	0	0	6
GOOGL	0	0	6
TOTAL	100	100	100

Architecture	Names	Expected total return %	Implied CAGR %	Bear/Base/Bull total return %	2.5-year delay return %	Top 3 %	HHI
Current diversified	24	61.5	12.2	-68.2 / 55.6 / 215.5	-0.0	28	0.0572
Concentrated — selected	8	134.8	22.8	-56.6 / 127.7 / 359.1	17.9	45	0.1292
Core-and-satellite	15	113.7	20.1	-56.6 / 107.3 / 311.6	13.2	32	0.0776

The architecture expected terminal wealth is the sum of each position's weighted expected wealth ratio. **This calculation does not assume independent stocks**: linearity of expectation holds despite correlation. However, correlation is critical to portfolio loss probabilities. No empirical covariance matrix or simulated portfolio-loss probability is claimed. The all-bear/base/bull columns are common-state stress endpoints, not three mutually exclusive portfolio outcomes with a single 25/50/25 distribution; individual-stock scenario probabilities differ.

5.2 What the numbers buy — and what they give up

Dimension	Current diversified bottlenecks	Concentrated eight — selected	Core-and-satellite fifteen
Company concentration	24 names; top three 28%; largest 11%	8 names; top three 45%; largest 15%	15 names; top three 32%; largest 12%
Effective holdings, 1/HHI	About 17.5	About 7.7	About 12.9
Economic allocation buckets	About 60% semiconductor/design/rack/fabric including INTC; 29% power; 5% security/operations; 6% other	77% compute/hardware/electrical delivery; 23% merchant generation/retail	69% hardware/design; 12% platform buyers; 19% power/EPC
Shared driver	Many tickers still rely on the same hyperscaler budgets, financing and grid build	Higher exposure to profitable conversion of AI capex; customer overlap is substantial	More platform cash-flow sources, but still largely the same AI investment cycle
Permanent-loss mechanism	Several expensive or speculative small holdings plus common-cycle exposure	A major customer/technology failure hits a much larger position; leverage/cash conversion matter at NRG/NVT/COHR/CLS	Lower single-name failure damage, but more lower-return capital and still meaningful common-cycle risk
Single-stock zero at maximum weight	−11% of EHF	−15% of EHF	−12% of EHF
Acceleration upside	Captures many links but dilutes the strongest entry valuations	Highest modeled upside; some component scarcity may migrate away from selected suppliers	Strong upside with less concentrated participation
Two-to-three-ye ar delay	More time for premium multiples and speculative financing to disappoint	Mature earnings and lower entry multiples help; still painful drawdown risk	More cash-generating platform exposure helps, but at lower expected upside
Main opportunity cost	Too much capital in strategic importance at inadequate prices	Gives up direct TSM/ASML/MU, platform, security, data, defense and bio upside	Gives up roughly 2.8 points of modeled annual return versus the eight

These are allocation buckets, **not measured percentages of company revenue derived from AI**. Calling NVT “electrical” and VRT “cooling” does not make them independent of the same customer budget. Likewise, owning VST and NRG does not provide a bond-like hedge against AI overbuilding.

5.3 Faster acceleration, delay and capex slowdown

Acceleration: the simultaneous bull endpoint is approximately +359% for the selected basket, versus +215% current and +312% core-and-satellite. It assumes successful deployment, high utilization, strong margins and supportive valuations. Faster capabilities can also accelerate substitution, regulation or obsolescence; the stock bull case is not synonymous with the fastest technological timeline.

A 2.5-year delay: a deliberately simple sensitivity compresses each stock's base terminal-wealth growth into the remaining 1.651 years and applies a 15% valuation haircut: 0.85 × (base wealth/current price)^((4.151−2.5)/4.151). The resulting cumulative outcomes are about **0% current, +17.9% concentrated and +13.2% core-and-satellite** through 2030. This is a mechanical wealth sensitivity, not a detailed production schedule or a new probability-weighted forecast. It does not mean the selected portfolio avoids a large interim drawdown. A delay

accompanied by funding failure is much worse than this orderly deferral.

A sharp AI-capex slowdown: model an immediate 35–60% price shock to selected holdings: NVDA −50%, AVGO −45%, VRT −55%, CLS −60%, NVT −40%, NRG −35%, VST −40%, COHR −60%. The weighted portfolio loss is **−47.9%** before any rebound, dividends or tax effects. These are explicit hypothetical shocks, not estimated betas or a confidence interval. A long-term recovery requires that customers still have profitable workloads and balance sheets can fund the wait. The current architecture's simultaneous bear endpoint is worse than the concentrated basket's, largely because current price-to-normalized-value gaps are worse—not because concentrated portfolios are inherently safer.

For an apples-to-apples immediate shock across all architectures, apply the same company shocks wherever a stock appears. The additional assumptions are TSM/ASML/AMAT/GEV −45%; MU −60%; SNPS/CCJ/MSFT −30%; ANET/DELL/INTC −50%; CEG/PWR −35%; CRWD/GOOGL −25%; PLTR/MP −40%; AVAV −15%; KTOS −20%; MRNA −5%; BE −60%. This yields **−43.55% current, −47.90% concentrated, and −44.70% core-and-satellite**. Thus concentration is worse in this near-term capex shock even though its better starting valuations produce a less negative modeled 2030 bear endpoint. MRNA's mild assumed decline is not a pandemic-hedge forecast. The shock vector is a transparent thought experiment, not a covariance estimate or a liquidation instruction.

Forecast fragility: reducing all selected expected terminal wealth by 20% lowers the implied CAGR from 22.8% to approximately **16.4%**. A 30% haircut lowers it to about **12.7%**. A uniform terminal-multiple reduction is not identical to this total-wealth haircut, particularly for EV models with debt. The comparison nevertheless demonstrates that small confidence errors can consume most of the apparent advantage over a broad equity benchmark. The opportunity is substantial; the precision is not.

5.4 Why choose eight, not twenty-four or fifteen?

The current 24 positions are excessive **for this objective and these valuations**, not in some universal portfolio-theory sense. Small allocations often add tracking work and narrative comfort without materially hedging the dominant factor. MRNA and INTC are the clearest examples of labels masquerading as reliable hedges. The alternative fifteen is defensible for an investor who values smoother single-name exposure more highly, but its additional names do not improve expected return in this model.

The selected eight deliberately accept higher idiosyncratic concentration and shared AI-demand exposure for stronger entry-price economics. The default 15% cap binds NVDA/AVGO/VRT. Execution/financing-sensitive CLS/NVT/NRG are capped at 12%; aggregate merchant power is kept at 23%, leaving VST 11%; COHR receives the residual 8% because of weaker cash-conversion confidence. These are **judgmental permanent-loss limits**, not results of an empirically calibrated optimizer. They explain why the arithmetic highest-ranked NRG is not automatically 15% and why the final weights are not a mechanical return-ranked fill.

The choice is decisive but conditional. If the investor cannot tolerate a plausible 50–60% EHF loss, the correct answer is not to pretend that fifteen AI-related names eliminate the risk. It is to revisit the overall EHF risk budget with the investor. This assignment retains the prescribed approximately 50% overall allocation, and does not silently expand Foundation or change Continuity.

5.5 Every selected stock must defeat another use of capital

For comparability, the table gives the estimated annual-return gap against a credible **excluded** replacement. The highest-return selected alternatives are already at judgmental risk caps, so they are not unlimited marginal destinations. A replacement is an underwriting destination, not a standing market order; if removal is caused by a common shock, replacing it with a correlated peer may be wrong.

Selected stock	Expected CAGR	Next-best relevant excluded contender	Estimated advantage	Why capital belongs here / destination if removed
NVDA	24.5%	MSFT 17.6%	+6.9 points	Better modeled cash growth per paid dollar, despite higher cycle risk. MSFT is the first replacement for a company-specific NVDA failure that leaves cloud economics intact.
AVGO	21.4%	LRCX 16.1%	+5.3 points	Custom silicon/software cash economics beat another indirect chip-capex exposure. LRCX receives the allocation if AVGO-specific program risk changes and WFE economics remain valid.
VRT	23.7%	MOD 19.7%	+3.9 points	Broader qualified deployment/service and better cash visibility. MOD is only a replacement after post-spin cash/perimeter verification; otherwise ETN.
CLS	23.5%	DELL 11.6%	+11.9 points	Growth-to-price advantage compensates for lower moat and customer concentration, capped at 12%. DELL receives capital only if its post-working-capital cash merits the replacement at then-current prices.
NVT	23.7%	ETN 14.8%	+8.8 points	Lower entry multiple with strong electrical growth; accept integration risk. ETN is the quality replacement if the NVT-specific acquisition thesis fails.
NRG	24.5%	TLN 17.2%	+7.3 points	Low entry earnings multiple and ordinary terminal EV assumptions; accept gas/retail/leverage exposure. TLN is the replacement after checking current debt and price.
VST	20.3%	TLN 17.2%	+3.1 points	Better modeled entry economics with a broader fleet/retail mix; NRG already at its cap. TLN is the first independent replacement, not an increase beyond NRG's limit.
COHR	19.1%	META 18.6%; MOD 19.7%	+0.4 versus META; **−0.7 versus MOD**	Optical exposure is retained on expected cash economics and confidence, not a claimed numerical edge over every challenger. META is the preferred replacement; MOD must first pass its lower-confidence post-spin cash test.

The negative COHR/MOD difference is intentional disclosure. An allocation decision can prefer a slightly lower central expected return when the higher number is much less credible. It would be misleading to “fix” the model to make every chosen stock outrank every unchosen one. The eight-stock basket's apparent advantage also depends heavily on the assumed high terminal margins at NVDA/AVGO and the successful conversion of CLS/NVT/VRT growth into cash; those are the most important assumptions to challenge.

5.6 November 9 forecast-driven challenge: should upstream scarcity change the allocation?

The deeper research makes a direct TSM/MU/ASML allocation more compelling **strategically**. It does not automatically make those stocks the best purchases at the November 6 prices. To avoid defending the original allocation merely by repetition, I tested a fourth, fully specified architecture and changed the upstream operating assumptions in a separate sensitivity.

Upstream challenge, not the recommendation: NVDA 15%, AVGO 15%, VRT 15%, CLS 12%, NVT 9%, NRG 8%, VST 7%, COHR 4%, TSM 5%, MU 5%, ASML 5% = **100%**. Relative to the selected eight, fund the 15% upstream addition from NVT −3, NRG −4, VST −4 and COHR −4 percentage points. This reduces direct generation from 23% to 15%, adds three names, keeps the top three at 45%, and reduces company HHI from 0.1292 to 0.1104. It does not diversify away the common AI-capex or Taiwan manufacturing drivers. All eleven stocks already have full underwriting in section 4.

Architecture	Original commercial expected TR / CAGR	50%-bull probability sensitivity: TR / CAGR	Stronger upstream operating assumptions only: TR / CAGR
Current 24	+61.5% / 12.2%	+119.4% / 20.8%	+71.9% / 13.9%
Selected eight	+134.8% / 22.8%	+219.3% / 32.3%	+134.8% / 22.8%
Core-and-satellite 15	+113.7% / 20.1%	+188.1% / 29.0%	+118.7% / 20.7%
Upstream challenge 11	+124.5% / 21.5%	+205.9% / 30.9%	+130.7% / 22.3%

Probability sensitivity: assign each stock 50% to its existing commercial bull endpoint, halve its original bear probability, and put the remainder in base. Thus 25/50/25 becomes 12.5/37.5/50 and 30/50/20 becomes 15/35/50. Every distribution sums to 100%. Expected terminal wealth is recomputed before CAGR; CAGRs are not averaged. This deliberately optimistic test is **not** a new estimate of AGI probability and is not adopted as the headline return forecast. Applying it to every stock prevents selectively making only the challengers bullish.

Operating sensitivity: without changing probabilities, shares, valuation multiples, dividends or excess cash, change TSM's bear/base/bull revenue to $140/$370/$520bn and net margins to 32/47/49%; MU to $120/$280/$450bn and 20/42/50%; ASML to $40/$90/$130bn and 24/37/40%. Other stocks are held unchanged. These are intentionally favorable analyst assumptions for upstream scarcity, not company guidance. The eleven-name challenger closes most of the gap but still trails by about 0.5 percentage point annualized. A further upstream-only rerating can reverse the ranking; the result is not robust enough to call eight mathematically optimal.

Do not confuse the three-stock package with the marginal stock decision. I also tested each upstream name separately at 5%, funded solely by reducing COHR from 8% to 3%, with the other seven weights unchanged. Under original assumptions, the resulting portfolio CAGRs are 22.60% for TSM, 22.58% for MU and 22.42% for ASML. Under the stronger upstream operating assumptions, they become 22.845%, 22.780% and 22.745%, respectively. Thus the TSM-only substitution **narrowly wins** in that sensitivity: TSM's own modeled CAGR rises to 19.36%, above COHR's 19.05%. The portfolio advantage over the selected eight's 22.831% is less than 0.02 percentage point annually—far below credible modeling precision. This demonstrates a genuine reversal, not an invariant ranking. The recommendation remains unchanged because these stronger operating cases are unadopted counterfactuals, not independently established forecasts; sustained evidence making them the better central estimates would reopen the TSM/COHR choice immediately.

Under original probabilities, the upstream challenge's simultaneous bear/base/bull terminal returns are −57.5% / +116.9% / +340.6%, versus −56.6% / +127.7% / +359.1% for the eight. Its mechanical 2.5-year-delay return is +15.4%, versus +17.9%; using the same immediate capex-shock vector as section 5.3 gives −48.80%, versus −47.90%. Restoring upstream holdings therefore does not automatically improve the modeled downside. It can improve exposure to a *different location of future rents*, which is a more defensible reason to own them than generic diversification.

The following thresholds show what would overturn a marginal exclusion. They hold all other assumptions fixed and compare against COHR's approximately 19.1% original modeled CAGR, the lowest among the chosen holdings. They are **relative-value reference points, not standing buy orders**; COHR's outlook and price must be refreshed simultaneously.

Challenger	Nov 6 price	Original modeled CAGR	Price at unchanged expected wealth matching COHR's CAGR	Increase in expected terminal wealth required at current price
TSM	$375.44	13.5%	About $308	+22%
MU	$690.00	13.0%	About $556	+24%
ASML	$1,424.75	8.6%	About $972	+47%
LRCX	$200.44	16.1%	About $180	+11%
MSFT	$530.61	17.6%	About $504	+5%
GOOGL	$284.78	10.2%	About $206	+38%

For the complete 15-point upstream substitution, a uniform approximately **43% increase in the three upstream stocks' expected terminal wealth**, with all displaced stocks unchanged, eliminates the original architecture's advantage. That is an earnings/shareholder-value hurdle, not 43% more FLOP or 43% more scanner utilization. Under the targeted operating sensitivity above, the required additional gap is much smaller. The deeper work therefore raises the priority of TSM/MU/ASML monitoring and lowers confidence in treating their zero weights as durable decisions.

Decision after the challenge: retain the eight weights, with architecture-superiority confidence 3/5 and little confidence in the exact return gap. The company scenarios already assume a very large buildout—NVDA $850bn base revenue, AVGO $250bn, VRT $30bn and CLS $65bn in 2030. Upstream strategic importance is not new enough, by itself, to warrant another blanket upward revision. The new evidence is more useful for identifying where the earnings assumptions could be wrong and when to revisit the exclusions. If realized EUV/HBM scarcity translates into sustained after-tax cash margins, profitable capacity expansion and stronger per-share earnings than these cases, restore upstream exposure through a fresh capital-competition calculation; do not wait for a scheduled annual review. Conversely, if algorithmic progress reduces physical intensity or policy captures the rents, strengthen the case for platform/distribution alternatives such as MSFT, rather than automatically buying more hardware suppliers.

This is a substantive update without an arbitrary trade: greater weight on rapid research automation; a direct upstream challenge with exact funding; stronger operating cases; explicit reversal thresholds; and acknowledgment that a narrow model advantage is not proof of a uniquely best portfolio. No changes have been made to the actual account or the active house allocation.

6. Final allocation + checksum (100.0%)

6.1 Strategic allocation

Ticker	Theme	EHF target %	Approx. total net investments %	Modeled CAGR %
NVDA	Accelerators/platform	15.0	7.5	24.5
AVGO	Custom silicon/networking/software	15.0	7.5	21.4
VRT	Power/thermal infrastructure	15.0	7.5	23.7
CLS	Systems/rack integration	12.0	6.0	23.5
NVT	Electrical distribution	12.0	6.0	23.7
NRG	Generation/retail	12.0	6.0	24.5
VST	Generation/retail	11.0	5.5	20.3
COHR	Optical components	8.0	4.0	19.1
TOTAL	Eight common-stock positions	100.0	50.0	22.8 portfolio

Exact checksum: 15 + 15 + 15 + 12 + 12 + 12 + 11 + 8 = 100.0%. Largest holding 15%; Moonshot 0%; strategic cash 0%; ETFs 0%; derivatives 0%; leverage 0%. Event Horizon remains approximately 50% of total net investments. The approximate total-investment weights in the table assume exactly 50% for illustration and exclude indirect overlap through Continuity funds.

6.2 Complete From / To / Delta and per-ticker decision framework

The “From” weights are the **supplied V9.1 targets**, not assumed live holdings. Actions and confidence apply to this independent proposal. All 19 removals are deliberate. There are no unchanged “Keep” positions and no partial “Reduce” positions in the final target; the five retained names all increase.

Ticker	Current % / From	Recommended % / To	Delta pp	Action	Confidence (1–5)	Why	What would falsify this
TSM	11.0	0.0	-11.0	Remove	4	Strategic foundry; 13.5% model CAGR loses capital competition.	Higher durable pricing/cash margins or lower price makes return superior.
NVDA	6.0	15.0	+9.0	Increase	4	24.5% model CAGR; scalable profits and substantial valuation reset.	Sustained margin erosion, unpaid receivables, major program substitution.
AVGO	3.0	15.0	+12.0	Increase	4	21.4% CAGR; custom silicon plus software cash at reduced multiple.	Anchor-customer loss, structurally sub-mid-40s common margin, leverage deterioration.
MU	6.0	0.0	-6.0	Remove	3	Peak-cycle cheapness is real but not durable enough; 13.0% CAGR.	HBM sustains high owner margins and cash beyond modeled normalization.
ASML	8.0	0.0	-8.0	Remove	3	Best lithography franchise, not best stock return; 8.6% CAGR.	Faster profitable shipments or lower price lifts return above selected names.
AMAT	4.0	0.0	-4.0	Remove	3	9.6% CAGR; LRCX is stronger tool alternative, both below final basket.	Higher process share/cash conversion supports materially higher owner value.
SNPS	5.0	0.0	-5.0	Remove	3	12.9% CAGR; acquisition/debt/dilution constrain attractive moat.	Integration produces stronger after-compensation cash with controlled shares.
ANET	4.0	0.0	-4.0	Remove	3	Operating success partly priced; 9.9% modeled CAGR.	Durable share gains and cash growth exceed 2030 assumptions.
VRT	6.0	15.0	+9.0	Increase	4	23.7% CAGR; improved earnings with sharply lower entry valuation.	Cancellations, weak book-to-bill and structural margin/cash deterioration.
DELL	2.0	0.0	-2.0	Remove	3	11.6% CAGR; CLS offers higher per-share upside with explicit risk caps.	Cash conversion and sustainable integration margins exceed model.
CEG	9.0	0.0	-9.0	Remove	3	12.7% CAGR; NRG/VST cheaper expected power cash flows.	Contracted nuclear/acquisition cash improves without offsetting liabilities.
VST	6.0	11.0	+5.0	Increase	3	20.3% CAGR; dispatchable generation and hedged cash bridge.	Outages, adverse rules or debt/integration undermine owner cash.
PWR	4.0	0.0	-4.0	Remove	3	10.9% CAGR; great backlog at a demanding earnings entry price.	Profitable backlog conversion and FCF support a superior return.
CRWD	3.0	0.0	-3.0	Remove	4	After-SBC valuation does not justify new capital despite strong growth.	Growth persists with much less dilution and much stronger owner FCF.

Ticker	Current % / From	Recommended % / To	Delta pp	Action	Confidence (1–5)	Why	What would falsify this
PLTR	2.0	0.0	-2.0	Remove	4	2.3% CAGR; excellent growth needs excessive future value at this price.	Durable common cash/margins substantially exceed modeled trajectory.
AVAV	1.0	0.0	-1.0	Remove	3	9.3% CAGR; procurement and integration do not justify token hedge.	Funded growth converts to cash with controlled dilution at attractive price.
KTOS	1.0	0.0	-1.0	Remove	3	Negative modeled CAGR; financing/cash burden outweighs autonomy story.	Sustained positive FCF and funded production scale beat the assumptions.
MRNA	1.0	0.0	-1.0	Remove	4	Rally rewards oncology, not reliable pandemic insurance; negative model CAGR.	Approved/commercialized oncology creates much higher sustainable owner cash.
MP	3.0	0.0	-3.0	Remove	3	8.5% CAGR; processing opportunity is not common-equity insurance.	Recurring magnet cash margins validate stronger independently economic returns.
INTC	3.0	0.0	-3.0	Remove	4	Development milestones do not offset foundry cash/dilution risk.	Competitive high-volume external foundry economics and FCF are verified.
GEV	5.0	0.0	-5.0	Remove	3	8.3% CAGR; strategic turbine scarcity already commands a premium.	Profitable deliveries and recurring cash exceed deposit-sensitive model.
CCJ	2.0	0.0	-2.0	Remove	3	3.4% CAGR; contract/production economics do not match scarcity premium.	Volumes and realized contract cash improve enough to beat power alternatives.
BE	3.0	0.0	-3.0	Remove	4	Negative modeled CAGR; growth/cash/financing demands exceed entry margin of safety.	Sustained unsubsidized owner cash and deployments exceed model with low dilution.
COHR	2.0	8.0	+6.0	Increase	3	19.1% CAGR; optical upside but cash demands cap weight at 8%.	Persistent negative operating cash, uncontrolled capex/debt or technology share loss.
CLS	0.0	12.0	+12.0	Add	3	23.5% CAGR; low guided-earnings multiple with substantial system growth.	2027 guide fails, margins compress or dilution/working capital consume upside.
NVT	0.0	12.0	+12.0	Add	3	23.7% CAGR; electrical distribution at better value than premium peers.	Organic orders/cash weaken or acquisition leverage defeats per-share growth.
NRG	0.0	12.0	+12.0	Add	3	24.5% CAGR; inexpensive normalized earnings and customer/power economics.	New results cut cash outlook; integration/leverage or retail margins deteriorate.

The 32 fully underwritten challengers not currently owned are also explicitly **Do not add / 0%** unless CLS, NVT or NRG is shown as Add above. Their price, valuation, scenarios, rationale and reconsideration triggers are documented in Section 4. A “Remove” target is not a direction to sell a tax lot without examining actual holdings and costs.

6.3 Exact funding map — every increase, including every new stock

All figures are percentage points of Event Horizon at the supplied target baseline. This is a budget map, not a lot-level trade ticket. Dollar proceeds in an actual account must be recalculated from live NAV, current drift and tax lots.

Existing source reduced to zero	Released %	Exact destination(s)
TSM	11	NVDA +9; AVGO +2
ASML	8	AVGO +8
AMAT	4	AVGO +2; CLS +2
SNPS	5	CLS +5
ANET	4	CLS +4
DELL	2	CLS +1; NVT +1
CEG	9	NRG +9
PWR	4	NVT +4
CRWD	3	NVT +3
PLTR	2	NVT +2
AVAV	1	NVT +1
KTOS	1	NVT +1
MRNA	1	NRG +1
MP	3	NRG +2; VST +1
INTC	3	VST +3
GEV	5	VST +1; VRT +4
CCJ	2	VRT +2
BE	3	VRT +3
MU	6	COHR +6
Total released	**77**	**NVDA 9 + AVGO 12 + VRT 9 + CLS 12 + NVT 12 + NRG 12 + VST 5 + COHR 6 = 77**

Starting retained capital is 23%; released/reinvested capital 77%; final 100%. Purchases equal sales before costs. New-stock funding is explicit: **CLS 12 = AMAT 2 + SNPS 5 + ANET 4 + DELL 1; NVT 12 = DELL 1 + PWR 4 + CRWD 3 + PLTR 2 + AVAV 1 + KTOS 1; NRG 12 = CEG 9 + MRNA 1 + MP 2.** No funding comes from Foundation or Continuity.

The 77% one-way target turnover is high. On the hypothetical unrebalanced May portfolio, the retained names would total only about 19.1%, so the actual reallocation would be closer to 80.9% of that portfolio's November NAV. Neither number is a broker-verified trade requirement. Tax drag can reduce the model's advantage: a one-time 5%

portfolio cost reduces the selected expected wealth by 5% and its implied CAGR to roughly 21.3%, before any market impact. That is an illustration, not an estimate of this investor's tax bill.

7. Risk, scenario, and invalidation dashboard

7.1 Top five risks

Risk	Why it threatens permanent capital	Observable trigger	Required response
1. AI investment earns inadequate end-user returns	Common customers cut spend across compute, networking, racks and electricity; backlog may be double-counted	Two major paying platforms materially cut comparable capex plans; utilization/pricing and supplier cash receipts weaken together	Freeze remaining buys, re-underwrite the whole basket, reduce expected terminal revenue rather than merely widening a price target. Do not replace one exposed supplier with another automatically.
2. Taiwan / export / geopolitical interruption	Direct holdings and apparently US-only suppliers depend on a global manufacturing system	Actual production/shipping interruption, binding restrictions affecting delivery, or a material change in insurability/counterparty access	Treat as a portfolio event; prioritize liquidity and loss containment. INTC/MP are not preapproved hedges. No unpriced assumption that government support rescues common shareholders.
3. Cash-conversion and financing failure	Profit without cash forces dilution, expensive debt or lost growth; greatest at CLS/COHR and acquisition-sensitive NVT/NRG	Two quarters of unexplained cash deterioration; covenant/refinancing stress; capital raise materially outside modeled shares/claims	Halt additions to the affected name; test liquidity runway and debt obligations; use the named replacement only if its independent thesis survives.
4. Technology or capacity eliminates scarcity rents	ASICs, networking integration, optical changes, improved utilization or new capacity compress price and margins	Falling lead times plus falling prices and declining customer urgency; lost design wins; sustained normalized margin below model	Reduce terminal margin/multiple first, not only revenue growth. A cheaper stock may still become a worse investment.
5. Power affordability, permitting and political intervention	A real shortage can prompt price caps, customer-funded solutions, slower approvals or demand relocation rather than windfall owner profit	Binding rules/contract changes that reduce returns; delayed commissioned MW; material outages; sustained poor project ROIC	Re-underwrite VST/NRG/NVT/VRT jointly. Distinguish proposed from enacted measures and capacity from actual delivered MWh.

These numerical triggers are review rules, not mechanical stop-loss orders. A price fall alone does not prove impairment; a price rise does not excuse a broken thesis. Conversely, waiting for two quarterly confirmations is inappropriate when there is a definitive liquidity, fraud, legal or operational event.

7.2 Scenario probabilities and uncertainty discipline

For long-horizon company underwriting, the default 25/50/25 bear/base/bull distribution is adjusted for fragile cases in the ledger. It represents a range of commercial outcomes, not an AGI-timeline probability. Published subjective forecasts **do** exist: section 3.4 now records the AI Futures authors' distributions and their no-imposed-slowdown conditioning. They should influence scenario design, but they cannot be copied directly into stock bull probabilities. Commercial outcomes also depend on financing, rents, dilution, competition and governance. No objectively established probabilities for a Taiwan blockade or engineered pandemic were obtained in this audit.

New explicit regime overlay. The 22.8% headline is conditional commercial underwriting. It is not an unconditional expected return across all futures described in AI 2027. The race ending includes an outcome in which conventional investor wealth ceases to be useful; a monetary expected-value calculation cannot meaningfully resolve human welfare in that world. More ordinary impairment of enforceable claims—confiscation, forced restructuring, prolonged market closure—can nevertheless be stressed financially. The table below adds a separate zero-recovery-by-2030 state, outside the existing commercial distribution. It is an illustrative valuation sensitivity, **not an extinction estimate**.

Expected wealth per $1 = (1 − q) × commercial expected wealth + q × recovery; here recovery = 0 and CAGR = expected wealth^(1/4.150582) − 1. Commercial probabilities retain their relative weights within the surviving regime,

so total probability remains 100%.

Additional zero-recovery regime q	Selected eight implied CAGR	Core-and-satellite CAGR	Current architecture CAGR
0% — original commercial model only	22.8%	20.1%	12.2%
5%	21.3%	18.6%	10.9%
10%	19.8%	17.1%	9.4%
20%	16.4%	13.8%	6.4%
30%	12.7%	10.2%	3.0%

A common proportional tail does not change the ranking: it reduces all architectures' expected wealth together. That is a property of this sensitivity, not proof their actual tail exposures are equal. A governable slowdown could favor profitable inference/platforms over frontier training; a Taiwan event would have different exposures; a tax or permit regime could transfer rents without reducing physical deployment. Those require differentiated company re-underwriting, not this uniform overlay. Do not add the same loss again if it is already incorporated in revised company bear cases.

Regime to monitor	Physical spending can…	Public equity implication / action
Fast capability gains with successful, funded adoption	Rise sharply as useful workloads expand	Positive commercial bull channel; increase assumptions only after mapping orders, margins and cash to the company
Fast software-led research on existing hardware	Grow slower than capabilities, while hardware designs change faster	Reassess unit demand, useful asset lives and software/platform substitutes; do not label this an AI failure
Governed frontier slowdown with continuing inference	Continue, but shift mix and financing sources	Separate training from inference exposure; examine actual rules before freezing the whole ecosystem
Loss of control or enforceable shareholder claims	Rise even as investor ownership becomes impaired	Treat as outside ordinary valuation; suspend deployment when a concrete event undermines ownership or market access

These are not mutually exclusive dated predictions. They describe causal regimes for monitoring. Taking the race scenario seriously increases attention to all four, not only the favorable spending branch.

A 100% common-equity loss is possible in a sufficiently severe tail. The model's positive bear values for mature companies are representative recession/competitive outcomes, not guaranteed floors. The zero-equity bears for selected excluded neoclouds demonstrate financing risk explicitly. To assess household resilience, stress both Event Horizon and Continuity together: a 60% EHF decline plus a 35% decline in the other 50% equity allocation would reduce net investments by approximately **47.5%**, before Foundation cash and any other assets. This is arithmetic, not a forecast.

The final portfolio is **not** an all-weather portfolio. It has no dedicated defense, privacy, bio-risk or geopolitical-insurance sleeve. That is a conscious opportunity-cost decision: those securities did not demonstrate reliable hedging payoffs or sufficiently attractive returns. The risk is accepted, not erased.

7.3 Monitoring and thesis invalidation

Review quarterly earnings, guidance, balance sheets, cash flow, fully diluted shares and customer concentration. Monitor order cancellations and conversion—not just announced orders—plus lead times, spot/contract pricing, utilization, credit terms and power commissioning. Separate organic growth from acquired revenue and product sales from financing income. Refresh the model after a material earnings/contract event or a 20% price change; there is no obligation to trade at every review.

For the portfolio thesis itself, three conditions would justify a broad downgrade by 2027: (a) capabilities keep improving but paid workloads fail to cover economic compute costs; (b) suppliers expand capacity so fast that

bottleneck margins normalize well below these assumptions; or (c) financing/policy constraints stop the physical build even though demand exists. A two-year delay is survivable for a profitable, conservatively financed business; it can destroy common equity at a highly levered builder.

Forecast-audit additions to monitoring: separately track (1) task performance at a stated reliability, cost and evaluation version; (2) measured research-output uplift, not just coding speed; (3) annualized revenue versus recognized revenue, external cash collection and funding commitments; (4) shipped, energized and utilized hardware; (5) HBM bits/yields, packaging capacity and EUV exposure intensity; and (6) training restrictions, verification requirements and changes to profit/ownership rights. A government-backed or strategic buildout can rationally run ahead of near-term customer ROI; retain it in the scenario set if funding is credible. It still must be translated into supplier cash receipts. Neither one quarter of weak lab profitability nor one announced sovereign commitment is decisive alone.

Acceleration trigger: sustained independent evidence of end-to-end research automation, faster useful experimental output and shortening iteration cycles requires an immediate model refresh, even if physical capex guidance has not yet changed. **Allocation trigger:** upstream margin/capacity evidence meeting section 5.6's per-share value hurdles requires reconsidering TSM/MU/ASML/LRCX. **Governance trigger:** distinguish an announced proposal, a binding training restriction, an inference restriction and an ownership impairment; these are not equivalent trade signals. No additional equity purchase is authorized simply because an AGI forecast moved earlier.

Rebalance at review dates if drift creates a position materially above 15%, using taxes and market conditions rather than a blind intraday rule. A deliberate target above 15% would require a new exceptional-evidence analysis; none is authorized by this report. Do not increase EHF's household share simply because the expected-return model looks attractive.

8. Targeted question reanalysis: all 15 May Part XIV questions

The question numbers preserve the May framework; the answers are independently reconsidered. Confidence concerns the stated investment conclusion, not certainty about an AGI timeline. References to current issuer results below point back to the dated, linked evidence in Sections 2 and 4. A strategic capability can matter greatly while the best associated stock still deserves 0%.

8.1 What is the complete AI datacenter bottleneck map?

Prior May conclusion. The physical stack was the organizing principle: leading-edge foundry, accelerators/custom silicon, memory, lithography and tools, EDA, networking, racks/cooling, power, electrical equipment, fuel and optics. The expanded portfolio treated many of these as simultaneous investable bottlenecks. The important insight was that model software cannot create operating megawatts, qualified chips or reliable interconnect on demand. The weakness was treating each bottleneck as deserving its own allocation.

New evidence since May. October/November operating releases show growth at several links simultaneously: TSM, ANET, VRT and COHR report substantial expansion; GEV's reservations/backlog indicate customer urgency; CLS provides a much larger 2027 outlook. But COHR's negative quarterly operating cash flow and heavy capex, IREN's remaining funding needs, and the equity financing at builders show that money and cash conversion are constraints too. The IEA's October 14, 2026 integration study adds current evidence that grid access, local policy and affordability shape deployment. These are fresher observations than the original AI timeline essays.

November 9 research update. The July Plan A compute model explicitly couples HBM and logic through EUV exposure capacity; ASML's October capacity commentary independently supports taking that upstream constraint seriously. The shared workbook's binding-HBM result shows why packaging growth alone is insufficient. The original map also needed a separate feedback edge: research automation improves algorithms and resource efficiency, which can change the map before new factories arrive. See sections 3.5–3.7 for source dates and limitations.

Updated answer. The full economic chain is: solvent end customer and useful paid workload, or credibly funded strategic demand → viable model/software → data rights, evaluation and security → financing and site rights → permits/water/grid or onsite generation → transmission/distribution/transformers/switchgear/UPS → thermal management → wafers/tools/materials → packaging/HBM → accelerators/custom silicon → racks/networking/optics/storage → commissioning/utilization → cash collection and reinvestment. Skilled labor, construction coordination and reliable suppliers cross several links. A missing link can idle everything downstream. Conversely, reducing compute per useful task can relieve several bottlenecks together. The bottleneck map is dynamic, not a permanent set of entitled holdings.

Rank each link on four tests: how scarce is *qualified* capacity, how long does incremental supply take, who can increase price, and how readily can customers redesign or substitute? A queue or reservation is weaker evidence than a funded contract; a funded contract is weaker than commissioned capacity and collected cash. Avoid counting a hyperscaler's announced campus, its developer's announcement and its equipment supplier's order as three independent demand sources. Also distinguish one-time construction rent from recurring operating rent and technology-specific scarcity from a durable franchise.

Portfolio implication. Own eight attractive exposures, not every box: NVDA/AVGO 30%, VRT/NVT 27%, CLS/COHR 20%, VST/NRG 23%. TSM, ASML, MU and others remain central to the map but receive no direct allocation at this opportunity cost. Financing scarcity supports rejecting debt-heavy neoclouds rather than assuming lenders will always fund growth.

Confidence: 4/5 in the dependency map; 3/5 in the location of 2030 excess profits. **Disconfirming evidence:** sustained falling equipment prices and delivery times alongside rising available capacity; project cancellations without replacement demand; or strong AI adoption with substantially less infrastructure spending per dollar of customer value than modeled. Any of those can leave the map technically correct and the stocks unattractive.

8.2 Hypothetical C15: What would an AGI/ASI do to maximize money?

Prior May conclusion. The answer emphasized compounding advantages from superior research, automation, financial decision-making and control of scarce physical inputs. Compute, power, data and real-world execution could turn cognitive superiority into economic leverage. That intuition supported owning infrastructure rather than trying to identify a single winning application.

New evidence since May. The October platform results show enormous commercial scale but also enormous cash commitments: Microsoft's reported cash spending on property/equipment rose sharply; Amazon's trailing free cash flow was negative; Meta's quarter also had negative free cash flow. These observations are not evidence that an autonomous system already maximizes wealth reliably. The METR time-horizon page also cautions against overinterpreting extrapolation; its September 8 update status and evaluation limitations make the original task-horizon slope an unsuitable proof of autonomous commercial competence. An older benchmark trend is not a live audit of a financially autonomous agent.

November 9 research update. The goals forecast makes the premise more conditional than the prior answer acknowledged: capability does not determine whose objectives are pursued, and neither written instructions nor commercial incentives guarantee robust alignment. Plan A's economics introduces another complication: even successful automation can shift value toward consumers, scarce permissions or governments. These are scenario mechanisms, not newly observed facts about autonomous systems. The August forecast distributions and current revenue records in section 3.4/3.6 make earlier commercial disruption more plausible, without establishing autonomous control of a business.

Updated answer. Conditional on an authorized, reliably aligned system tasked with lawful commercial wealth creation, it would seek the highest risk-adjusted return on incremental capital: automate expensive work, improve products, shorten discovery/design cycles, negotiate efficient resource contracts, and reinvest where competitors cannot quickly eliminate the advantage. It would distinguish accounting revenue from owner cash and avoid building capacity whose replacement cost exceeds its economic yield. It might prefer renting compute, acquiring an existing cash generator or selling useful software over owning a datacenter. Ownership of every physical bottleneck is not a logical necessity. If alignment, authorization or enforceable property rights fails, that commercial model is no longer the right framework; section 7 treats that separately.

For an outside shareholder, the central issue is the division of surplus. Customers can capture productivity gains through lower prices; employees, governments, suppliers, creditors and new competitors can capture other portions. A software company can become vastly more useful while its incremental returns decline. Likewise, a supplier's strategic importance can induce its customers to vertically integrate or subsidize competitors. Agency, access to capital, authorization and execution remain meaningful constraints; this report assumes neither unconstrained autonomous action nor inevitability of a single winner.

Portfolio implication. Prefer businesses with demonstrable monetization and an attractive entry value. The eight-stock basket is a practical claim on selected cash flows, not a claim to own what an AGI would own. NVDA and AVGO win on current economics plus modelled normalized growth; CLS/NRG win partly on price. MSFT remains a credible 17.6% modeled alternative but loses the marginal comparison after capex and valuation. No trading bots, leverage, crypto or private model-lab interests are added.

Confidence: 3/5. Economic incentives are clearer than the behavior of hypothetical advanced systems. **Disconfirming evidence:** applications retain much more durable owner cash than infrastructure while customers commoditize hardware; or autonomous systems demonstrably deliver commercial work cheaply enough to reduce aggregate investment demand. Reallocate toward the actual cash-capturing layer after valuation analysis, not toward the most impressive demonstration.

8.3 C14: What are the top stocks for the 100 GW datacenter buildout?

Prior May conclusion. The portfolio favored the broad physical stack: TSM/NVDA/ASML/MU, cooling and rack integration, electrical equipment, nuclear/gas power, and optical networking. The 100 GW framing highlighted how demand at that scale could overwhelm current supply chains. It did not establish that all the required capacity would be financed, built and utilized before 2030.

New evidence since May. GEV's October backlog/reservation disclosure, VRT's October order/revenue expansion, CLS's October 2027 guidance, and the IEA's October integration report strengthen the case that buildout is real. They do not convert a multi-year pipeline into delivered demand. As a scale cross-check, the IEA's April 16, 2026 report, **STALE and not new since May**, projects roughly 950 TWh of worldwide datacenter electricity use in 2030. It must not be relabeled a fresh November forecast.

Updated answer. First define the units. A 100 GW *facility* load running at 85% utilization consumes about 745 TWh annually. A 100 GW *IT* load at 1.2 power-usage effectiveness and 85% utilization consumes about 894 TWh. At continuous nameplate operation, 100 GW alone is 876 TWh before any additional facility overhead. Therefore an incremental 100 GW IT buildout in one country by 2030 is an aggressive scenario relative to that older global energy projection, not the base case. Nameplate generating capacity, facility connection capacity, installed accelerator capacity and average delivered load are different quantities.

November 9 interpretation update. Being above an IEA central estimate is not sufficient to dismiss a rapid-progress scenario. The trillion-dollar-cluster argument and the later compute models provide a reason to explicitly stress much larger buildouts. They do not refer to identical things: one extreme campus, global peak AI capacity and incremental US IT load have different financing and engineering requirements. Section 3.5 also corrects the inherited Palo Verde comparison's year and scope. I retain 2/5 confidence in the specified incremental US 100 GW outcome, while assigning the broader high-demand branch materially more analytical attention.

The best strategic suppliers and the best stocks are different rankings. TSM and ASML remain exceptionally difficult to substitute, but their modeled returns of 13.5% and 8.6% do not beat the chosen basket. Section 5.6 now tests stronger upstream earnings rather than assuming those original estimates settle the issue. VRT/NVT benefit before and during commissioning; CLS participates in system integration; NVDA/AVGO capture compute/networking economics; COHR benefits if optical intensity expands profitably; NRG/VST sell electricity and customer solutions. GEV has strategic turbine exposure but a much more demanding price and deposit-sensitive cash profile.

Portfolio implication. The valuation-adjusted shortlist is the selected eight, with the largest targets capped at 15%. The top modeled CAGRs are NRG/NVDA near 24.5%, followed by VRT/NVT/CLS around 23.5–23.7%; this ordering is too close to justify false precision. COHR has the smallest weight because its cash-conversion risk is more immediate. No allocation is justified solely by a 100 GW headline. A slower 30–50 GW pathway must still support owner returns.

Confidence: 4/5 in the unit arithmetic and physical requirements; **2/5** in realization of an incremental 100 GW IT scenario by 2030. **Disconfirming evidence:** funded commissioning slips materially; average utilization remains low; power contracts are canceled; or customers achieve their useful-output targets with far less hardware. Those would reduce revenue assumptions even if announced campus capacity continued rising.

8.4 C16: What would an AGI/ASI do to maximize its own intelligence?

Prior May conclusion. More compute, better algorithms, stronger research tools and iterative self-improvement could form a reinforcing loop, making chips, power and infrastructure unusually valuable. The investment inference leaned toward physical scarcity as the constraint on recursive progress.

New evidence since May. The latest company releases support greater spending on compute and interconnect, but they do not demonstrate an uncontrolled recursive capability loop. Current Epoch trend material separates training-compute growth, installed capacity and price-performance improvements; its underlying Q2/earlier observations are **STALE (>30 days)**. METR's September update status and limits to measuring longer tasks caution against extending a short benchmark curve into an assured schedule for autonomous research.

Improvements in model capability, reliable completion, inference efficiency and commercial usefulness are related but not identical.

November 9 research update. The takeoff forecast's fixed-hardware/software-driven mechanism is the most important addition to this answer. Superior experiment selection can increase useful discoveries per experiment, not merely perform the same coding steps faster. Thus a simple fixed-task Amdahl's-law argument or multi-year factory lead time does not establish a hard ceiling on research acceleration. The August revision appropriately adds training lags, and measured whole-research-process uplift remains less certain than coding capability. These distinctions warrant a serious early takeoff case, not an assured date.

Updated answer. Conditional on a system pursuing improved research capability, it would allocate resources to the highest marginal improvement in reliable capability: better algorithms and experiments, higher-quality feedback, useful tools and memory, robust evaluation, specialized hardware where justified, and carefully selected additional compute. More parameters or more electricity need not be the best next investment. Scientific validity, error correction and reproducibility become more important as research tasks lengthen. Scarce experimental data and access to real-world laboratories can matter alongside digital compute. Digital research and physical experimentation can have very different bottlenecks; neither clock should be imposed on the other without evidence.

This creates two competing investment effects. Cheaper inference can broaden use enough to increase total infrastructure demand; it can also reduce the amount of infrastructure required for a fixed task volume. Which effect dominates is an empirical question about demand elasticity, prices and new applications. A claim that efficiency *must* increase total energy demand is no stronger than a claim that it *must* destroy it. The portfolio needs paid utilization or credibly funded strategic demand that converts into supplier cash, not simply faster benchmark improvement. If algorithmic progress eventually makes current infrastructure easy to replace, scarcity rents can peak well before technological progress does.

Portfolio implication. Hold both general-purpose and custom-silicon exposure, rather than betting that one architecture serves all workloads forever. NVDA/AVGO total 30%, but both share exposure to customer budgets, TSM manufacturing and networking substitution. VRT/NVT/NRG/VST are less tied to a specific chip design, although still exposed to aggregate deployment. Do not add EDA or proprietary-data stocks merely because they participate in the intelligence feedback loop; their lower expected returns currently lose the capital competition.

Confidence: 3/5. The marginal-resource framework is robust; the future balance between algorithms, data and physical compute is not. **Disconfirming evidence:** algorithmic improvements consistently outpace new useful demand, depressing aggregate spend and utilization; a dominant architecture substantially bypasses today's networking/cooling intensity; or evaluations show capability progress stalling despite much greater spending. Each would require a different earnings path rather than a cosmetic change to the narrative.

8.5 C17: What strategy would an AGI/ASI use to beat China in the race to ASI?

Prior May conclusion. The geopolitical answer centered on a US/allied advantage in advanced compute, energy, capital, talent and supply-chain coordination. Domestic foundry and strategic materials were treated partly as insurance against dependence on China or Taiwan. It risked conflating national strategic value with attractive common-equity returns.

New evidence since May. Intel's October announcement of a 14A process-design-kit milestone is evidence of development progress, not proof of commercially competitive high-volume external foundry output. TSM's October operating results demonstrate continuing manufacturing strength. MP's November customer qualification and financial disclosure add concrete processing/magnet progress but also expose dependence on contract economics and non-product income. The GAO July 22 critical-minerals review underscores that substitution and recycling timelines differ substantially by application. These developments make a simplistic US-versus-China ticker basket less persuasive.

November 9 research update. The security forecast and Plan A expose two different pathways: an intensified race with high strategic spending and secrecy, or an agreement that slows frontier research while allowing inference. The August scorecard itself says the original mid-2026 Chinese consolidation narrative had not materialized as depicted.

Do not treat that narrative as a verified government program. The investment-relevant uncertainty is the policy response to capability gains, not a presumption that a unilateral rush is the only possible response.

Updated answer. A lawful national competitiveness strategy would build reliable energy, high-quality research and training, resilient allied manufacturing, secure infrastructure, efficient permitting and capital markets, and verifiable standards for safe deployment. It would diversify critical dependencies, preserve international scientific and commercial cooperation where appropriate, and measure actual output rather than symbolic announcements. Competitive advantage depends on execution and institutions, not just access to a model. Preserving human control and preventing catastrophic escalation are distinct objectives from being first on a benchmark. No confident current ranking of hypothetical ASI capability follows from public company earnings.

For investors, a successful industrial policy can still produce poor shareholder returns. Subsidies may support customers, jobs or strategic capacity without covering the cost of capital for minority owners. National-security intervention can restrict exports, cap profits, force investment or dilute shareholders. Conversely, a globally competitive supplier can remain profitable while its home government faces strategic vulnerabilities. These are different objective functions and should not be collapsed into one score.

Portfolio implication. INTC falls from 3% to 0%; MP from 3% to 0%. Neither is accepted as a reliable Taiwan or geopolitical hedge. Direct TSM falls from 11% to 0% on return competition, but this does **not** remove Taiwan dependence: NVDA, AVGO and other suppliers still rely on that ecosystem. The final portfolio openly accepts that correlated exposure. A US listing or headquarters is not evidence of a domestic-only supply chain.

Confidence: 3/5 in the investment distinction; **2/5** in geopolitical outcomes. **Disconfirming evidence:** Intel wins economically material external volume at competitive yields with improving free cash flow and controlled dilution; MP demonstrates durable unsubsidized product margins sufficient for a superior forward return; or major trade/production restrictions change the suppliers' cash-flow outlook. Reconsider after those facts and prices change, not because the political narrative intensifies.

8.6 C20: Are critical minerals the ultimate Achilles' heel?

Prior May conclusion. Concentrated processing and supply chains were a potential hidden constraint, with MP retained at 3% as a strategic-materials exposure. The broad thesis correctly recognized that extraction alone is insufficient; refining, separation, qualification and manufacturing matter. It was less precise about which public stock monetizes which specific material shortage.

New evidence since May. GAO's July 22, 2026 analysis, **STALE as a current observation but genuinely post-May evidence**, describes application-specific limits to substitution and recycling. Semiconductor uses can be harder to replace quickly than certain battery uses. MP's November results include a smaller price-protection contribution and a one-time item, while qualified magnet sales were still expected to begin in the fourth quarter. Those facts are more informative for equity value than the generic statement that rare earths are strategic.

Updated answer. Critical minerals are one class of potential Achilles' heels, not a universal final constraint. Separate ore availability from economically recoverable reserves, processing capacity, purity, qualified components, logistics and geopolitical access. Gallium, germanium and indium are not interchangeable with magnet rare earths, and owning MP is not a comprehensive semiconductor-materials hedge. A small-volume material can stop production, but a high market price may also accelerate recycling, substitution, redesign and new supply. A shortage's social importance and the shareholder profit captured by one miner are not proportional.

There are also two clocks: a sudden export interruption can bind immediately, while new processing and qualification can take years. A four-year investor needs both interruption risk and the supply response. Government-backed price floors can improve project financing, but they do not guarantee the common stock's valuation, protect against dilution, or assure customers buy every unit produced. The contractual beneficiary and economic obligations matter more than the headline subsidy amount.

Portfolio implication. MP goes to 0%; the 3% funds NRG 2% and VST 1%. CCJ also goes to 0% for separate uranium valuation and contract reasons, not because uranium and rare earths share identical economics. No small basket of miners is added as a vague insurance sleeve. Materials remain on the dependency dashboard; if a

specific publicly investable processor obtains durable high-return contracts at an attractive valuation, it can reenter the capital competition.

Confidence: 4/5 that material-specific analysis is necessary; **3/5** in MP's exclusion at the current opportunity cost. **Disconfirming evidence:** substantial recurring magnet revenue with strong cash margins and limited new equity, enforceable contract economics that support superior owner returns, or a verified shortage for which an investable supplier has protected capacity and pricing. A rising commodity price alone is insufficient.

8.7 C6: How big a deal is the need for power, and which source benefits most?

Prior May conclusion. Power was the leading physical constraint. The portfolio spread exposure across existing nuclear and gas generation, grid construction, equipment, uranium and grid-bypass fuel cells. The conventional power-related sleeves totaled 29%, before counting VRT's thermal/electrical role. Nuclear scarcity and onsite alternatives received substantial narrative weight.

New evidence since May. VST's November release includes the completed Cogentrix acquisition and 2027 guidance, giving a clearer near-term operating perimeter. NRG's August outlook, **STALE pending November 9 results**, shows a lower earnings entry valuation and conditional onsite-power opportunities; the latter are not all signed, commissioned megawatts. GEV's October results show substantial customer commitments but also cash boosted by deposits. The October 14 IEA study documents the importance of policy, flexible connections and customer/system cost allocation. It is newer evidence than the April demand projection.

Updated answer. Power remains a major constraint, but location and timing dominate a simple national capacity total. Existing dispatchable capacity near usable interconnections can be valuable quickly. Gas can provide flexible supply where turbines, fuel and permits are available; existing nuclear offers high utilization but not instant new construction; renewables and storage can contribute rapidly in favorable locations, without being identical to continuous firm supply. Grid upgrades, demand flexibility, storage duration and onsite systems determine how these sources combine. No single technology supplies every incremental workload on the same timetable.

The most valuable asset is often *deliverable power at the right place and time*, not nameplate generation. Announced non-firm connection potential should not be counted as constructed generation or guaranteed energy. Retail and hedging contracts also determine which generator captures higher market prices; VST's largely hedged 2027 generation limits simplistic spot-price extrapolation. Affordability constraints can encourage new supply or limit rents. A genuine shortage does not ensure uncapped producer profits.

November 9 research update. The workbook's kW/GW display, stock-efficiency and nameplate-versus-energy issues make a single headline electricity forecast insufficient for valuation. A cohort calculation can increase estimated power demand, while retirement, utilization, deployment delays or more efficient software can lower it. Fast capability gains also need not wait for another generation of power plants. This strengthens the case for existing deliverable power in a mobilization branch but weakens any assumption of permanently rising merchant rents. NRG/VST's 23% combined weight remains a commercial valuation decision; it is not a pure bet on a guaranteed 100 GW outcome.

Portfolio implication. Use NRG 12% and VST 11% for merchant/retail generation economics. Add NVT 12% and retain VRT at 15% for the electrical/thermal chain: together these four are 50% of EHF, although only 23% is classified as generation. Remove CEG/PWR/GEV/CCJ/BE because their current prices and business-specific risks lose the return comparison, not because their products cease to matter. The accepted risk is much greater dependence on gas/merchant/retail economics and less pure nuclear/fuel exposure.

Confidence: 4/5 in power's importance; **3/5** in which owners capture the surplus. **Disconfirming evidence:** persistent weak realized prices or customer margins, poor acquired-asset returns, outages, adverse binding policy changes, falling onsite-system costs that bypass the selected generators, or a clear reduction in required electricity per paid workload without compensating demand growth.

8.8 C9: Should the portfolio invest in Department of Defense and war stocks?

Prior May conclusion. AVAV and KTOS at 1% each offered exposure to autonomous, lower-cost systems and a potentially important defense procurement shift. Their small weights limited direct financial damage but also made their supposed geopolitical-hedge contribution immaterial. Strategic relevance and budget growth were doing much of the argumentative work.

New evidence since May. AVAV's September quarter, **STALE**, showed modest revenue growth rather than a clean explosive cash-flow ramp. KTOS's November disclosure still points to negative free cash flow and substantial investment needs despite revenue/EBITDA growth. These are real commercial businesses with contracts, but the timing and economics of procurement remain material. The relative stock weakness since May does not automatically make them cheap; the scenario model already reflects a more favorable entry price.

Updated answer. The sector deserves research, but not a permanent entitlement to capital. Distinguish technological promise from qualified production, funded procurement, delivery acceptance, and cash collection. Small contractors can face customer concentration, fixed-price execution risk and dilution; larger primes may have more stable cash flows but less incremental exposure to the relevant systems. A conflict can help some contracts while simultaneously damaging markets, supply chains and fiscal conditions. Defense stocks are not a reliable hedge against every conflict scenario.

The screen compares AVAV/KTOS with diversified primes and other autonomy/security exposures. A mature contractor could win on cash yield even with slower growth; a newer supplier could win if funded scale and margins exceed the price-implied requirements. Neither is selected merely because war risk has increased. The analytical task is expected owner return under lawful procurement, not prediction of battlefield outcomes.

Portfolio implication. Remove AVAV 1% and KTOS 1%; both proceeds fund NVT. Modeled CAGRs of approximately 9.3% and −0.3% do not justify preserving token weights against the selected 19–25% range. AXON is separately considered under public safety, not conflated with a defense prime. No dedicated war sleeve remains, and the report explicitly accepts the loss of any favorable defense-stock payoff in a conflict.

Confidence: 3/5. Procurement and execution outcomes are uncertain, but the current opportunity-cost case is clear. **Disconfirming evidence:** sustained funded order conversion, positive free cash flow after required capacity investment, controlled dilution and a price that produces a superior independently modeled return. A large unfunded program announcement or a single procurement headline is not enough; neither is a stock-price decline without improved economics.

8.9 C12: Can the portfolio invest in proprietary data as a bottleneck?

Prior May conclusion. Proprietary data and enterprise context were potentially durable complements to AI, but no dedicated data-stock sleeve was adopted. PLTR supplied an indirect operating-context exposure. The difficulty was finding a public company whose legally usable, hard-to-recreate information translated into superior stock returns rather than simply a compelling narrative.

New evidence since May. RELX's October update reports continued underlying growth, including a stronger legal segment; VRSK's November results show recurring growth but a weaker quarterly cash-flow comparison. PLTR's November release shows exceptional revenue/profit growth, yet much slower growth in total contract value and material stock compensation. These are distinct evidence sets: a workflow provider does not automatically own its clients' underlying data or an unrestricted right to reuse it.

Updated answer. Yes, proprietary data is investable through public equities, but the moat must be decomposed. The valuable combination is exclusive or difficult-to-replicate data, permission to use it, freshness, verification, embedded distribution, and a workflow that produces a measurable customer outcome. A static archive that a model can cheaply reconstruct is less attractive than continuously updated decision-critical information with auditability. Rights, privacy rules, licensing costs and customer bargaining power can determine who captures the value.

RELX and VRSK are credible candidates because they already sell information and decision support in recurring workflows. Their defensibility is not equivalent to a guarantee of accelerated growth. AI may improve products and margins, but it may also reduce retrieval costs and intensify competition. PLTR's strength is integration and operating software; it should not be valued as though it owns every customer's private dataset. That distinction matters when assigning terminal margins and multiples.

Portfolio implication. RELX and VRSK are fully underwritten but receive 0% at modeled CAGRs of approximately 13.2% and 14.4%. PLTR is removed from 2% to 0% because its entry valuation overwhelms the business improvement under this model. The absence of a data allocation is an opportunity-cost result, not a conclusion that data is free or irrelevant. If the final hardware basket becomes substantially more expensive, these recurring-cash businesses are plausible replacements.

Confidence: 4/5 in the existence of investable data moats; **3/5** in their AI-driven acceleration. **Disconfirming evidence:** loss of exclusive rights, customer churn, commoditized access or weak pricing despite rising use; conversely, verified AI-driven retention, organic growth and owner cash materially above the model would falsify the current exclusion. Distinguish a better product from demonstrably better incremental returns.

8.10 C2: Should the portfolio invest in identity, provenance, and compliance rails?

Prior May conclusion. More autonomous software should increase the need for identity, permissioning, audit trails and security. CRWD and PLTR provided broad exposure; specialized identity and privileged-access vendors remained candidates. The earlier framework tended to treat the necessity of the control layer as supportive of its leading equities without a sufficiently hard price test.

New evidence since May. CRWD's August results, **STALE**, show continued subscription/ARR growth and cash generation, but GAAP profit and stock compensation still matter. PANW's September guidance, also **STALE**, offers an integrated security-platform comparison. CyberArk's acquisition closed before the May baseline; the post-May evidence is integration and guidance, **not a newly announced November independent CYBR opportunity**. PLTR's November expansion supports demand for operating control, not proof that it will dominate all identity or provenance standards.

Updated answer. The requirement is real, but likely spans several vendors and embedded platform capabilities. Identity verifies the actor; authorization limits permitted actions; provenance records origin/history; monitoring detects problems; compliance demonstrates adherence to applicable obligations. None alone guarantees trustworthy output. Agent proliferation can expand the number of identities while procurement consolidates spending into broad platforms. Standards may commoditize one layer and strengthen another. Open protocols or mandated controls do not by themselves create monopoly rents for the most recognizable ticker.

Investment analysis must charge for economic dilution and customer acquisition, not capitalize adjusted cash flow while ignoring the shares used to pay employees. Here CRWD and PANW are valued using owner free cash flow after an explicit economic stock-compensation charge. This is deliberately different from applying a premium multiple to management's unadjusted free-cash-flow headline. A high-quality business can still deliver poor stock returns when the starting price assumes too much growth.

November 9 research update. AI 2027's security supplement distinguishes protection of model weights, algorithmic secrets and deployed systems; its goals supplement raises a different problem, whether an increasingly capable agent actually pursues intended objectives. Ordinary enterprise identity and endpoint products do not establish frontier-model containment or alignment. The race scenario increases the urgency of security spending but can also concentrate it within labs or governments, beyond these vendors' addressable products. Neither buying CRWD nor labeling a portfolio “secure” hedges loss of control. Revisit security vendors when contracts, product effectiveness and owner cash support the thesis, not simply because the scenario makes security socially essential.

Portfolio implication. CRWD moves from 3% to 0%; PLTR from 2% to 0%. No dedicated identity/provenance sleeve replaces them. PANW is fully underwritten but excluded. OKTA and related candidates remain screened alternatives; an acquired or unavailable standalone listing is not treated as an actionable public stock. The selected basket accepts security-provider opportunity cost rather than keeping token exposure to complete the thematic

diagram.

Confidence: 4/5 in demand for controls; **3/5** in the exclusion decision because terminal software economics are uncertain. **Disconfirming evidence:** sustained growth with materially lower economic dilution, stronger after-SBC cash margins, and/or a price reset that lifts modeled return above the marginal selected position. A major cyber incident can demonstrate need without proving that any specific vendor gains profitable business.

8.11 C1: Should the portfolio invest in domestic surveillance and public safety?

Prior May conclusion. AXON was the most direct candidate for connected public-safety workflows, with MSI a more established communications alternative. The portfolio did not establish a dedicated allocation. The earlier thesis associated AI capability with expanded sensing, evidence management and operational coordination, but the investable benefit remained price- and procurement-dependent.

New evidence since May. AXON's November release reports growth while also identifying delayed federal activity, substantial projected stock compensation, and a large Skydio investment. MSI's November update raises or supports its earnings outlook and shows meaningful backlog. Those facts make the comparison richer than a simple high-growth-versus-old-hardware distinction: one offers potentially greater expansion with more execution and dilution risk, the other a more established installed base and cash profile.

Updated answer. Public safety is investable, but broad surveillance adoption should not be assumed inevitable or socially costless. Budgets, procurement rules, civil liberties, accuracy, data retention, liability and public acceptance can constrain deployment. The durable business may be secure evidence management, emergency communications and accountable workflows rather than unrestricted collection. Recurring software can improve economics, but its reported growth must be separated from hardware cycles, acquisitions and customer financing.

AXON and MSI therefore compete on normalized owner cash, not on the emotional salience of crime or security. A large strategic investment can expand optionality while lowering near-term cash available to shareholders. Stock compensation is a real claim on future value. Delayed contracts may recover later, but they reveal the mismatch between long-term demand narratives and quarterly cash realization. Neither equity is a dependable hedge against domestic disorder.

Portfolio implication. Both receive 0%. AXON's modeled CAGR of approximately 10.9% is too low for its risk and MSI's approximately 8.2% loses on opportunity cost. The final portfolio has no surveillance sleeve and no need to invent one to cover all 15 questions. These companies remain reasonable research candidates if price or cash economics materially change.

Confidence: 3/5. Demand is credible, but procurement, valuation and social constraints are material. **Disconfirming evidence:** a durable acceleration in recurring revenue and after-SBC free cash flow without comparable increases in dilution/capital commitments; or a materially cheaper entry price. On the negative side, contract cancellations, liability or restrictions on use can impair the thesis even while technological capabilities improve.

8.12 C7: How should the portfolio invest for a world where bio-engineered pandemics become easily accessible?

Prior May conclusion. MRNA's 1% allocation was treated as a small bio-risk/rapid-response exposure. The strategic premise was that more powerful AI could increase biological risk and the value of fast vaccine development. The weakness was describing an operating company's common stock as insurance without a defined, reliable payout in the feared state.

New evidence since May. The major MRNA development is oncology: Merck and Moderna's August phase-3 announcement and October 22 detailed melanoma results materially improve the clinical evidence for an investigational therapy. The reported primary benefit concerns recurrence or death, not a demonstrated reduction in overall mortality alone; overall-survival follow-up remains incomplete. November financial guidance also updates liquidity and revenue. None of this establishes that engineered pandemics are now easily accessible or that MRNA stock reliably hedges one.

Updated answer. Treat the premise as a serious conditional risk, not an established universal fact. Model capability, access to physical resources, tacit expertise, safeguards and effective response are separate determinants. This report does not attempt to operationalize harmful biological activity. The relevant investment question is whether public suppliers of prevention, detection, treatment or manufacturing can earn attractive returns while meeting public-health needs.

A successful public-health response may deliberately constrain prices, involve multiple technologies, require public financing and take time. A vaccine company could face trial failure, manufacturing delays, liability, dilution or a market-wide selloff even if demand eventually rises. Diagnostic and diversified healthcare companies may be more stable businesses, but they are not exact pandemic insurance either. Household liquidity, resilience and appropriate insurance arrangements are distinct from this equity-selection exercise and are not changed here.

November 9 research update. Taking the race ending seriously strengthens this distinction. Some systemic outcomes destroy the ordinary connection between business activity, enforceable financial claims and investor welfare; no small listed-healthcare sleeve is demonstrated to pay reliably in that state. Section 7.2 now separates such regime risk from the ordinary commercial bear case. Its zero-recovery calculation is a financial sensitivity, not a prediction of an engineered pandemic, a probability of extinction, or a valuation of human life. This report proposes no operational biological activities and no claim that a stock allocation solves the underlying safety problem.

Portfolio implication. Remove MRNA's 1%; fund NRG. MRNA's approximately 368% historical total return was valuable, but it validates neither the original insurance mechanism nor the new valuation. The current model allows substantial oncology success and still produces a negative expected CAGR at the November price. BNTX, ABT and other health-related alternatives fail the final capital competition or lack a sufficiently specific investable advantage; no replacement token hedge is added. Moonshot stays at 0%.

Confidence: 4/5 that common-stock exposure is not reliable insurance; **2/5** in the timing/severity of future bio-risk. **Disconfirming evidence:** verified commercial approvals, reimbursement, scalable production and sustained owner cash that materially exceed the model, or a much lower price. A new outbreak by itself would not falsify the valuation conclusion or prove that a particular stock pays off in the required state.

8.13 C8: Should the portfolio invest in anonymity, privacy, VPNs, crypto, or blockchain?

Prior May conclusion. Privacy demand could rise as AI expands surveillance and data processing, but there was no sufficiently compelling standalone stock allocation. The portfolio did not own crypto. Consumer VPN economics, enterprise security and decentralized infrastructure were recognized as different exposures rather than one uniform market.

New evidence since May. GEN's October 28 release reports 12% quarterly revenue growth and a higher FY27 adjusted earnings outlook; its low guide-based multiple is explicitly evaluated in Section 4.6, while a verified comparable consensus series remains unavailable. PANW/CRWD's post-May releases and RELX's October update reinforce that identity, security and data-rights spending can grow without a blockchain-specific architecture. They do not establish an investable monopoly in anonymity. No fresh primary evidence reviewed here demonstrates that token ownership is necessary to capture AI-related privacy demand.

Updated answer. Privacy is a product attribute, a governance requirement and sometimes a business model; it is not synonymous with anonymity, VPN usage or blockchain. A VPN addresses a particular network layer and does not make an application, device or user unobservable. Enterprise demand may favor access controls, encryption, confidential processing, permissioned data use and auditable handling. Value can accrue to platform vendors and customers rather than an independent privacy stock. A blockchain can provide an audit structure in some applications but can also increase public visibility; its use does not automatically improve confidentiality or unit economics.

Evaluate public equities on retention, acquisition cost, pricing, incremental margins, debt and dilution. GEN's consumer franchise is more directly investable than an abstract privacy theme, but its modeled return is about 5.3%, below the selected opportunities. Other screened names such as CLBT, CGNT and ARLO have materially different investigative, security or consumer-device exposures; they are not interchangeable proxies for individual privacy.

Current listing and security identity must be checked before any future trade.

Portfolio implication. Keep the sleeve at 0%. Crypto, tokens and derivatives remain outside the authorized common-stock/ADR universe regardless of their narrative appeal. No blockchain company is added merely to create exposure. The opportunity cost of excluding a future privacy winner is accepted; it is preferable to pretending that a broad label establishes an attractive valuation.

Confidence: 4/5 in the scope and economic distinction; **3/5** in the stock exclusions. **Disconfirming evidence:** a public company demonstrates durable privacy-related pricing power, improving owner cash and an attractive entry valuation; or a mandated standard creates a verifiable, defensible commercial toll rather than a competitive commodity. User growth without cash conversion does not meet that test.

8.14 What could prove the bottleneck thesis wrong by 2027?

Prior May conclusion. Watch supply expansion, shorter lead times, weaker orders, capability disappointment, policy constraints and demand that fails to match buildout. The thesis was conditional, but the broad architecture risked interpreting almost every new constraint as a reason to add another sleeve instead of reducing the overall forecast.

New evidence since May. Many stocks fell despite better reported earnings, proving that fundamental direction alone was insufficient at May prices. COHR's cash demands, neocloud funding requirements, and the MRVL/LITE transaction add financing, ownership and substitution risks. The latest platform cash-flow releases make end-customer economics more important, while METR's measurement caveats weaken calendar certainty. These are not proof that the thesis has failed; they are evidence against treating physical demand as a sufficient investment thesis.

Updated answer. The strong version fails if profitable *paid demand* cannot support the capacity, if competitive supply eliminates supplier rents, or if financing/physical constraints prevent delivery within the investment horizon. AI can succeed socially while this portfolio fails financially. Conversely, a two-year delay need not destroy a conservatively financed franchise but can invalidate a highly levered capacity builder. Review the following decision rules rather than relying on a single benchmark or stock price:

By-2027 test	Specific disconfirming observation	Model/action consequence
End-user economics	Two major customers cut comparable AI-related capex plans by at least 15%, alongside weakening paid utilization or monetization	Reduce supplier revenue paths; freeze basket additions pending review. The 15% threshold is a chosen alert, not a sourced prediction.
Scarcity rents	Two quarters of falling delivery times and prices with rising available capacity and weaker margins	Normalize margins/multiples faster; backlog growth alone no longer supports the bull case.
Cash conversion	Two quarters of unexplained cash underperformance or material funding beyond modeled shares/claims	Rebuild the financing bridge; do not label equity dilution a temporary accounting issue.
Commissioning	Major funded projects slip more than a year, without compensating demand elsewhere	Shift revenue later and apply a lower terminal value; distinguish delay from permanent cancellation.
Technology substitution	A verified architecture materially reduces required components per useful workload and customers actually adopt it	Cut affected unit content/pricing assumptions, not just total AI spending.
Model capability/usefulness	Reliability and monetizable task completion stagnate despite rising spending	Lower demand elasticity and growth assumptions; reassess the entire architecture.
Fast progress, weaker hardware rents	Autonomous research improves rapidly while useful-output-per-dollar rises faster than spending and incumbent suppliers lose margins	The AI forecast may be right while this stock thesis is wrong; re-underwrite platform/distribution alternatives and physical content per workload.
Governance / security regime	Binding training limits, compulsory access/pricing, loss of protected IP or material impairment of shareholder claims	Separate inference demand from frontier training; revise cash flows and the regime-risk overlay rather than merely lowering the terminal multiple a little.

November 9 research update. The forecast audit adds two-sided tests: failure can arise from capability stagnation **or from successful AI progress changing the source and ownership of profits faster than expected**. Conversely, strategic competition can sustain credibly financed demand despite weak near-term lab profits. That is not an exemption from cash collection, but it means customer ROI alone is an incomplete short-run falsifier. Distinguish announced strategic intent from appropriated funding, enforceable commitments and delivered equipment.

Portfolio implication. These are review triggers for the whole eight-stock recommendation, not instructions to rotate blindly from one exposed name into another. Under a broad failure, the independent proposal would need revision; the report does not authorize leverage, new asset classes, or changes to Foundation/Continuity. If only one company fails, test its named replacement and the remaining positions independently before redeploying.

Confidence: 4/5 that these are useful falsification tests; **3/5** that the current base case survives all of them. **Disconfirming evidence for the skeptical case:** several quarters of strong paid adoption, high utilization, cash collection and profitable incremental capacity without excessive dilution. A valid process must permit both upward and downward revisions. Neither a price rally nor a model benchmark alone settles the issue.

8.15 Is the portfolio architecture optimal versus alternative models?

Prior May conclusion. A broad bottleneck basket, with larger core holdings and smaller emerging exposures, was preferred to a pure single-layer bet. It preserved 24 holdings across many thesis categories. The principal benefit was coverage; the cost was capital tied to lower-return or weakly justified positions and an impression of diversification that understated common AI-capex exposure.

New evidence since May. The full accountability exercise shows a −12.27% hypothetical portfolio return, including major losses at companies whose operating outlook improved. New challenger guidance and lower prices alter the opportunity set. The present model specifies all weights and uses the same terminal-value method across the current, concentrated and core-and-satellite architectures. This is a new quantitative comparison, not merely a renamed version of the May allocation.

Updated answer. Under the stated objective and these assumptions, select eight holdings. The current 24-name allocation has an implied 12.2% CAGR, the 15-name core-and-satellite 20.1%, and the concentrated eight 22.8%. The concentrated portfolio's approximate 2.8-point annual advantage over core-and-satellite is meaningful but smaller than plausible model error. It wins because the user prioritizes credible expected return, not because eight is a universally optimal number.

Concentration increases the cost of being wrong about individual margins, financing and customer budgets. Top-three weight rises to 45%; hardware/thermal/electrical exposure is 77%; generator exposure is 23%. Eight independent ticker symbols do not mean eight independent demand sources. The all-bear terminal loss is roughly 57%, and a sufficiently severe tail can be worse. Core-and-satellite offers greater platform and process-layer breadth, but its 15 names do not eliminate the common AI investment factor either.

The result is a judgment-based constrained choice, not an unconstrained mathematical maximizer. With linear expected terminal wealth and only a 15% cap, a purely mechanical optimizer would gravitate toward the highest modeled returns and be extremely sensitive to small forecast changes. Here execution-sensitive names receive lower caps, COHR is limited to 8%, and MOD's higher modeled CAGR does not earn a position because its post-spin cash/operating evidence is weaker. Those constraints are explicit costs paid to reduce permanent-loss risk.

November 9 research update. Section 5.6 adds an 11-stock upstream challenge with 5% each in TSM, MU and ASML, funded by reductions elsewhere. It yields 21.5% under the original commercial assumptions; explicitly stronger upstream revenue/margin assumptions raise it to 22.3%, close to the retained eight's 22.8%. A separate 50%-bull sensitivity still favors the eight, but does not prove that the chosen terminal earnings or probabilities are right. These narrow gaps warrant humility and prompt re-underwriting when evidence changes. Section 7.2 also shows how a common zero-recovery branch lowers absolute modeled returns without changing the ranking;

differential policy, geopolitical and financing losses could change the ranking. A single common haircut is not a complete regime-specific optimization.

Portfolio implication. Final allocation: NVDA 15, AVGO 15, VRT 15, CLS 12, NVT 12, NRG 12, VST 11, COHR 8. Nineteen names exit, three enter and five increase. Foundation/Continuity and the approximately 50% household EHF share remain unchanged; Moonshot stays zero. Do not mechanically average the three portfolios or preserve a small position merely to avoid admitting a changed view.

Confidence: 3/5 in architecture superiority; 5/5 in the arithmetic, conditional on inputs. **Disconfirming evidence:** more realistic cash/dilution assumptions erase the return premium; actual tax/transaction costs make the transition uneconomic; the household already has unacceptable overlapping exposure; or selected-company evidence deteriorates enough that the core-and-satellite portfolio offers a superior credible return. Those are reasons to revise the recommendation, not to claim the initial ranking was guaranteed.

9. Near-term deployment strategy summary

Use the separate **Deployment Strategy 2026-11-08.md / .pdf** in Analysis 2026-11-08/Deployment Strategy/. It is a conditional implementation proposal, not an order or record of execution. Its three windows are November 9–13, November 16–20 and November 23–25, subject to fresh prices, issuer results, liquidity and tax-lot checks. This November 9 forecast-research revision retains the November 6 price snapshot and original company-data cutoff; it does not incorporate or presume knowledge of November 9 issuer results.

On the supplied target-weight baseline, release 77 percentage points from the 19 removed positions. Sell 50%/30%/20% of each exit across the three windows; purchase 40%/35%/25% of each positive allocation gap. That means sales of 38.50/23.10/15.40 points, purchases of 30.80/26.95/19.25, and temporary closing reserves of 7.70/3.85/0.00 points. This reserve is inside EHF, not a withdrawal from Foundation, and is not a permanent strategic cash allocation. Actual drifted holdings require a fresh dollar calculation.

Scenario A, 55% planning weight: fundamentals remain intact and prices remain below the updated valuation ceilings; follow the schedule after event checks. **Scenario B, 30%:** a 10–20% risk-off decline occurs without a commensurate deterioration in company value; deploy the reserve earlier within remaining target gaps after re-underwriting. **Scenario C, 15%:** earnings, funding or liquidity evidence breaks a thesis; halt affected purchases, retain the applicable reserve, and revise the recommendation. These are subjective execution planning weights, not the long-horizon company scenario probabilities.

Do not complete the calendar mechanically. NRG's expected November 9 release and NVDA's scheduled November 17 event require updated checks; schedules must be reconfirmed before trading. A stock that rises above its 18% model-implied return ceiling is not automatically chased. Likewise, a large fall accompanied by broken cash economics is not automatically bought. Do not trade with unverified security identity, unavailable funds, abnormal spreads/halts, unresolved material filings, or an unassessed material tax cost. Unused reserve can remain past three weeks if a no-trade condition persists.

The forecast audit adds a regime check before each tranche: material changes to training permission, financing enforceability, security or shareholder rights require re-underwriting affected purchases. Faster coding benchmarks alone do not authorize buying above the valuation ceiling. Verified stronger upstream cash economics can justify replacing a selected position, but only after an updated funded allocation and valuation comparison—not an unmodeled extra sleeve. No weights or tranche amounts change merely because this research revision exists.

Verification and research record

The deliverable set includes this user-requested independent report, an identical conventionally named analysis copy, both analysis PDFs, and the separate deployment Markdown/PDF. The canonical filenames are packaging conveniences, not an adopted V9.2 allocation. Supporting market observations, the 56-stock scenario model, generated tables and accountability CSV are retained under inputs/research/ so another reader can reproduce arithmetic and change assumptions.

Required item	Result / location
All required foundational inputs	September V7.3 and May analysis read in full; all four specified background sites reviewed; November 9 deep audit of requested forecast pages, formulas and material side-link datasets in Sections 3.3–3.7
Current holdings	All 24 freshly researched; all 24 priced, performance-attributed and fully modeled
Sleeve competition	All 18 sleeves have at least two explicitly ranked alternatives in Section 4.1
Cross-sleeve candidates / prior watchlists	Dedicated candidate register in Section 4.2; listing/acquisition issues and screened-out names identified
Full valuation coverage	56 companies; includes every architecture constituent and the ten strongest excluded challengers; probabilities total 100% for each

Required item	Result / location
Valuation computation	Expected terminal wealth first, then implied CAGR; dividends, dilution and financing claims stated; no average of scenario CAGRs
Architectures	Three original full weight sets plus an 11-stock upstream challenge; expected returns, downside, common factors, acceleration/delay/capex stress, stronger upstream rents and opportunity costs
Forecast-to-investment bridge	Distinguishes capability, research uplift, physical demand and shareholder returns; adds explicit commercial-versus-regime-risk sensitivity
Allocation	15 + 15 + 15 + 12 + 12 + 12 + 11 + 8 = **100.0%**; largest position 15%; eight holdings
Funding	77 points released = 77 points of increases/additions; retained capital 23%; no external funding
Decisions	All 24 baseline names plus three additions receive explicit action, rationale, confidence and falsifier
Strategic questions	All 15 reanswered with prior view, new evidence, updated answer, allocation implication, confidence and disconfirmation
Execution	Separate three-tranche plan, explicit scenario branches, no-trade and reserve rules
Boundaries	No actual trades; no README/active-allocation changes; no unsupported claim of guaranteed optimal returns

Reproducibility. valuation_model.py regenerates valuation_outputs.json, analysis_tables.md and accountability.csv from saved public price observations. build_report.py inserts numerical tables and validates portfolio/funding identities. render_reports.py generates PDFs with embedded fonts, wrapped table cells and clickable source links. verification.json records the final structural, arithmetic and PDF checks. The assumptions in the code are analyst scenarios, not hidden consensus estimates.

Forecast-audit trail. inputs/research/forecast_audit/ contains dated public-source snapshots, extracted link inventories, the full Plan A PDF, bounded native Sheets values/formulas/notes, and audit_calculations.py / audit_results.json. The calculations reproduce workbook outputs, check grade aggregation, vary HBM growth/price escalation, and compare architecture and regime sensitivities. README.md records scope and limitations; audit_verification.json records coverage and numerical checks. The pre-revision Markdown/PDF and deployment materials are preserved in pre_revision/. No source workbook or external document was edited.

Evidence hierarchy. Dated issuer releases/filings are primary operating evidence; public market histories provide prices; IEA/GAO/METR/Epoch provide specialized observations with their own measurement limitations. AI 2027 / AI Futures provide explicit forecasting methods, distributions and conditional scenarios that materially inform the analysis; AI 2040 adds a policy-contingent alternative; Situational Awareness provides a strategic industrial-mobilization argument. None is downgraded merely for being a forecast, and none substitutes for evidence of per-share cash value. Own-history valuation comparisons use the May report's observations and verified contemporaneous guidance where comparable; unavailable full historical percentiles and archived consensus remain explicitly unavailable. The original November 8 analysis is supplemented by the November 9 research audit, using source observation/publication dates explicitly. Retrieval dates do not turn old data into current data. Recheck prices, filings, corporate actions and event dates before acting.

Bottom line: the deeper forecasting work strengthens the need to take rapid AI progress seriously and changes the risk framework materially. It does not, under the tested valuations, overturn the eight-stock recommendation. Retain that independent proposal with only moderate confidence in its superiority, revisit upstream exclusions when cash evidence warrants, and distinguish AI success from shareholder success throughout.